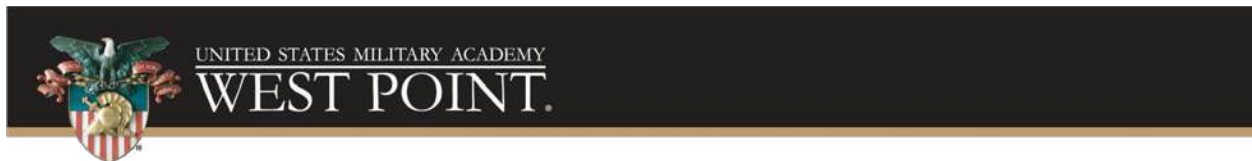




NE300

Fundamentals of Nuclear Engineering



NE300 FUNDAMENTALS OF NUCLEAR ENGINEERING

ABET eCOURSEBOOK

AY2025-2026

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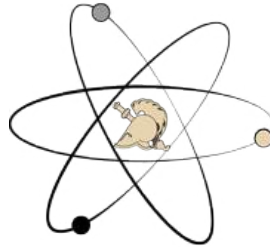
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UNITED STATES MILITARY ACADEMY



Nuclear Engineering Program Course Syllabus

NE300 Fundamentals of Nuclear Engineering

AY 26-1

June 28, 2025

Nuclear Engineering Program Director:

Professor Kenneth Allen, PhD, PE

Course Director:

Major Dave Pert

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1 Course Introduction

Fundamentals of Nuclear Engineering (NE300) is the introductory course to both the nuclear engineering major (NEN1) and nuclear engineering core engineering sequence (NECES). Nuclear engineering is a diverse field of study which applies radiation and radioactive material in ways which benefit mankind. Among the benefits of nuclear engineering are commercial power production, medical diagnoses and treatments, spacecraft power generation, radiation shielding for biological protection, nuclear weapons for strategic national defense, and myriad types of cutting-edge research. In order to solve complex, technical problems found in these areas, one must first grasp the fundamental principles which govern them.

The NE300 course is designed to establish a foundation for such understanding through study of nuclear physics, radiation interactions with matter, nuclear reactor theory, radiation detection, and various nuclear engineering applications. Students who successfully complete NE300 will possess the requisite tools to explore topics such as *Nuclear Reactor Engineering* (NE355), *Nuclear Weapons Effects* (NE450), *Nuclear Instrumentation and Shielding* (NE452), and *Radiological Safety* (NE474).

2 Conduct and Expectations

2.1 What can I expect from the Course Director?

The course director will provide engaging and relevant lecture material, resources, and evaluation tools to build and assess cadets' mastery of fundamental nuclear engineering concepts and skills as outlined in this syllabus. While the basics of nuclear engineering are well-established, technology in our field grows daily. Accordingly, the course director will introduce new and cutting-edge discoveries to the curriculum as points of discussion throughout the semester. Cadets are encouraged to bring these topics to the attention of their instructor or the course director as they arise.

The course director, with the guidance of the program director, will select points of emphasis to evaluate throughout the course using board problems, writs, WPRs, and the TEE. You should expect timely and objective assessment of your performance on graded events with the complementary goals of teaching you how analyze and synthesize solutions to complex problems while continually building your knowledge of fundamental nuclear engineering topics. These points of emphasis support areas in which graduates have shown need for improvement in the past and serve to underpin the accomplishment of Course Objectives.

2.2 What does the Course Director expect from me?

Cadets enrolled in NE300 are expected to execute their duties, both implicit and explicit, with the intellect and deportment of the commissioned officers they aspire to soon become. Always live the Cadet Honor Code and conduct yourself in a professional manner.

You are expected to arrive at each class meeting wholly prepared to discuss the day's material, including any pertinent questions you may have over the assigned reading or study

problems. During class, you should remain engaged in the discussion at hand by taking meaningful notes, constructively contributing to lecture through positing or answering questions as applicable, putting forth genuine thought and effort at boards, and confidently delivering briefings or presentations as required.

As professionals and future commissioned leaders of character, you are expected to respect the instructor and the time allotted for each class meeting. As such, refrain from using laptops, iPads, etc. to view unrelated material during class. Leave cell phones, materials unrelated to NE300, and any other potential distractors outside the classroom. Stand and move to the rear of the classroom if you become drowsy. Quietly excuse yourself and return quickly from any necessary trip to the latrine.

2.3 Sexual Harassment Policy

Prevention and elimination of sexual harassment and sexual assault is the number one priority at the United States Military Academy. The Army "I. A. M. STRONG" campaign combats sexual harassment and sexual assault by engaging all Soldiers to "Intervene," "Act," and "Motivate" others in preventing sexual assaults before they occur. Sexual harassment and sexual assault are reprehensible, hurt other team members, and wound our Army. It is a betrayal of the trust inherent in serving in the Profession of Arms. Your instructor is dedicated to fostering a positive SHARP environment.

If you need to speak to anyone about SHARP, contact your instructor, the unit representative for Cadets Against Sexual Harassment and Assault (CASHA), or one of the following resources:

- West Point Sexual Assault 24/7 Helpline: (845) 659-7467
- DoD Safe Helpline: (877) 995-5247
- SHARP Victim Advocate: (845) 938-3532 or (845) 938-1938.

2.4 Chain of Command

Below is the chain of command for NE300: Fundamentals of Nuclear Engineering. The Department of Physics and Nuclear Engineering has an open-door policy. Feel free to address questions, problems, or concerns through the chain of command.

- **Your instructor, NE300**

Name: _____

Room #: _____, Phone #: _____

- **Course Director, NE300**

MAJ Dave Pert

BH472A, x4297

- **Program Director, Nuclear Engineering**

Prof. Kenneth Allen

BH472B, x3548

- **Department Head, Physics and Nuclear Engineering**

COL Peter Chapman

BH480B, x3901

3 Course Objectives

Upon successful completion of NE 300, a Cadet should be able to accomplish the following:

1. Define and discuss the various fundamental particle types.
2. Comprehend the various nuclear decay processes, types of nuclear reactions, and wave-particle duality.
3. Apply appropriate conservation laws to solve quantitative problems involving radioactive decay, nuclear reactions, mass-energy equivalence, and reaction energy.
4. Solve quantitative problems radiation interactions with matter, to include: number density, neutral and charged particle attenuation, scattering interactions, neutron lethargy, and reaction rates.
5. Explain the basic function of a nuclear reactor including the fission process, the neutron life cycle, and the nuclear fuel cycle.
6. Solve quantitative problems involving nuclear reactor theory to include: fuel enrichment, multiplication factor calculations, fuel consumption rates, conversion and breeding, and system efficiency.
7. Comprehend and communicate professional and ethical knowledge required of nuclear engineers when solving contemporary nuclear engineering problems with particular focus on relevant military applications of nuclear energy.
8. Search out relevant scholarly publications and other credible published literature to synthesize and communicate (in written and/or oral form) the significance of a body of work to a specified topic in nuclear engineering.

These objectives are part of a broader set of Nuclear Engineering Program Educational Objectives which satisfy both the Accreditation Board for Engineering and Technology

(ABET) requirements and the USMA Academic Program Goals (APGs).

3.1 Nuclear Engineering Program Educational Objectives (NE PEOs)

The nuclear engineering program educational objectives describe the career and professional accomplishments that the NE program prepares Cadets to achieve in a few years after graduation. The NE PEOs are designed to meet the needs of our primary constituency: the United States Army. The NE student outcomes are what Cadets are expected to know and be able to do upon graduation. They are given below.

1. As Army leaders of character, graduates solve complex, multi-disciplinary problems for the Army and the Nation.
2. Graduates demonstrate the necessary leadership and teamwork skills to lead technical combined-arms Army teams solving problems in support of the Army mission.
3. Graduates are prepared to provide appropriate nuclear and radiological engineering expertise to the Army.
4. Graduates communicate effectively, orally and in writing; providing clear instructions to subordinates and clear feedback to supervisors.
5. Graduates and continue to develop and grow their intellect, professionalism, and character – as Army officers and as engineers.

3.2 Course Objectives Mapped to ABET Student Outcomes

The Nuclear Engineering Program Student Outcomes describe what Cadets are expected to know and be able to do at the time of graduation. The student outcomes relate to

the skills, knowledge, and behaviors that a Cadet acquires during progression through the Nuclear Engineering program, as defined by ABET. Below are the stated NE program outcomes (1 - 7) which align directly with ABET accreditation student outcomes.

Cadets must attain:

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics;
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors;
3. an ability to communicate effectively with a range of audiences;
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental , and social contexts;
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives;
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions; and,
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Student outcomes 1, 3 and 7 are best supported by NE300. These outcomes are shown in the matrix below in Table 1 along with the embedded indicators used to assess the outcomes against the establish rubrics.

Embedded Indicators	1	2	3	4	5	6	7
Term End Exam	X						
Problem Sets	X						X
Written Partial Review	X						X
FNEK Writ			X				X
Group Project	X		X		X		X

Table 1: NE300 course objectives mapped to ABET student outcomes.

3.3 Course Objectives Mapped to Academic Program Goals

The NE 300 course is used to assess Cadet attainment of the USMA Academic Program Goals (APGs) and What Graduates Can Do (WGCD) statements for the Core Engineering Sequence (CES); the NE ABET SOs are nested with the APGs/WGCD statements.. The APGs and WGCD statements introduced and/or reinforced by NE300 are listed below.

- 2.1 (Critical Thinking and Creativity) Identify the essential aspects of a situation and ask relevant questions.
- 2.4 (Critical Thinking and Creativity) Reason both quantitatively and qualitatively.
- 2.5 (Critical Thinking and Creativity) Think innovatively and accept risk to pursue solutions in the face of ambiguity.
- 5.1 (Science, Technology, Engineering, and Mathematics) Apply mathematics, science, and computing to model devices, systems, processes or behaviors.
- 5.4 (Science, Technology, Engineering, and Mathematics) Apply an engineering design process to create effective and adaptable solutions.
- 7.4 (Disciplinary Depth) Synthesize knowledge and concepts from across their chosen disciplines.
- 7.5 (Disciplinary Depth) Contribute disciplinary knowledge and skills as a part of a collaborative effort engaging challenges that span multiple disciplines.

The embedded indicators associated with each WGCD statement are shown in Table 2. In the table, a letter “I” indicates the goal is introduced in NE 300 while a letter “R” indicates that NE 300 reinforces the goal.

Embedded Indicators	2.1	2.4	2.5	5.1	5.4	7.4	7.5
Term End Exam		R				R	
Problem Sets		R	I	R			
Written Partial Review		R				R	
FNEK Writ		R					
Group Project	R	R	R			R	R

Table 2: NE300 course objectives mapped to Academic Program Goals.

4 Course Requirements and Grading

4.1 Course Point Allocation

The point allocation by event is shown in Table 3. Of note, several events are identified as requirements for course credit. If a cadet fails or refuses to complete any or all these events, a grade of 'F' or 'NC' will be awarded at the end of the term regardless of points earned, current letter grade, or course standing.

A short synopsis of each graded event is given below. While these serve as guidelines for planning purposes, minor changes may be made at the discretion of the course director upon approval of the program director.

Events	Possible Marks	Fraction of Total [%]	Required for Course Credit
Written Partial Reviews	500 (2 @ 250)	31	Y
Term End Exam	500	31	Y
Problem Sets	300 (6 @ 50)	18	N
Group Project	120	8	Y
Instructor Marks	100	6	N
FNEK Writ	80	5	N
Total	1600	100	

Table 3: NE300 course point allocation by event.

Term End Exam (TEE): The term end exam is a comprehensive reinforcement event which evaluates key learning objectives from each of the five blocks of instruction. Problems posed on TEE are both qualitative (explain, sketch, define) and quantitative (calculate, derive, solve) in nature. Authorized references are the NERD-P, NE300 Reference Card, and an approved calculator.

Problem Sets: The six assigned problem sets in NE300 may serve as reinforcement and/or extension events. These problems may be more complex than text problems and may require synthesis of multiple concepts, principles, and ideas. Each set is worth 50 marks.

Due dates and times will be identified with each assignment.

Collaboration on problem sets is anticipated and should be used to refine one's understanding of the learning objectives encompassed by the problems as noted in each assignment. Accordingly, documentation is required IAW the Documentation of Academic Work guidebook and the memorandum "You and the Department of Physics and Nuclear Engineering." Both references are available on Canvas.

Written Partial Review (WPR): The two WPRs are reinforcement events which assess key learning objectives from corresponding blocks of instruction. Problems posed on the WPRs will be both qualitative (explain, sketch, define) and quantitative (calculate, derive, solve) in nature. Authorized references are the NERD-P, NE300 Reference Card, and an approved calculator.

Fundamental Nuclear Engineering Knowledge (FNEK) Writ: The FNEK writ is administered to assess retention of key nuclear engineering facts and equations which should be committed to memory and available for recall without reference materials. Accordingly, the FNEK writ is administered without references or calculators permitted. No make-up FNEK writs are given. Cadets with authorized absences are excused without penalty and their total marks available are reduced accordingly.

Group Project: The group project event assesses each cadet's ability to research, process, and effectively communicate technical information verbally in a clear, concise manner. Each student will work as a member of a group to plan, develop, and deliver one 15-minute-long presentation on a chosen topic. These presentations are intended to serve as a thorough introduction to practical applications of Nuclear Engineering. An established rubric is provided prior to the presentation.

Bonus Points: Bonus marks may be offered at the instructor's discretion. The total bonus marks awarded to any one cadet may equal no more than 16 (1% of course grade). If assistance is received in completion of a bonus opportunity, documentation must conform

to that in the DAW. No marks are awarded for copied bonus problems.

4.2 Grade Cutoffs

Each cadet will earn numerical marks (points) in accordance with his or her performance on each graded event throughout the term. Based on the total percentage of earned marks in the course, Cadets will be awarded a letter grade. The maximum cutoff percentage for each letter grade is shown in the Table 4.

Grades are usually available in CIS/Canvas as soon as they are posted by the instructor. In the case of major graded events (WPRs and TEE), grades will be visible to cadets after approval by the head of the Department of Physics and Nuclear Engineering.

A+	94.0%	C+	73.0%
A	90.5%	C	69.5%
A-	87.0%	C-	66.0%
B+	83.5%	D	60.0%
B	80.0%	F	< 60.0%
B-	76.5%		

Table 4: NE300 letter grade cutoffs by percentage

4.3 Late Assignment Policy

A Cadet who believes he/she will fail to meet a suspense for submission of a graded assignment will inform the instructor prior to the missed deadline. Failure of a Cadet to notify the instructor of such an event may result in a negative Cadet Observation Report (COR). **Additionally, turning in homework late without prior notification to the instructor will result in a 25% reduction per calendar day.**

5 Course Resources

5.1 Course Text

The textbooks used for this course are:

1. (Required) Shultis, J. Kenneth and Richard E. Faw. *Fundamentals of Nuclear Science and Engineering*, 3rd ed., CRC Press, 2017.
2. (Optional) Knolls Atomic Power Laboratory. *Nuclides and Isotopes: Chart of the Nuclides*, 17th ed., Bechtel National, Inc., 2010.

A variety of supplemental texts may be referenced throughout the course at the discretion of the instructor. As required, notes, photocopies, or online access will be provided for these materials.

5.2 Course Outline

All lesson assignments are found in Lesson Summaries (published separately). For each assignment, Cadets may be tasked to **Scan**, **Read**, or **Study** certain material(s). Cadets are required to understand the learning objectives, study and know the lesson notes and vocabulary, and are provided a list of additional problem(s) that they can choose to do for practice and reinforcement of concepts. Specific definitions of these tasks and items are given below.

1. **Scan** tasks are those which require Cadets to become familiar with a given set of material prior to a lecture. Often this material will be review from NE300 or fundamental knowledge. The cadet should be able to understand how it supports concepts in follow on NE courses.

2. **Read** tasks are those which require Cadets to read a given set of material prior to a lecture in order to gain insights which will facilitate better understanding of the material as well as any “study” requirements. Reading assignments are typically given to provide depth, clarity, and examples beyond those provided during lecture and support specific learning objectives.
3. **Study** tasks are those which require Cadets to read and recall certain material in order to demonstrate the ability to comprehend, analyze, and synthesize material by solving assigned problems and/or discussing and evaluating common applications.
4. **Lesson Learning Objectives (LOs)** are key principles which guide Cadets’ study and preparation for each lesson. Studying nuclear engineering requires much more than merely reading text(s). Contained within each text reading assignment are derivations and example problems. After understanding the LOs from the text, Cadets review model problems and attempt to solve study problems. The LOs use two types of highlighted verbs which are defined in the following paragraphs.
 - (a) **List, explain, sketch, draw,** and/or **describe** are used to indicate an objective which is evaluated by short-answer questions. In response to such questions, Cadets must correctly state pertinent facts about a given situation and then provide a conclusion based on those facts. Cadets may be required to sketch a graph or process to help illustrate the conclusion.
 - (b) **Calculate, construct, solve,** and/or **derive** are used to indicate an objective which is evaluated by methods used above or by performing algebraic, diagrammatic, and/or numerical calculations. Cadets may be required to synthesize multiple concepts in order to solve a problem.

5.3 Calculators

The only calculators authorized for use during in-class, graded events are those approved for use on the Fundamentals of Engineering Examination. A current list is available at [NCEES Website](#). If in doubt, use the issued TI-30XIIS calculator.

5.4 Canvas

Course material for NE300 can be accessed online using the Canvas interface. Canvas will provide you valuable information about assignments, grades, etc. and is updated frequently. You will find all course administrative publications (i.e., syllabus, course outline, course schedule, reference cards, NERD-P, and other supplementary materials), lesson information, links to other related materials, and more on the NE300 Canvas site. Instructions for accessing Canvas will be provided by your instructor.

5.5 Examination Materials

A sterile NE300 Reference Data Card and NERD-P will be provided to you for each examination. The reference card and NERD-P contain the equations, data, and charts that you will need as reference for all examinations.

It is recommended you study, do your homework, and do your in-class board work with your copy of the reference cards and the NERD-P. We encourage you to make notes (identify parameters, write conditions, write units, etc.) on your reference cards and NERD-P as you study. This will provide you a greater degree of familiarity with these materials when you receive a clean copy for the examinations.

5.6 Additional Instruction (AI)

While not required, Cadets are encouraged to schedule additional instruction (AI) with their instructor. The primary purpose of AI is to provide answers to specific questions a Cadet may have after attempting to work a problem set on their own. Additional instruction may also be scheduled to assist in clarifying concepts from the reading or answers from returned, graded assignments.

Once an AI appointment is requested and confirmed by the instructor, it becomes that Cadet's place of duty unless rescheduled in advance. The Cadet should come to AI prepared with detailed questions in order to facilitate efficient use of AI time.

6 Documentation of Academic Work (DAW)

The purpose of documentation is to give credit where credit is due by delineating original work from portions which originated elsewhere. Proper documentation identifies the source(s) of borrowed ideas, quotations, and assistance in the preparation of assignments (DAW, paragraph II.A.1). Proper documentation reflects good scholarship and personal integrity.

Any homework assignment prepared outside of class that Cadets turn in for grade must be documented in accordance with the DAW and must include a cover sheet. Documents submitted with improper documentation and/or missing a signed cover sheet will be returned back to the cadet ungraded for clarification. Furthermore, improperly documented assignment submissions will be marked “late” in accordance with the policy for late assignments until the assignment is resubmitted.

Collaboration is highly encouraged; however, each Cadet must document all substantive assistance received. Endnotes should clearly distinguish the author’s personal effort from that of his/her collaborators, such as in the following example:

Jerrad P. Auxier, D-1, '20; Matthew T. Cook, D-2, '19; George K. Schweitzer, H-3, '21; collaboration with the author, verbal and written discussion, West Point, NY, 22 September 2018. We worked through the derivation of the flux for the diffusion equation as an informal group. CDT Cook set up the general equation. CDT Schweitzer added the proper form of the Laplacian. CDT Auxier determined the correct source boundary condition.

Properly documented copying is subject to an academic cut of no less than 50% of the achieved score for that portion of the work that is copied. For bonus problems, copying will result in no credit for the problem(s) copied.

If you are ever in doubt about how to correctly document your work, refer to the most

current edition of the *Documentation of Academic Work* (DAW) guidebook published by the Dean's office. The DAW is also posted on the course Canvas site. If still uncertain after consulting the DAW, ask your instructor for further guidance.

6.1 Documentation of Previous Work

If a cadet reuses or resubmits work in accordance with course policies, he or she must still take efforts to avoid what is known as “self-plagiarism”: the reuse of your own previous work in another venue while presenting it as new work. Cadets are required to cite and document that work like any other source cited or assistance received IAW the *Documentation of Academic Work* [1].

6.2 Course Common Knowledge

Documentation is not required for ideas and materials regarded as common knowledge. A list of such material for the NE300 course follows.

1. Any equation, numerical value, or diagram listed on the NE300 Reference Data Card or contained within the Nuclear Engineering Reference Data Pamphlet (NERD-P).
2. Any equation or numerical value found in the required course textbook. The figures and text contained within the textbooks are not considered common knowledge and must be appropriately documented.
3. Material provided by the instructor, to include: board notes taken by the Cadet during a formal class meeting; example problem solutions worked out in a formal class meeting; transcribed discussions, explanations, or definitions given during a formal class.

4. Any discussions with an NE300 instructor are considered common knowledge; discussions with other faculty NOT identified as an NE300 instructor must be documented.

Materials and ideas outside of those directly listed above are not common knowledge and must be documented IAW the DAW. Furthermore, the use of any diagram outside of the NERD-P is not common knowledge and must be documented. Finally, any notes copied from another Cadet (e.g., CDT Smith is absent during lesson 3 and copies CDT Jones' notes) are not considered common knowledge and must be cited appropriately.

7 Works Cited

- [1] Office of the Dean Academic Affairs and Registrar Services. *Documentation of Academic Work*. United States Military Academy, 2017.

ENCLOSURES

Enclosure 1: NE300 AY26-1 Schedule and Calendar

Enclosure 2: NE300 AY26-1 Learning Objectives

Enclosure 3: NE300 AY26-1 Lesson Outlines

Enclosure 4: NE300 AY26-1 RDC



UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION I

COURSE ASSESSMENTS



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB A

AY2026 COURSE ASSESSMENT

NE 300 Fundamentals of Nuclear Engineering
AY 26-1



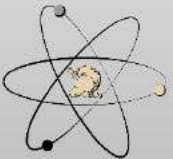
ARMY
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NUCLEAR ENGINEERING

MAJ Dave Pert
NE300 Course Director

Agenda

AY 26-1 NE300 CAT Briefing

- Previous Guidance
- NE Program Student Outcomes
- NE300 Assessment Process
- AY26 Course Assessment
- Recommended Changes for AY27



AY25 Program Director's Recommendations (for AY26) (1 of 3)

Adopt CAT recommendations and

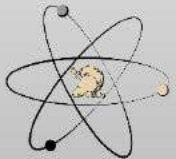
- Properly complete ABET record year data collection
- Integrate new junior and senior faculty into program
- Continue to work on assessment tools that correctly identify any lack of fundamental skills to include time management (i.e. late turn-ins of homework)
- Continue assessment SO (1)-(7)

Standardization of briefings

- Continue standardized briefs and reports for NE-specific topics
- Continue use of Overleaf

Effective use of NE labs and spaces

- Finish cleaning and organizing subcrit, BH50 and accelerator spaces
- Integrate PROLAB into curriculum (especially NE355)
- Continue registration for cluster (VM) with RSIC and meet all compliance for use
- Work with IT for replacement for VMware



AY25 Program Director's Recommendations (for AY26) (2 of 3)

Complete updates and integration of text for NE450

Integrate new version of *Nuclear Engineering Design Supplement*

- Use document for NE350 and NE495/496 and note errors
- Complete instruction for Project Management using document

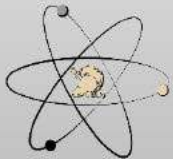
Sustain internal and external relations:

- Include ANS Club, 1 activity per month
- Trip sections for courses (STERIS)

Continue to develop and engage lab tech in the program

Other Items:

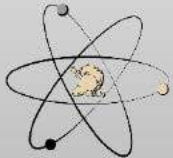
- Execute ABET advisory board
- Improve a lab bench stock of materials to order and have on hand to assist with the construction and development of lab, research and capstone projects
- Continue to improve FE exam preparation between NE400 and NE496



AY25 Program Director's Recommendations (for AY26) (3 of 3)

NE Working Group Topics

- Shift leadership responsibilities between senior faculty to include new AP
- ABET record year
- Work on future manning for near and sustained term with department leadership
- Continue work with OCE and other outside organizations for research and manning
- Continue to improve use of advanced computational tools such as Attila4MC throughout the program
- Annual update of internal and external websites – ensure compliance with ABET APPM

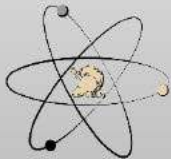


Program Director's Areas of Focus

Academic Year 2026

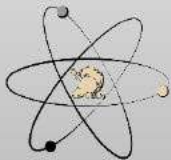
NE300

- ✓ NE Outcomes are: **1, 3, 7**
- ✓ Implement CAT recommendations **Partially Complete**
- ✓ Improve course content and pre-work all homework **Complete**
- ✓ Standardize Briefs **Complete**
- ✓ Use Overleaf for course material **Partially Complete**
 - ✓ Migrated major graded events to Gradescope



NE Student Outcomes

1. **an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics**
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. **an ability to communicate effectively with a range of audiences**
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. **an ability to acquire and apply new knowledge as needed, using appropriate learning strategies**



NE300 Course Concept Map (AY 26)

FUNDAMENTALS

Nuclear Fundamentals

Fundamental Concepts

Modern Physics Concepts

Atomic and Nuclear Models

Nuclear Energetics

Nuclear Reactions

Types of Radioactive Decay

Binary Nuclear Reaction Kinematics

Neutron-Induced Binary Reactions and Fission

Characteristics of Radioactive Decay

Binary Nuclear Reaction Threshold Energy

Fusion

APPLICATIONS

Detection Applications

Neutral Particle Attenuation I: Photons

Neutral Particle Attenuation II: Neutrons

Charged Particle Interactions

Detection and Measurement of Radiation

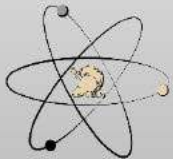
Reactor Applications

Neutron Considerations in a Nuclear Reactor

Neutron Life Cycle in a Thermal Reactor

Nuclear Power Reactor Overview

The Nuclear Fuel Cycle

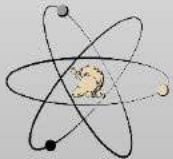


4-Step Evaluation & Assessment Process

1. Identify the connections between:
 - Course block objectives and student outcomes
 - Lesson learning objectives and course block objectives by developing a concept map
2. Select multiple embedded indicators to measure against established rubrics that will:
 - Assess the cadets' retention of previously learned concepts and procedures (reinforcement-level instruments)
 - Assess the cadets' ability to synthesize multiple concepts and procedures (extension-level instruments)
3. Assess the level of attainment of student outcomes based on the population distribution of cadet performance on the embedded indicators as:

<u>Descriptor</u>	<u>Color Code</u>	<u>Likert Rating</u>	<u>Criteria</u>
Excellent	Black	5	most frequent grade is A and B, no D or F
Good	Green	4	most frequent grade is A and B, some D or F
Acceptable	Yellow	3	most frequent grade is C or better, with < 20% D or F
Marginally Acceptable	Orange	2	most frequent grade is C or better, with at least 20% D or F but < 40% D or F
Unacceptable	Red	1	most frequent grade is D or F

4. Analyze results and propose recommendations for change if necessary or needed



Evaluation Instrument Sets

Reinforcement Tools

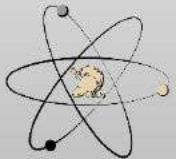
Know the base-line knowledge outlined in the lesson learning objectives, lesson vocabulary, and lesson notes.

1. *Homework and Instructor Writs*
2. *Written Partial Reviews (WPR)*
3. *Term End Examination (TEE)*

Extension Tools

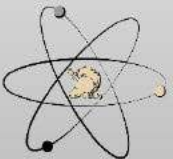
Synthesize multiple concepts and principles beyond reinforcement demonstrating a higher understanding of the material. Requires creative and critical thinking to solve more complex problems.

1. *Board Problems (Coach classes)*
2. *Extension Problems (EP)*
3. *Research Paper*



Embedded Indicators

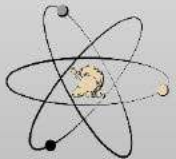
Embedded Indicators (Tools)	NE Student Outcomes						
	1	2	3	4	5	6	7
Problem Sets	X	X					X
FNEK Writ							X
WPRs	X					X	
Practical Application			X	X	X		X
TEE	X					X	



Assessment Instruments

(for each student outcome)

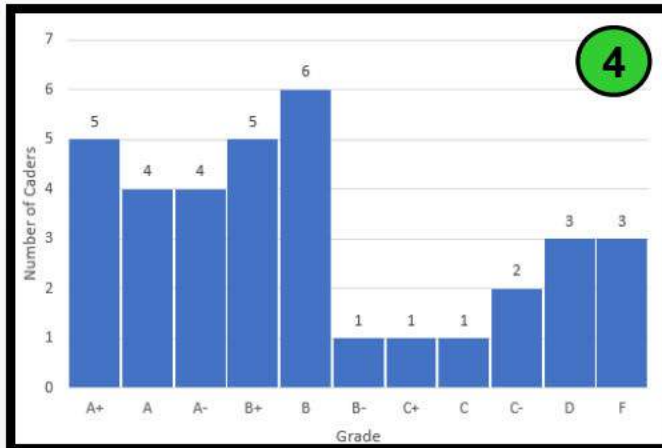
- SO 1 – TEE Problem 1: Neutron Lifecycle
– (90 marks)
- SO 3 – Practical Application Presentation
– (120 marks)
- SO 7 – Problem Set #4
– (50 marks)



NE Student Outcome Assessment – Embedded Indicators

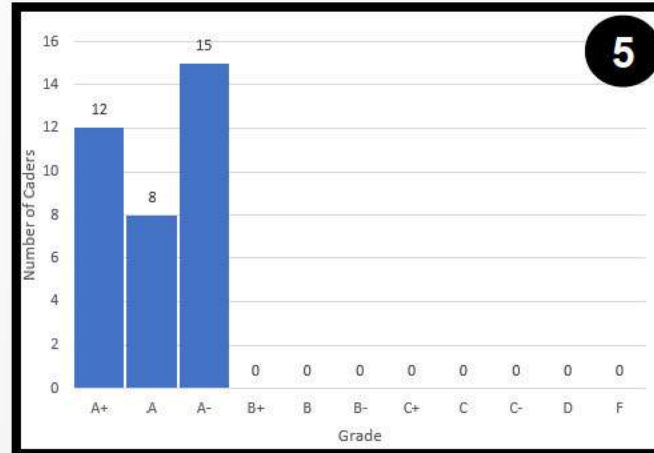
SO 1: TEE Q2 (90)

25-2



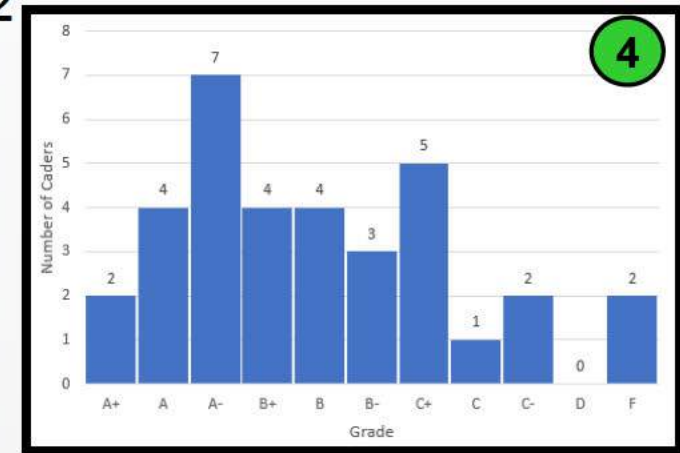
SO 3: PA Brief (120)

25-2

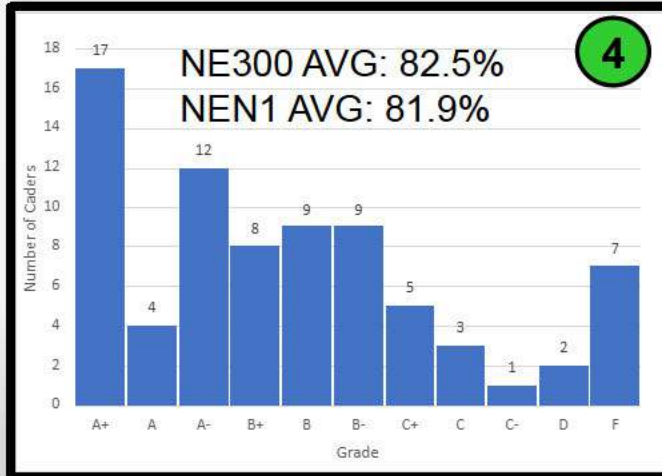


SO 7: Problem Set 4 (50)

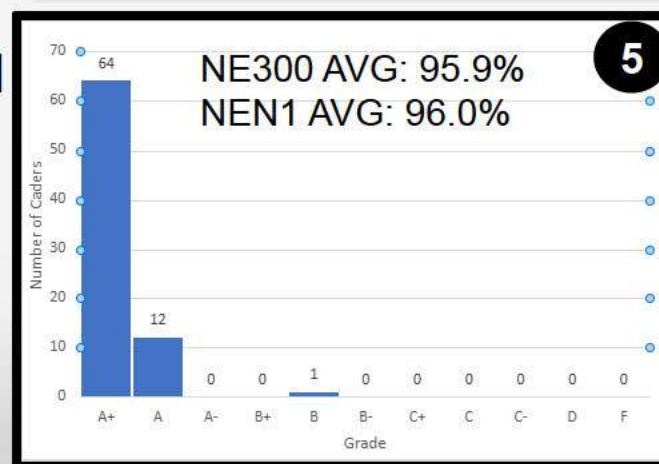
25-2



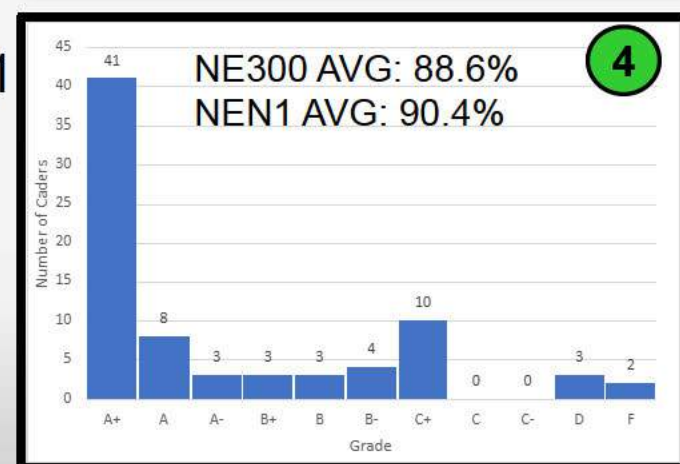
26-1



26-1



26-1



AY 25-2: 0/35 Cadets NEN1 majors ----- AY 26-1: 23/77 Cadets NEN1 majors



NE Student Outcome Assessment – Cadet Surveys (AY 25)

Question	SA	A	N	D	SD	Responses	Mean	S.D.
I am confident in my ability to identify, formulate, and solve engineering problems by applying principles of engineering, science, and mathematics (NE Outcome 1).	20	46	7	1	0	74	4.15	0.63
I am confident in my ability to communicate effectively with a range of audiences (NE Outcome 3).	27	40	6	1	0	74	4.26	0.66
I am confident in my ability to acquire and apply new knowledge as needed, using appropriate learning strategies (NE Outcome 7).	26	45	3	0	0	74	4.31	0.55

NE300 Course Director's Assessment – Assessments for AY25

Student Outcomes	NE300	Indicators	Surveys	CD
1. Solve problems	X	4.00	4.15	4.00
2. Engineering design				
3. Communicate effectively	X	5.00	4.26	4.50
4. Ethical responsibilities				
5. Function on team				
6. Conduct experiments				
7. Apply knowledge	X	4.50	4.31	4.00

Redbook Entry

NE300 (Version: 2022 1) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
NE300	FUNDAMENTALS OF NUCLEAR ENGR	2022	1	Physics and Nuclear Engineering	3.0 (BS=0.0, ET=3.0, MA=0.0)

SCOPE

This course provides the student with an understanding of the fundamental physical principles involved in radioactive decay, radiation interaction with matter, nuclear fission, and nuclear fuel material properties. The course covers neutron interactions with matter, fission, neutron lifecycle, neutron moderation, and reactor criticality. This course is essential for the nuclear engineer and is an excellent choice for the applied scientist.

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

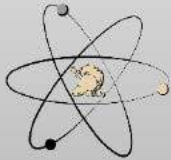
SPECIAL REQUIREMENTS:

None

NE300 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
CO REQUISITE	PH202	2007	2	1	Y
	PH252	2007	2	2	Y
	PH205	2016	1	3	Y
	PH255	2016	1	4	Y
	PH201X	2022	1	5	Y
	PH201	2023	1	6	Y
	PH251	2023	1	7	Y
	PH275	2025	1	8	Y

No changes to the Redbook suggested.

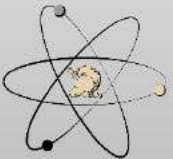


NE300 CAT Briefing AY26-1

Department of Physics & Nuclear Engineering, USMA

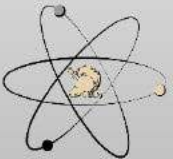
AY26-1 Recommended Changes (For AY-27)

- Sustain live demos (cloud chamber, detectors)
- Sustain iRDC
- Sustain Gradescope for major graded events
- Recommend Gradescope for problem sets
- Recommend simplification of practical application rubrics
- Recommend adding critical energy of fission and a fusion problem to problem sets

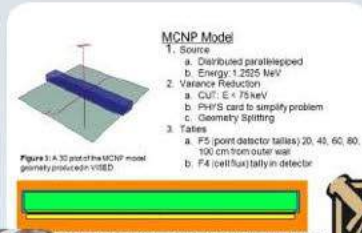


Resource Requirements

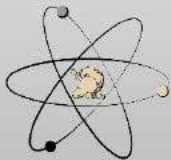
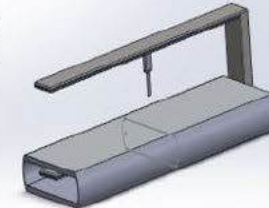
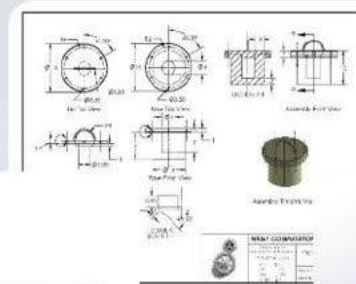
- New NERD-P order for AY27 (50 exam copies, 120 issue copies)



Questions?



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NE300 CAT Briefing AY26-1

Department of Physics & Nuclear Engineering, USMA



UNITED STATES MILITARY ACADEMY
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TAB B

AY2025 COURSE ASSESSMENT

NE300: Fundamentals of Nuclear Engineering AY25



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NUCLEAR ENGINEERING

Agenda: AY25 NE300 CAT Briefing

- Previous Guidance
- NE Program Student Outcomes
- NE300 Assessment Process
- AY25 Course Assessment
- Recommended Changes for AY26 and beyond

AY24 Program Director's Recommendations (for AY25)

(1 of 3)

Adopt CAT recommendations and

- Fully use Canvas to its highest potential
- Continue to work on assessment tools that correctly identify any lack of fundamental skills to include time management (i.e. late turn-ins of homework)
- Continue assessment SO (1)-(7)

Standardization of briefings

- Continue standardized briefs and reports for NE-specific topics
- Continue use of Overleaf

Effective use of NE labs and spaces

- Finish cleaning and organizing subcrit, BH50 and accelerator spaces
- Continue Orientation for 2nd class for BH347 as collaborative space
- Continue Orientation for 1st class for BH66B as collaborative space
- Continue registration for cluster (VM) with RSIC and meet all compliance for use
- Work with IT for replacement for Vmware

AY24 Program Director's Recommendations (for AY25)

(2 of 3)

Complete updates and integration of text for NE450

Update *Nuclear Engineering Design Supplement*

- Include format for a standard template for peer-reviewed journal and conference poster and presentation
- Change “restrictions” to “standards”
- Change IPR2 and remove “new start”

Sustain internal and external relations:

- Include ANS Club, 1 activity per month
- Trip sections for courses (STERIS)

Continue to develop and engage lab tech in the program

Other Items:

- Execute ABET advisory board
- Improve a lab bench stock of materials to order and have on hand to assist with the construction and development of lab, research and capstone projects
- Continue to improve FE exam preparation between NE400 and NE496

AY24 Program Director's Recommendations (for AY25)

(3 of 3)

NE Working Group Topics

- Divide leadership responsibilities between senior faculty
- Begin preparations for ABET record year
- Work on future manning for near and sustained term with department leadership
- Continue work with OCE and other outside organizations for research and manning
- Continue to improve use of advanced computational tools such as Attila4MC throughout the program
- Annual update of internal and external websites – ensure compliance with ABET APPM

NE Program Director's Assessment – Assessments for AY25

Student Outcomes	NE300	NE350	NE355	NE361	NE400	NE450	NE452	NE461	NE474	NE495/6
1. Solve problems	X		X	X			X		X	
2. Engineering design		X		X				X		X
3. Communicate effectively	X						X		X	X
4. Ethical responsibilities					X	X			X	X
5. Function on team		X	X				X	X		X
6. Conduct experiments			X				X			
7. Apply knowledge	X		X		X	X				X

No change from AY22



NE Student Outcomes

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. An ability to communicate effectively with a range of audiences
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies



Program Director's Areas of Focus AY24

NE300:

- NE Outcomes are: 1, 3, 7
- Improve course content and pre-work all homework – Complete
- Implement CAT recommendations – Complete
- Standardize briefs – Complete
- Use Overleaf for course material – Complete

Recommended Changes for AY 25-1

Administrative

- Implement Block Objectives/Scrub Lesson Objectives Complete (AY 25-1)
- Add 6th PS/Remove Rad PE Complete (AY 25-1)
- Add PWR/BWR to FNEK concepts card – Complete (AY 25-1)
- Review and Update NERD-P – Complete (AY 25-1)
- Add Example Problems for each lesson to CANVAS – Complete (AY 25-1)
- Move CAT Assessments away from averaged/group assignments– In-Progress (AY 25)

Course Content

- Add L.O. concerning specific fusion reactions Complete (AY 25-1)

COURSE	TITLE	CREDIT HOURS
NE300 2022-1	FUNDAMENTALS OF NUCLEAR ENGR	3.0 (BS=0.0, ET=3.0, MA=0.0)
Scope		Offerings
SCOPE This course provides the student with an understanding of the fundamental physical principles involved in radioactive decay, radiation interaction with matter, nuclear fission, and nuclear fuel material properties. The course covers neutron interactions with matter, fission, neutron lifecycle, neutron moderation, and reactor criticality. This course is essential for the nuclear engineer and is an excellent choice for the applied scientist.		2024 2
		2024 8
		2024 9
		2025 1
		2025 2
		2025 3
		2025 8
		2026 1
		2026 2
		2026 8
		2026 9
		2027 1
		2027 2
		2027 8
		2027 9



4-Step Evaluation & Assessment Process

1. Identify the connections between:
 - Course block objectives and program outcomes
 - Lesson learning objectives and course block objectives by developing a concept map
2. Select multiple embedded indicators to measure against established rubrics that will:
 - Assess the cadets' retention of previously learned concepts and procedures (reinforcement-level instruments)
 - Assess the cadets' ability to synthesize multiple concepts and procedures (extension-level instruments)
3. Assess the level of attainment of program outcomes based on the population distribution of cadet performance on the embedded indicators as:
4. Analyze results and propose recommendations for change if necessary or needed

Descriptor	Color Code	Likert Rating	Criteria
Excellent	Black	5	most frequent grade is A and B, no D or F
Good	Green	4	most frequent grade is A and B, some D or F
Acceptable	Yellow	3	most frequent grade is C or better, with < 20% D or F
Marginally Acceptable	Orange	2	most frequent grade is C or better, with at least 20% D or F but < 40% D or F
Unacceptable	Red	1	most frequent grade is D or F



NE300 Course Concept Map (AY 25)

FUNDAMENTALS

Nuclear Fundamentals

Fundamental Concepts

Modern Physics Concepts

Atomic and Nuclear Models

Nuclear Energetics

Nuclear Reactions

Types of Radioactive Decay

Binary Nuclear Reaction Kinematics

Neutron-Induced Binary Reactions and Fission

Characteristics of Radioactive Decay

Binary Nuclear Reaction Threshold Energy

Fusion

APPLICATIONS

Detection Applications

Neutral Particle Attenuation I: Photons

Neutral Particle Attenuation II: Neutrons

Charged Particle Interactions

Detection and Measurement of Radiation

Reactor Applications

Neutron Considerations in a Nuclear Reactor

Neutron Life Cycle in a Thermal Reactor

Nuclear Power Reactor Overview

The Nuclear Fuel Cycle

Embedded Indicators

Embedded Indicators (Tools)	NE Student Outcomes						
	1	2	3	4	5	6	7
Problem Sets	X	X					X
FNEK Writ							X
WPRs	X					X	
Practical Application			X	X	X		X
TEE	X					X	

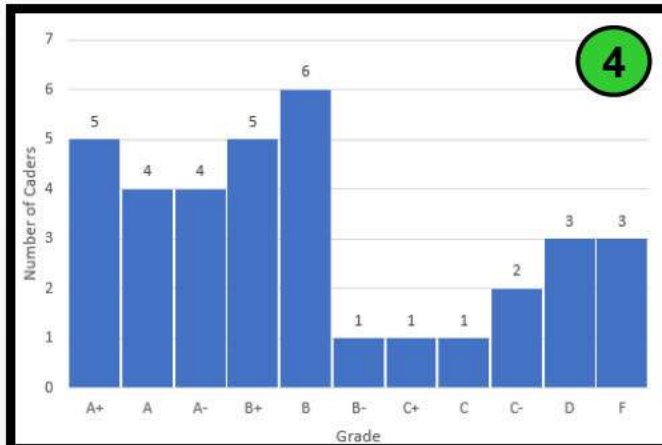
Best Assessment Instrument

- SO 1 – TEE Problem 2: Neutron Lifecycle
– (90 marks)
- SO 3 – Practical Application Presentation
– (120 marks)
- **SO 7 – Problem Set #4 (Change from previous assessments)**
– **(50 marks)**

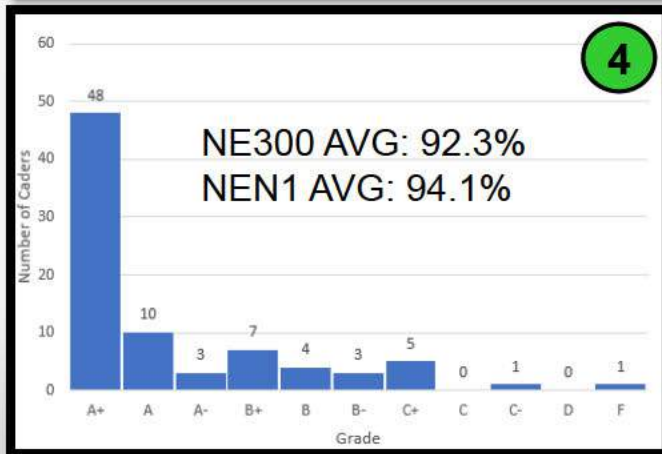
NE Student Outcome Assessment – Embedded Indicators

SO 1: TEE Q2 (90)

25-2

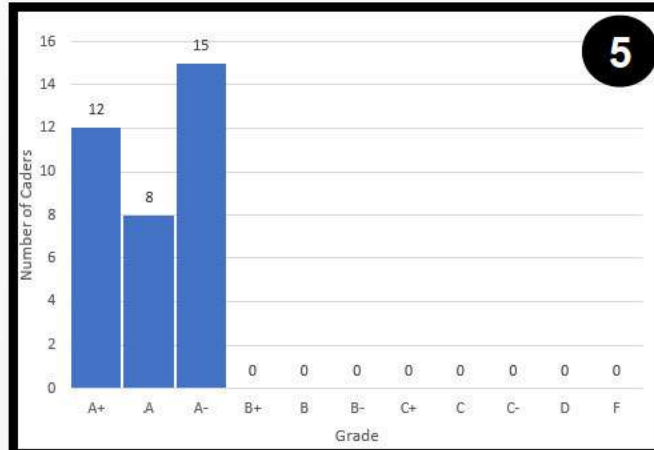


25-1

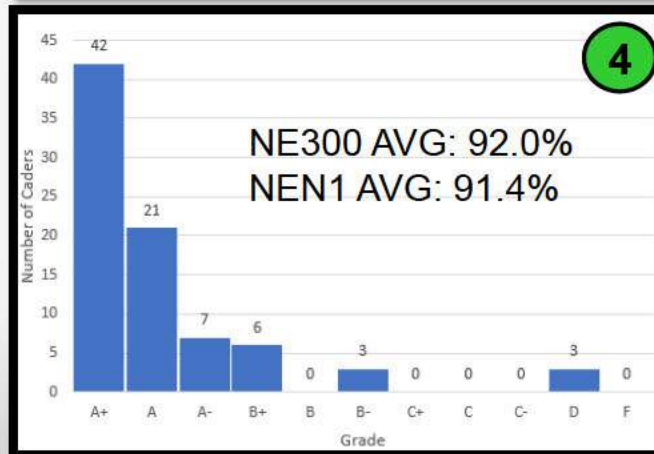


SO 3: PA Brief (120)

25-2

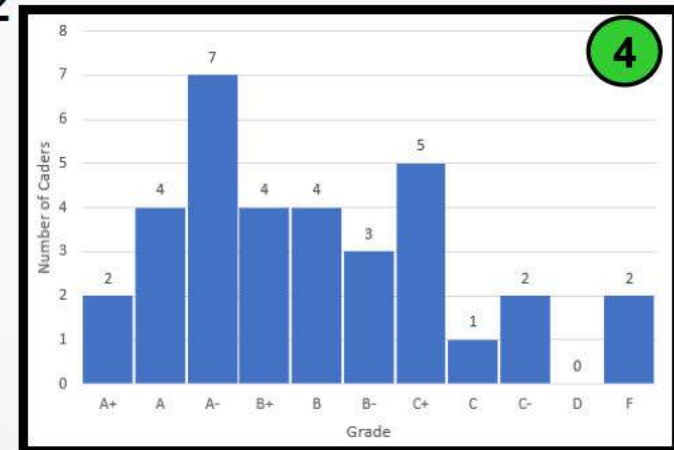


25-1

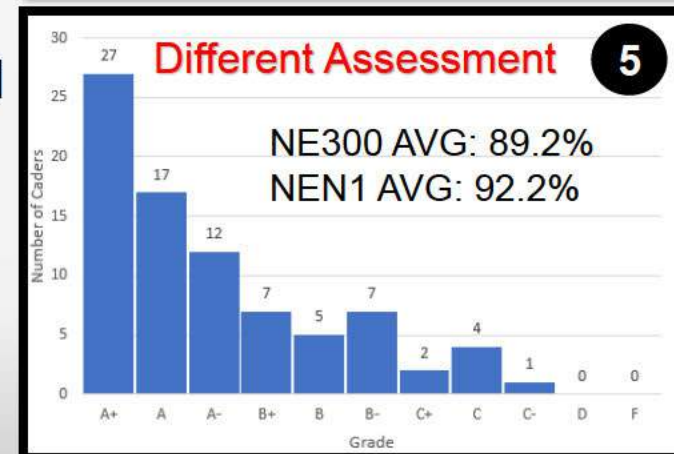


SO 7: Problem Set 4 (50)

25-2



25-1



AY 25-2: 0/35 Cadets NEN1 majors ----- AY 25-1: 28/82 Cadets NEN1 majors



NE Student Outcome Assessment – Cadet Surveys (AY 25)

Question	SA	A	N	D	SD	Responses	Mean	S.D.
I am confident in my ability to identify, formulate, and solve engineering problems by applying principles of engineering, science, and mathematics (NE Outcome 1).	15	48	18	9	0	90	3.77	0.85
I am confident in my ability to communicate effectively with a range of audiences (NE Outcome 3).	25	42	20	2	1	90	3.98	0.83
I am confident in my ability to acquire and apply new knowledge as needed, using appropriate learning strategies (NE Outcome 7).	20	53	15	2	0	90	4.01	0.69



NE300 Course Director's Assessment – Assessments for AY25

Student Outcomes	NE300	Indicators	Surveys	CD
1. Solve problems	X	4.00	3.77	4.00
2. Engineering design				
3. Communicate effectively	X	4.50	3.98	4.50
4. Ethical responsibilities				
5. Function on team				
6. Conduct experiments				
7. Apply knowledge	X	4.50	4.01	4.50

Recommended Changes for AY 26 and beyond

Administrative

- Assess utility of newly added 6th Problem Set
- Assess Fusion material assessment methods
- Map NE300 concepts to NE350/450 concept maps

Course Content

- Expand Practice Problems per Lesson
- Consider adding SWU Calculations to Nuclear Fuel Cycle Lesson
- Consider in-class lab-like activities

Resource Requirements

- Acquisitions this AY
 - New NERD-Ps (60 exam copies, 120+ issue copies)
- Requirements for next AY
 - N/A

Questions?

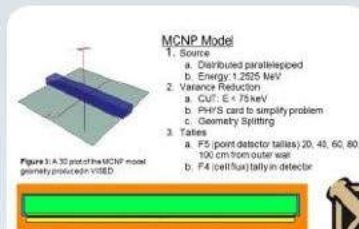


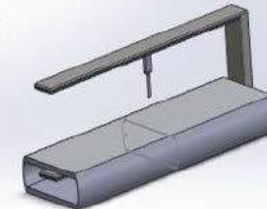
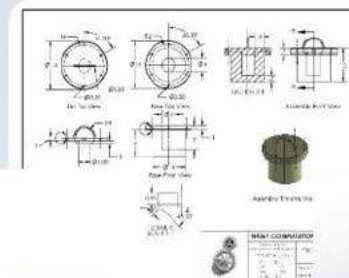
Figure 1: A 3D plot of the MCNP model geometry produced in VISOED.

Point view of the model geometry produced in VISOED. Note that green and orange in blue denotes lead and yellow is the detector box.

Left: Point view of the model geometry produced in VISOED. Note that green and orange in blue denotes lead and yellow is the detector box.



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TAB C

AY2024 COURSE ASSESSMENT

NE300: Fundamentals of Nuclear Engineering AY24



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Agenda: AY24 NE300 CAT Briefing

- Previous Guidance
- NE Program Student Outcomes
- NE300 Assessment Process
- AY24 Course Assessment
- Recommended Changes for AY25 and beyond

AY23 Program Director's Recommendations for AY24 (1 of 3)

Adopt CAT recommendations and:

- Fully use Canvas to its highest potential
- Implement Canvas assessment into courses and program
- Work on assessment tools that correctly identify any lack of fundamental skills to include time management (i.e. late turn-ins of homework)
- Continue assessment SO (1)-(7)

Standardization of briefings:

- Continue to work with Core Program and CD Academy for standard WPR and grades briefs
- Continue standardized briefs and reports for NE-specific topics
- Continue use of Overleaf

Effective use of NE labs and spaces:

- Finish cleaning and organizing subcrit, BH50 and accelerator spaces
- Continue Orientation for 2nd class for BH347 as collaborative space
- Continue Orientation for 1st class for BH66B as collaborative space
- Complete registration for cluster (VM) with RSIC and meet all compliance for use



AY23 Program Director's Recommendations for AY24 (2 of 3)

Work on a comprehensive text for NE450

Update *Nuclear Engineering Design Supplement*:

- Include format for a standard template for peer-reviewed journal and conference poster and presentation

Sustain internal and external relations:

- Include ANS Club, 1 activity per month
- Trip sections for courses (STERIS)

Integrate new lab tech into program

Other Items:

- Execute ABET advisory board
- Improve a lab bench stock of materials to order and have on hand to assist with the construction and development of lab, research and capstone projects
- Coordinate a better FE exam preparation between NE400 and NE496

AY23 Program Director's Recommendations for AY24 (3 of 3)

NE Working Group Topics:

- Work on future manning for near and sustained term with department leadership
- Continue and expand sponsored research on accelerator with outside organizations
- Continue to improve use of advanced computational tools such as Attila4MC throughout the program
- Annual update of internal and external websites – ensure compliance with ABET APPM

NE Program Director's Assessment – Assessments for AY24

Student Outcomes	NE300	NE350	NE355	NE361	NE400	NE450	NE452	NE461	NE474	NE495/6
1. Solve problems	X		X	X			X		X	
2. Engineering design		X		X				X		X
3. Communicate effectively	X						X		X	X
4. Ethical responsibilities					X	X			X	X
5. Function on team		X	X				X	X		X
6. Conduct experiments			X				X			
7. Apply knowledge	X		X		X	X				X

No change from AY22



NE Student Outcomes

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. An ability to communicate effectively with a range of audiences
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies



Program Director's Areas of Focus AY24

NE300:

- NE Outcomes are: 1, 3, 7
- Improve course content and pre-work all homework – complete
- Implement CAT recommendations – complete
- Standardize briefs – complete
- Use Overleaf for course material – complete

Recommended Changes for AY24

- Add one or two additional concepts covered to further prepare for NE350 or other courses (Build-up flux, secular equilibrium) - **Complete (added secular/transient equilibrium coverage)**
- Modify further reduction in Practical Applications emphasis - **Complete**
- Continue to follow the book format - **Complete**
- Sustain two WPRs and TEE graded event template - **Complete**
- Continue to utilize Overleaf to create course products – **Complete (except TEE)**
- ~~Change the practical exercise~~ – **Complete (removed from course)**
- Introduce fissile vs fissionable as a condition necessary to fission – **Complete**
- Expand use of Canvas LMS - **Complete**
- Add PWR/BWR to FNEK concepts card – **Ongoing (AY 25-1)**

COURSE	TITLE	CREDIT HOURS
NE300 2022-1	FUNDAMENTALS OF NUCLEAR ENGR	3.0 (BS=0.0, ET=3.0, MA=0.0)
Scope		Offerings
SCOPE This course provides the student with an understanding of the fundamental physical principles involved in radioactive decay, radiation interaction with matter, nuclear fission, and nuclear fuel material properties. The course covers neutron interactions with matter, fission, neutron lifecycle, neutron moderation, and reactor criticality. This course is essential for the nuclear engineer and is an excellent choice for the applied scientist.		2024 2
		2024 8
		2024 9
		2025 1
		2025 2
		2025 3
		2025 8
		2026 1
		2026 2
		2026 8
		2026 9
		2027 1
		2027 2
		2027 8
		2027 9



4-Step Evaluation & Assessment Process

1. Identify the connections between:
 - Course block objectives and program outcomes
 - Lesson learning objectives and course block objectives by developing a concept map
2. Select multiple embedded indicators to measure against established rubrics that will:
 - Assess the cadets' retention of previously learned concepts and procedures (reinforcement-level instruments)
 - Assess the cadets' ability to synthesize multiple concepts and procedures (extension-level instruments)
3. Assess the level of attainment of program outcomes based on the population distribution of cadet performance on the embedded indicators as:
4. Analyze results and propose recommendations for change if necessary or needed

Descriptor	Color Code	Likert Rating	Criteria
Excellent	Black	5	most frequent grade is A and B, no D or F
Good	Green	4	most frequent grade is A and B, some D or F
Acceptable	Yellow	3	most frequent grade is C or better, with < 20% D or F
Marginally Acceptable	Orange	2	most frequent grade is C or better, with at least 20% D or F but < 40% D or F
Unacceptable	Red	1	most frequent grade is D or F



NE300 Course Concept Map (AY 24)

FUNDAMENTALS

Nuclear Fundamentals

Fundamental Concepts

Modern Physics Concepts

Atomic and Nuclear Models

Nuclear Energetics

Nuclear Reactions

Energetics of Radioactive Decay

Binary Nuclear Reaction Kinematics

Neutron-Induced Binary Reactions and Fission

Characteristics of Radioactive Decay

Binary Nuclear Reaction Threshold Energy

Fusion

APPLICATIONS

Detection Applications

Neutral Particle Attenuation

Charged Particle Interactions

Detection and Measurement of Radiation

Radiation Detection Practical Exercise

Reactor Applications

Neutron Considerations in a Nuclear Reactor

Neutron Life Cycle in a Thermal Reactor

Nuclear Power Reactor Overview

The Nuclear Fuel Cycle

NE300 Course Concept Map (AY 25)

FUNDAMENTALS

Nuclear Fundamentals

Fundamental Concepts

Modern Physics Concepts

Atomic and Nuclear Models

Nuclear Energetics

Nuclear Reactions

Types of Radioactive Decay

Binary Nuclear Reaction Kinematics

Neutron-Induced Binary Reactions and Fission

Characteristics of Radioactive Decay

Binary Nuclear Reaction Threshold Energy

Fusion

APPLICATIONS

Detection Applications

Neutral Particle Attenuation I: Photons

Neutral Particle Attenuation II: Neutrons

Charged Particle Interactions

Detection and Measurement of Radiation

Reactor Applications

Neutron Considerations in a Nuclear Reactor

Neutron Life Cycle in a Thermal Reactor

Nuclear Power Reactor Overview

The Nuclear Fuel Cycle



Embedded Indicators

Embedded Indicators (Tools)	NE Student Outcomes						
	1	2	3	4	5	6	7
Problem sets	X	X					X
FNEK writ							X
WPRs	X					X	
Practical Exercise					X	X	X
Practical Application			X	X	X		X
TEE	X					X	

Best Assessment Instrument

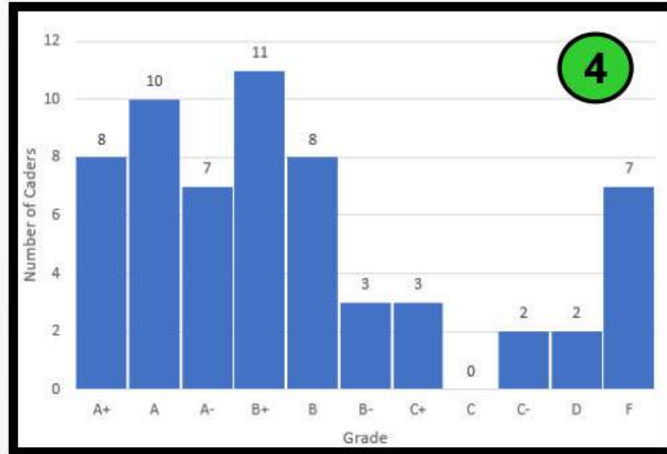
- SO 1 – TEE Problem 1/2: Neutron Lifecycle
– (90 marks)
- SO 3 – Practical Application Presentation
– (120 marks)
- SO 7 – Problem Sets
– (250 marks)



NE Student Outcome Assessment – Embedded Indicators

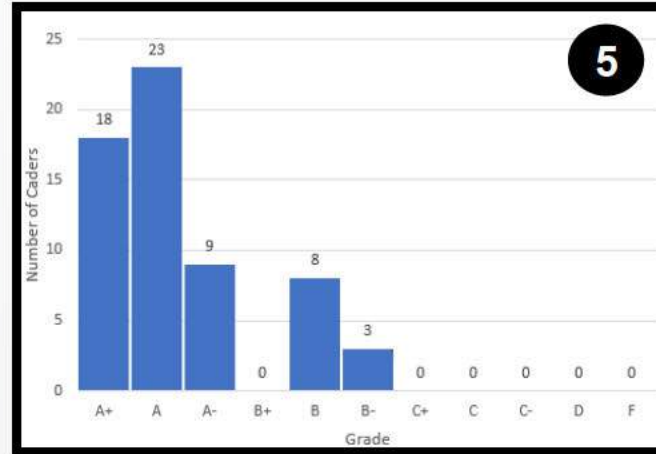
SO 1: TEE Q1 (90)

24-1



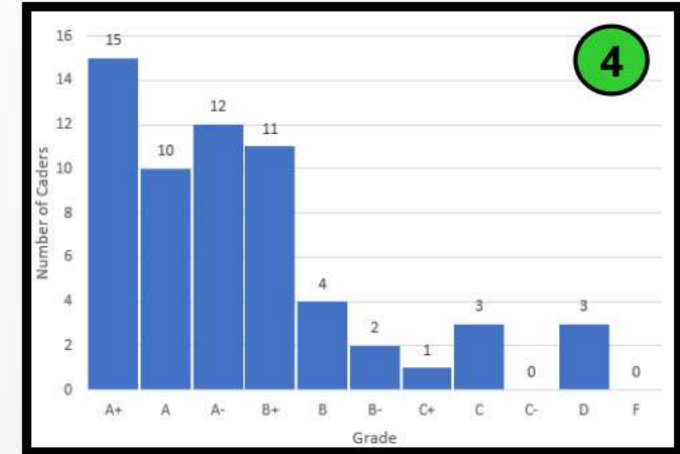
SO 3: PA Brief (120)

24-1

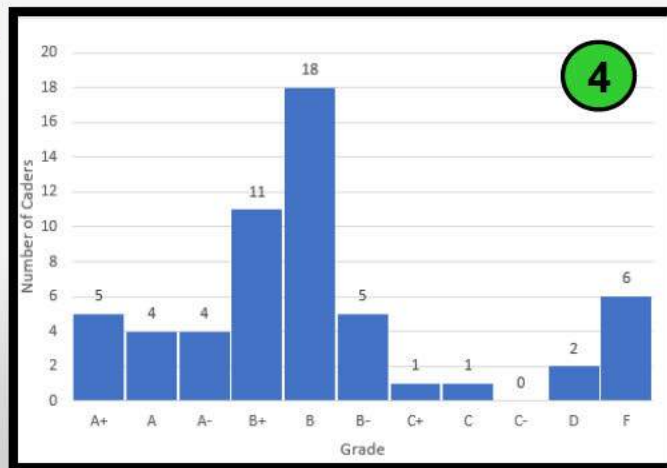


SO 7: Problem Sets (250)

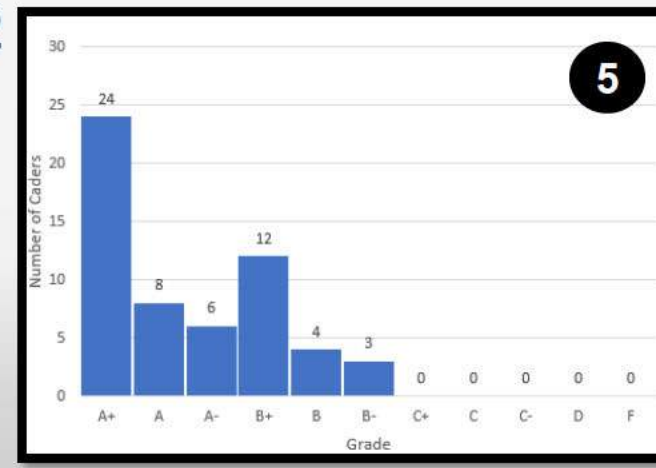
24-1



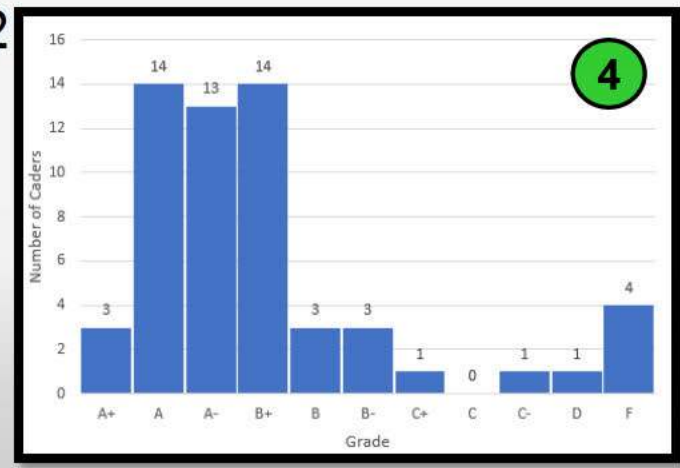
24-2



24-2



24-2



AY 24-1: 21/61 Cadets NEN1 majors----AY 24-2: 0/57 Cadets NEN1 majors



NE Student Outcome Assessment – Cadet Surveys (AY 24)

Question	SA	A	N	D	SD	Responses	Mean	S.D.
I am confident in my ability to identify, formulate, and solve engineering problems by applying principles of engineering, science, and mathematics (NE Outcome 1).	14	64	23	11	0	112	3.72	.81
I am confident in my ability to communicate effectively with a range of audiences (NE Outcome 3).	33	56	13	10	0	112	4.00	.88
I am confident in my ability to acquire and apply new knowledge as needed, using appropriate learning strategies (NE Outcome 7).	17	78	16	1	0	112	3.99	.58

NE300 Course Director's Assessment – Assessments for AY24

Student Outcomes	NE300	Indicators	Surveys	CD
1. Solve problems	X	4.0	3.7	4.0
2. Engineering design				
3. Communicate effectively	X	5.0	4.0	4.5
4. Ethical responsibilities				
5. Function on team				
6. Conduct experiments				
7. Apply knowledge	X	4.0	4.0	4.0

Recommended Changes for AY24-2/AY 25

Administrative

- Continue to utilize Overleaf to create course products **Complete (AY 24-1)**
- Implement Block Objectives/Scrub Lesson Objectives **Complete (AY 25-1)**
- Add Reaction Rate equation to Reference Card **Complete (AY 24-2)**
- Add 6th PS/Remove Rad PE **Ongoing (AY 25-1)**

Course Content

- ~~Update the radiation detection practical exercise~~
- Add L.O concerning prompt and delayed energy releasing processes **Complete (AY 24-2)**
- Add L.O. concerning specific fusion reactions **Ongoing (AY 25-1)**
- Add L.O. concerning charged particle ranges (CSDA ranges) **Complete (AY 24-2)**
- Explore consolidating L3 and L4 in AY25 **Considered and rejected (AY 24-2)**



Resource Requirements

- Acquisitions this AY
 - N/A
- Requirements for next AY
 - N/A



Questions?

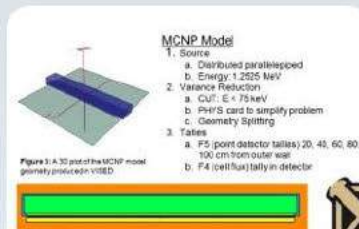
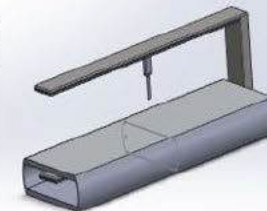
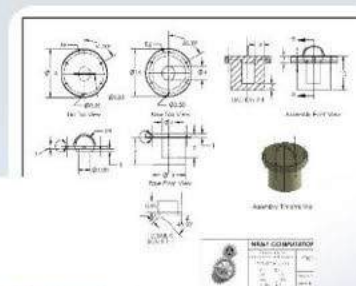


Figure 1: A 3D plot of the MCNP model geometry produced in VISOED.

Point view of the model geometry produced in VISOED. Note that green and orange in blue denotes lead and yellow is the detector tube.



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TAB D

AY2023 COURSE ASSESSMENT

NE300: Fundamentals of Nuclear Engineering AY23



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Agenda: AY23 NE300 CAT Briefing

- Previous Guidance
- NE Program Student Outcomes
- NE300 Assessment Process
- AY23 Course Assessment
- Recommended Changes for AY24



AY22 Program Director's Recommendations for AY23 (1 of 3)

Adopt CAT recommendations and:

- Assess and work on improving basic skills lost to COVID
- Work on assessment tools that correctly identify any lack of fundamental skills to include time management (i.e. late turn-ins of homework)
- Continue assessment SO (1)-(7)

Standardization of briefings:

- Continue to work with Core Program and CD Academy for standard WPR and grades briefs
- Continue standardized briefs and reports for NE-specific topics
- Continue use of Overleaf

Effective use of NE labs and spaces:

- Work on cleaning and organizing subcrit, BH50 and accelerator spaces
- Orientation for 2nd class for BH347 as collaborative space
- Orientation for 1st class for BH66B as collaborative space
- Cadets and faculty leverage remote use VMs

AY22 Program Director's Recommendations for AY23 (2 of 3)

Emphasize assumptions in design process and assess the impact to a design throughout the program

Update *Nuclear Engineering Design Supplement*:

- Include format for a standard template for peer-reviewed journal and conference poster and presentation

Sustain internal and external relations:

- Include ANS Club, 1 activity per month
- Trip sections for courses (STERIS)

Maintain flexibility and continue level of instruction during period of reduced faculty and missing lab tech

Other Items:

- Execute ABET advisory board
- Improve a lab bench stock of materials to order and have on hand to assist with the construction and development of lab, research and capstone projects
- More external review within courses when possible
- Coordinate a better FE exam preparation between NE400 and NE496

AY22 Program Director's Recommendations for AY23 (3 of 3)

NE Working Group Topics:

- Work on future manning for near and sustained term with department leadership
- Refine NE program concept map
- Perform sponsored research on accelerator with outside organizations
- Continue to improve use of advanced computational tools such as Attila4MC throughout the program
- Annual update of internal and external websites – ensure compliance with ABET APPM

NE Student Outcomes

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. An ability to communicate effectively with a range of audiences
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies



NE Program Director's Assessment – Assessments for AY23

Student Outcomes	NE300	NE350	NE355	NE361	NE400	NE450	NE452	NE461	NE474	NE495/6
1. Solve problems	X		X	X			X		X	
2. Engineering design		X		X				X		X
3. Communicate effectively	X						X		X	X
4. Ethical responsibilities					X	X			X	X
5. Function on team		X	X				X	X		X
6. Conduct experiments			X				X			
7. Apply knowledge	X		X		X	X				X

No change from AY22



Program Director's Areas of Focus AY23

NE300:

- NE Outcomes are: 1, 3, 7
- Improve course content and pre-work all homework – complete
- Implement CAT recommendations – complete
- Standardize briefs – complete
- Use Overleaf for course material – complete*

AY22-2 Recommended Changes for AY23

- ~~Consolidation of NE major cadets into a single section (NE351)~~
- One or two additional concepts covered to further prepare for NE350 or other courses (Build-up flux, secular equilibrium) *Continuing to Analyze*
- ~~Removal of FNEK WRITs from NE300~~
- Additional in class graded events – self graded if necessary *Continuing to Analyze*
- Further reduction in Practical Applications emphasis *Complete*
- Follow the book format *Complete*
- Sustain two WPRs and TEE graded event template *Complete*
- Add a fifth PS, one due every two weeks *Complete*

NE300 (Version: 2022 1) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
NE300	FUNDAMENTALS OF NUCLEAR ENGR	2022	1	Physics and Nuclear Engineering	3.0 (BS=0.0, ET=3.0, MA=0.0)

SCOPE

This course provides the student with an understanding of the fundamental physical principles involved in radioactive decay, radiation interaction with matter, nuclear fission, and nuclear fuel material properties. The course covers neutron interactions with matter, fission, neutron lifecycle, neutron moderation, and reactor criticality. This course is essential for the nuclear engineer and is an excellent choice for the applied scientist.

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

None

NE300 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
CO REQUISITE					
	PH202	2007	2	1	Y
	PH252	2007	2	2	Y
	PH205	2016	1	3	Y
	PH255	2016	1	4	Y
	PH201X	2022	1	5	Y
	PH201	2023	1	6	Y
	PH251	2023	1	7	Y



4-Step Evaluation & Assessment Process

1. Identify the connections between:
 - Course block objectives and program outcomes
 - Lesson learning objectives and course block objectives by developing a concept map
2. Select multiple embedded indicators to measure against established rubrics that will:
 - Assess the cadets' retention of previously learned concepts and procedures (reinforcement-level instruments)
 - Assess the cadets' ability to synthesize multiple concepts and procedures (extension-level instruments)
3. Assess the level of attainment of program outcomes based on the population distribution of cadet performance on the embedded indicators as:

Descriptor	Color Code	Likert Rating	Criteria
Excellent	Black	5	most frequent grade is A and B, no D or F
Good	Green	4	most frequent grade is A and B, some D or F
Acceptable	Yellow	3	most frequent grade is C or better, with < 20% D or F
Marginally Acceptable	Orange	2	most frequent grade is C or better, with at least 20% D or F but < 40% D or F
Unacceptable	Red	1	most frequent grade is D or F

4. Analyze results and propose recommendations for change if necessary or needed



NE300 Course Concept Map

FUNDAMENTALS

Nuclear Fundamentals

Fundamental Concepts

Modern Physics Concepts

Atomic and Nuclear Models

Nuclear Energetics

Nuclear Reactions

Energetics of Radioactive Decay

Binary Nuclear Reaction Kinematics

Neutron-Induced Binary Reactions and Fission

Characteristics of Radioactive Decay

Binary Nuclear Reaction Threshold Energy

Fusion

APPLICATIONS

Detection Applications

Neutral Particle Attenuation

Charged Particle Interactions

Detection and Measurement of Radiation

Radiation Detection Practical Exercise

Reactor Applications

Neutron Considerations in a Nuclear Reactor

Neutron Life Cycle in a Thermal Reactor

Nuclear Power Reactor Overview

The Nuclear Fuel Cycle



Embedded Indicators

Embedded Indicators (Tools)	NE Student Outcomes						
	1	2	3	4	5	6	7
Problem sets	X	X					X
FNEK writ							X
WPRs	X					X	
Practical Exercise					X	X	X
Practical Application			X	X	X		X
TEE	X					X	

Best Assessment Instrument

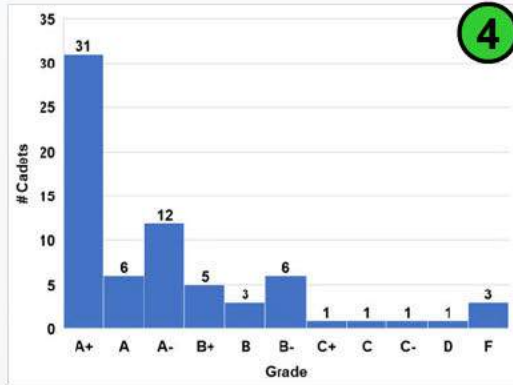
- SO 1 – TEE Problem 1: Neutron Lifecycle
– (90 marks)
- SO 3 – Practical Application Presentation
– (120 marks)
- SO 7 – Problem Sets
– (250 marks)



NE Student Outcome Assessment – Embedded Indicators

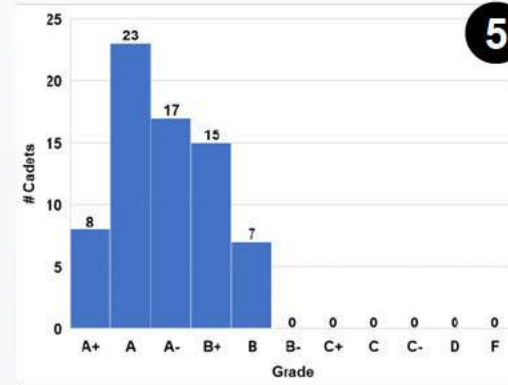
SO 1: TEE Q1 (90)

23-1



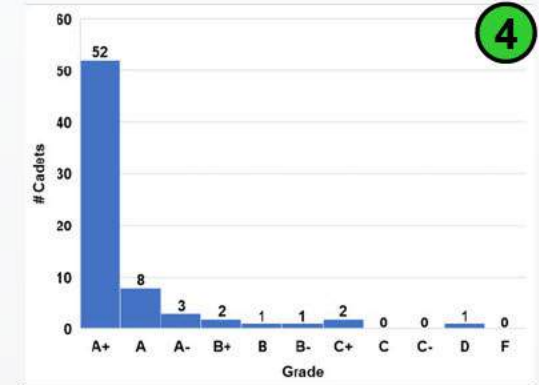
SO 3: Presentation (120)

23-1

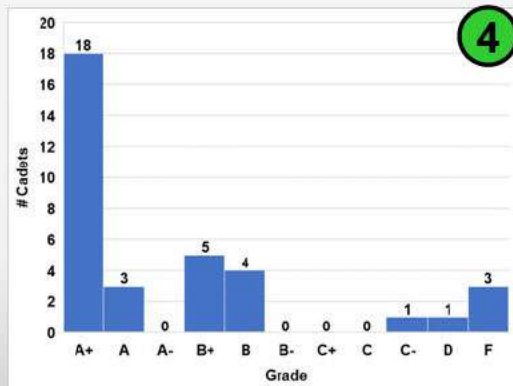


SO 7: Problem Sets (250)

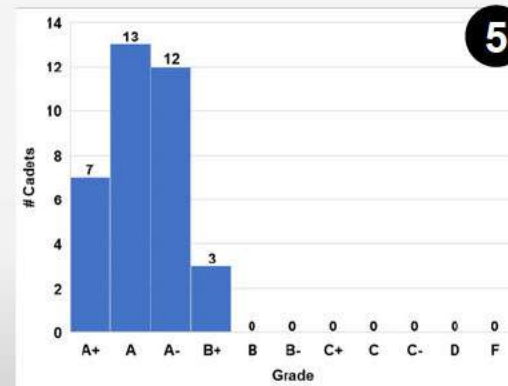
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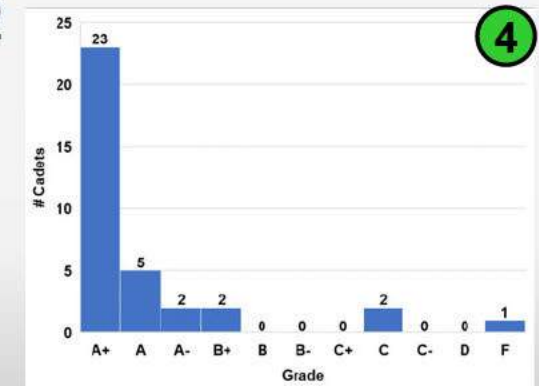
23-2



23-2



23-2



NE Student Outcome Assessment – Cadet Surveys

#	Question	SA	A	N	D	SD	N	Mean	Std Dev
Q9	I am confident in my ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics (NE Outcome 1).	20	30	2	0	0	52	4.3	0.6
Q13	I am confident in my ability to communicate effectively with a range of audiences (NE Outcome 3).	15	33	4	1	0	53	4.2	0.6
Q15	I am confident in my ability to recognize the ongoing need for additional knowledge and locate, evaluate, integrate, and apply this knowledge appropriately (NE Outcome 7)	16	35	2	0	0	53	4.3	0.5

Aggregate data from whole AY

NE300 Course Director's Assessment – Assessments for AY23

Student Outcomes	NE300	Indicators	Surveys	CD
1. Solve problems	X	4.0	4.3	4.0
2. Engineering design				
3. Communicate effectively	X	5.0	4.2	4.0
4. Ethical responsibilities				
5. Function on team				
6. Conduct experiments				
7. Apply knowledge	X	4.0	4.3	4.0

Recommended Changes for AY24

- Add one or two additional concepts covered to further prepare for NE350 or other courses (Build-up flux, secular equilibrium) Ongoing
 - Modify Further reduction in Practical Applications emphasis
 - Continue to follow the book format
 - Sustain two WPRs and TEE graded event template
 - Continue to utilize Overleaf to create course products – all but TEE complete
-
- Change the practical exercise
 - Introduce fissile vs fissionable as a condition necessary to fission
 - Expand use of Canvas LMS
 - Add PWR/BWR to FNEK concepts card

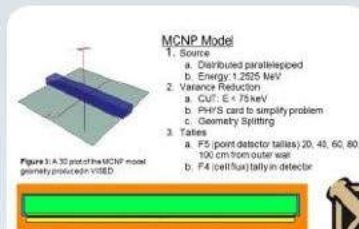


Resource Requirements

- Acquisitions this AY
 - Chadwick the Roomba
- Requirements for next AY
 - N/A



Questions?



- MCNP Model**
1. Source
 - a. Distributed parallelepiped
 - b. Energy: 1.2525 MeV
 2. Variance Reduction
 - a. CUT: $E < 75$ keV
 - b. PHITS card to simplify problem
 - c. Geometry Splitting
 3. Tally
 - a. P5 (point detector tallies) 20, 40, 60, 80, 100 cm from outer wall
 - b. F4 (cell flux) tally in detector

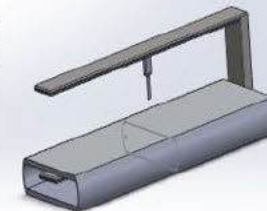
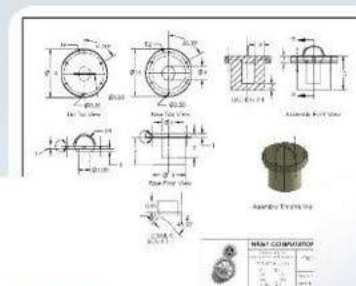
Figure 1: A 3D plot of the MCNP model geometry produced in VISOED.



Point view of the model geometry produced in VISOED. Note that green and orange in blue denotes lead and yellow is the detector tube.



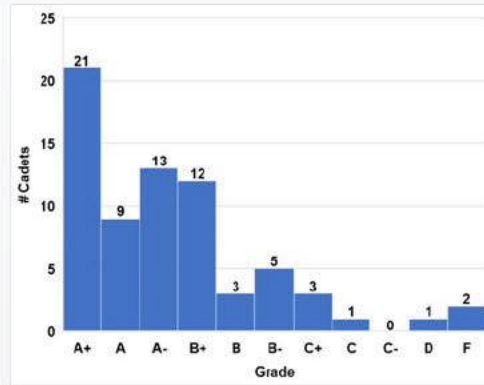
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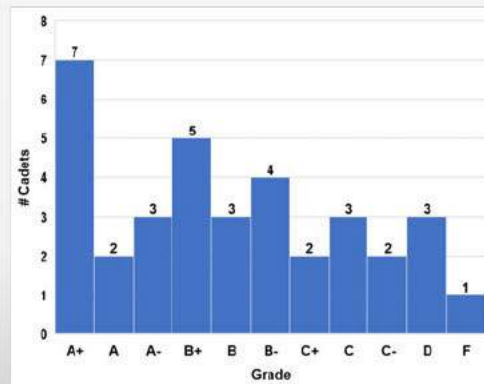
NE300 TEE Performance in AY23

TEE Grades

23-1



23-2



Grading Scale		23-1	23-2	T
A+	94.0%	21	7	28
A	90.5%	9	2	11
A-	87.0%	13	3	16
B+	83.5%	12	5	17
B	80.0%	3	3	6
B-	76.5%	5	4	9
C+	73.0%	3	2	5
C	69.5%	1	3	4
C-	66.0%	0	2	2
D	60.0%	1	3	4
F	0.0%	2	1	3
Total		70	35	105





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TAB E

AY2022 COURSE ASSESSMENT

NE300 Fundamental of Nuclear Engineering AY22



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**LTC Jonathan Baker
NE300 Course Director**

Agenda

AY22 NE300 CAT Briefing

- Previous Guidance
- NE Program Student Outcomes
- NE300 Assessment Process
- AY22 Course Assessment
- Recommended Changes for AY23

AY21 Program Director's Recommendations (for AY22) (1 of 3)

Adopt CAT recommendations and

- Continue to assess effect of COVID response for students in follow-on courses
- Assess impact of MC311 taken in STAP for Class 22
- Continue assessment SO (1)-(7)

Standardization of briefings

- Work with Core Program and CD Academy for standard WPR and grades briefs
- Continue standardized briefs and reports for NE-specific topics
- Work on use of Overleaf

Standard use of Academic Terms

- Where appropriate use terms standard in academia (i.e. course-wide writ vs. PLW)

Effective use of NE labs and spaces

- Orientation for 2nd class for BH347 as collaborative space
- Orientation for 1st class for BH66B as collaborative space
- Cadets and faculty leverage remote use VMs

AY21 Program Director's Recommendations (for AY22) (2 of 3)

Emphasize assumptions in design process and assess the impact to a design throughout the program.

Update *Nuclear Engineering Design Supplement*

- Include format for a standard template for peer-reviewed journal and conference poster and presentation

Sustain internal and external relations:

- Sustain ANS Club, 1 activity per month
- Sustain relations with D/Math (MA255 and MA364) and D/CME
- Reestablish trip sections for courses (STERIS)

Identify a backup instructor for every course and ensure they are current in the material

Other Items:

- Execute ABET advisory board
- Improve a lab bench stock of materials to order and have on hand to assist with the construction and development of lab, research and capstone projects
- Continue to develop laboratory tech assigned to the program
- More external review within courses when possible

AY21 Program Director's Recommendations (for AY22) (3 of 3)

NE Working Group Topics

- Develop funded research to help bring research dollars to department in support of Department Head guidance
- Refine NE program concept map.
- Get accelerator operational and back in program.
- Continue to expand use of accelerator and sub-crit in curriculum beyond research.
- Annual update of internal and external websites – ensure compliance with ABET APPM

NE Student Outcomes

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. an ability to communicate effectively with a range of audiences
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

NE Program Director's Assessment

(Recommended Assessments for AY22)

NE Student Outcomes	NE300	NE350	NE355	NE361	NE400	NE450	NE452	NE461	NE474	NE495/6
1. Solve Problems	XX		XX	XX			XX		XX	
2. Engineering design		XX		XX				XX		XX
3. Comm. Effectively	XX						XX		XX	XX
4. Ethical Responsibilities					XX	XX			XX	XX
5. Function on Team		XX	XX				XX	XX		XX
6. Conduct Experiments			XX				XX			
7. Apply Knowledge	XX		XX		XX	XX				XX

No change from AY21

Program Director's Areas of Focus

Academic Year 2022

NE300

- NE Outcomes are: 1, 3, 7
- Improve course content and pre-work all homework Complete
- Implement CAT recommendations Complete
- Prepare for in-person, remote or hybrid classes for AY22 Complete
- Assess use of remote technology to enhance course even when taught in-person such as asynchronous video lessons for “flipped classrooms” or video AI for cadets who are on quarters or on trips. Complete

AY21-2 Recommended Changes (For AY22)

- Continue to expand content and deepen Cadet ownership of Practical Applications block; Develop new ways to scaffold critical concepts by helping them link the Physics to prior lesson concepts and learning objectives *Complete*
- Improve long-term retention of FNEK materials and concepts across the course and program *Complete*
- Review the Radiation Practical exercise with NE350 Course Director to build in hands-on exercises to reinforce key concepts *Complete*
- ~~Update Nuclear Reactor Supplement and Standards for Writing Technical Reports~~
- Maintain the Chart of Nuclides as an optional text, and expand the use of digital database resources in class and problem sets to allow Cadets to gain familiarity with the resources available *Complete*
- Maintain the use of the new Lesson Supplements and continue to expand or improve the content and organization to maximize their use as a study resource *Complete*
- Re-evaluate the inclusion of individual presentation topics, extension problems, and technical reading assignments to determine where they may add value to course concepts and the practical applications block *Complete*

Redbook Entry

NE300 (Version: 2022 1) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
NE300	FUNDAMENTALS OF NUCLEAR ENGR	2022	1	Physics and Nuclear Engineering	3.0 (BS=0.0, ET=3.0, MA=0.0)

SCOPE

This course provides the student with an understanding of the fundamental physical principles involved in radioactive decay, radiation interaction with matter, nuclear fission, and nuclear fuel material properties. The course covers neutron interactions with matter, fission, neutron lifecycle, neutron moderation, and reactor criticality. This course is essential for the nuclear engineer and is an excellent choice for the applied scientist.

LESSONS: 30 @ 75 min (2,000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

None

NE300 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
CO REQUISITE					
	PH202	2007	2	1	Y
	PH252	2007	2	2	Y
	PH205	2016	1	3	Y
	PH255	2016	1	4	Y
	PH201X	2022	1	5	Y
	PH201	2023	1	6	Y
	PH251	2023	1	7	Y

4-Step Evaluation & Assessment Process

1. Identify the connections between:
 - Course block objectives and program outcomes
 - Lesson learning objectives and course block objectives by developing a concept map
2. Select multiple embedded indicators to measure against established rubrics that will:
 - Assess the cadets' retention of previously learned concepts and procedures (reinforcement-level instruments)
 - Assess the cadets' ability to synthesize multiple concepts and procedures (extension-level instruments)

3. Assess the level of attainment of program outcomes based on the population distribution of cadet performance on the embedded indicators as:

<u>Descriptor</u>	<u>Color Code</u>	<u>Likert Rating</u>	<u>Criteria</u>
Excellent	Black	5	most frequent grade is A and B, no D or F
Good	Green	4	most frequent grade is A and B, some D or F
Acceptable	Yellow	3	most frequent grade is C or better, with < 20% D or F
Marginally Acceptable	Orange	2	most frequent grade is C or better, with at least 20% D or F but < 40% D or F
Unacceptable	Red	1	most frequent grade is D or F

4. Analyze results and propose recommendations for change if necessary or needed

Embedded Indicators

Embedded Indicators (Tools)	NE Student Outcome						
	1	2	3	4	5	6	7
Problem Sets	X	X					X
FNEK WRIT							X
WPR	X					X	
Practical Exercise					X	X	X
Practical Application			X	X	X		X
TEE	X					X	

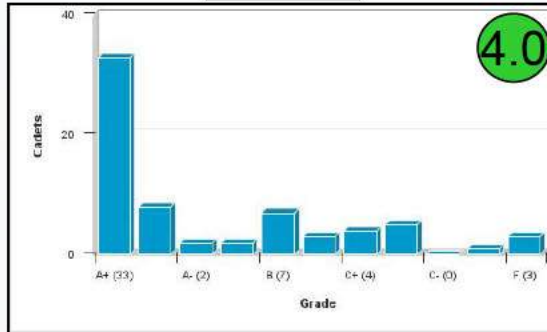
Best Assessment Instrument

- SO 1 – TEE Problem 1: Neutron Lifecycle
 - (75 marks)
- SO 3 – Practical Application Presentation
 - (250 marks)
- SO 7 – Problem Sets
 - (250 marks)

Outcome 1

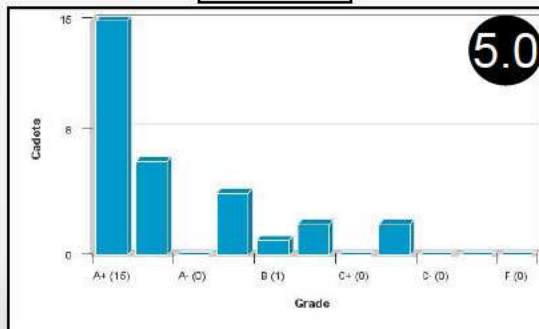
TEE Q1 (75)

AY 22-1



Course Director: 4.5
CDT Survey: 4.2

AY 22-2

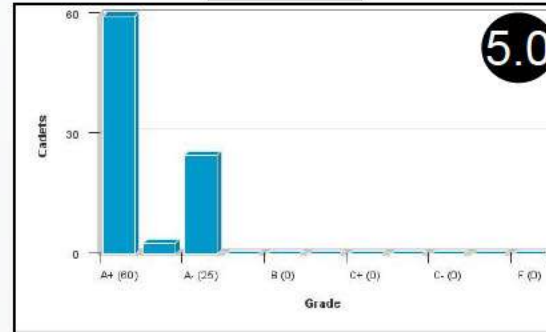


Course Director: 4.5
CDT Survey: 4.3

Outcome 3

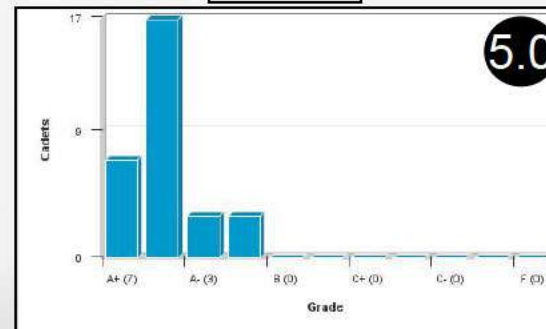
Presentation (250/120)

AY 22-1



Course Director: 4.5
CDT Survey: 4.3

AY 22-2

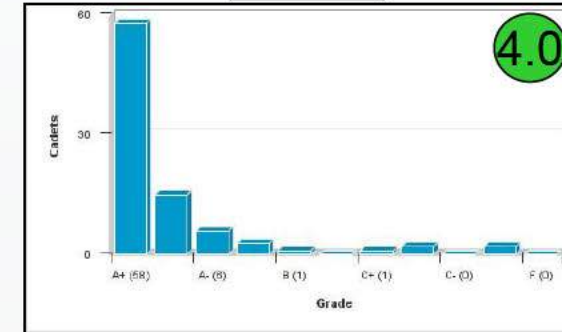


Course Director: 4.5
CDT Survey: 4.2

Outcome 7

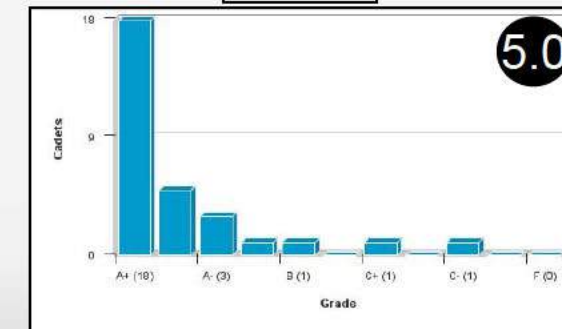
Problem Sets (250)

AY 22-1



Course Director: 4.0
CDT Survey: 4.2

AY 22-2



Course Director: 4.0
CDT Survey: 4.3

AY22 Recommended Changes (For AY23)

- ~~Consolidation of NE major cadets into a single section (NE351)~~
- One or two additional concepts covered to further prepare for NE350 or other courses (Build-up flux, secular equilibrium)
- ~~Removal of FNEK WRITs from NE300~~
- Additional in class graded events – self graded if necessary
- Further reduction in Practical Applications emphasis
- Follow the book format
- Sustain two WPRs and TEE graded event template
- Add a fifth PS, one due every two weeks

Resource Requirements

- Acquisitions this AY
 - N/A
- Requirements for next AY
 - N/A

Questions?

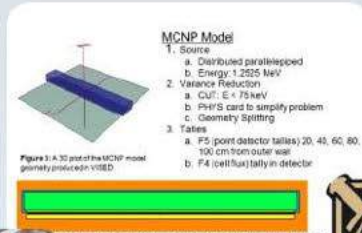
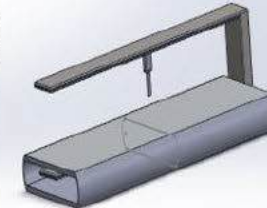
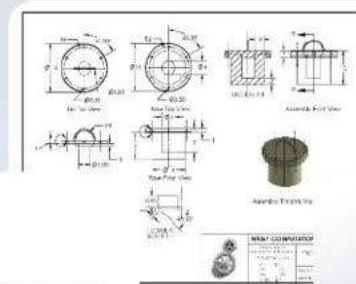


Figure 1: A 3D prototype MCNP model geometry produced in VISOED.

Front view of the model geometry produced in VISOED. Note that green and orange red lines denotes lead, and yellow is the detector box.



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TAB F

AY2021 COURSE ASSESSMENT

NE 300 Fundamental of Nuclear Engineering

AY21



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MAJ Anthony Severson
NE 300 Course Director

Agenda

AY 21-2 NE300 CAT Briefing

- Previous Guidance
- NE Program Student Outcomes
- NE300 Assessment Process
- AY21 Course Assessment
- Recommended Changes for AY22

AY20 Program Director's Recommendations (for AY21) (1 of 3)

Adopt CAT recommendations and

- Continue to assess effect of remote classes for students in follow-on courses
- Continue assessment SO (1)-(7)

Standard use of Blackboard

- Look to see if there are some of the resources used in the remote environment that will benefit the course content even when taught in person

Upgrades for new BH177B computer center and teaching of NE361 and use in other courses

- Finish upgrade this space NLT beginning of 21-1
- Every NE cadet has a unique IP address for a remote machine
- Cadets and faculty leverage in-class and remote use of BH177B and BH347 to continue to build on computational thread within program

Integrate NE461 as a required course in the NE program

Create a Program Drop Schedule to post to NE SharePoint

AY20 Program Director's Recommendations (for AY21) (2 of 3)

Update *Nuclear Engineering Design Supplement*

- Include format for a standard template for peer-reviewed journal and conference poster and presentation

Sustain internal and external relations:

- Sustain ANS Club, 1 activity per month
- Sustain relations with D/Math (MA255 and MA364) and D/CME
- DTRA,MSKCC, NDU, STERIS (continue modified trip sections because of 75-min classes)

Identify a backup instructor for every course and ensure they are current in the material

Other Items:

- Execute ABET visit in Nov remotely
- Develop a lab bench stock of materials to order and have on hand to assist with the construction and development of lab, research and capstone projects
- Train and establish a relationship with new laboratory tech assigned to the program

AY20 Program Director's Recommendations (for AY21) (3 of 3)

NE Working Group Topics

- Refine NE program concept map.
- Look at overlap and redundancy and where NE350 fits in program
- Continue to expand use of accelerator and sub-crit in curriculum beyond research.
- Complete all administrative edits to Redbook
- Complete update of internal and external websites – ensure compliance with ABET APPM

AY20-2 Recommended Changes (For AY21)

- Sustain order and type of major graded events (MGEs): PLW, WPR, PLW, TEE **Complete**
- Based on cadet feedback, include more experiments/visual demonstrations of concepts in class as available. **Complete**
- Develop program concept/course map. **Complete**
 - Particularly instructive for junior rotators who need to understand how their course(s) fit into the program as a whole.
- Sustain block and lesson structure as much as possible to focus assessment on recent course changes. **Complete**
- ~~Sustain reading quizzes while building a larger repository of pertinent, peer-reviewed, technical articles and report of the appropriate level detail.~~
- Update FNEK cards to reflect course material changes in NE300/NE350. **Complete**
- Update Nuclear Reactor Supplement and Standards for Writing Technical Reports. **Ongoing**

Redbook Entry

NE300 (Version: 2022 1) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
NE300	FUNDAMENTALS OF NUCLEAR ENGR	2022	1	Physics and Nuclear Engineering	3.0 (BS=0.0, ET=3.0, MA=0.0)

SCOPE

This course provides the student with an understanding of the fundamental physical principles involved in radioactive decay, radiation interaction with matter, nuclear fission, and nuclear fuel material properties. The course covers neutron interactions with matter, fission, neutron lifecycle, neutron moderation, and reactor criticality. This course is essential for the nuclear engineer and is an excellent choice for the applied scientist.

LESSONS: 30 @ 75 min (2,000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

None

NE300 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
CO REQUISITE					
	PH202	2007	2	1	Y
	PH252	2007	2	2	Y
	PH205	2016	1	3	Y
	PH255	2016	1	4	Y

Changes: Replaced “neutron diffusion” with “neutron lifecycle” and eliminated the research paper requirement.

Program Director's Areas of Focus

Academic Year 2021

NE300

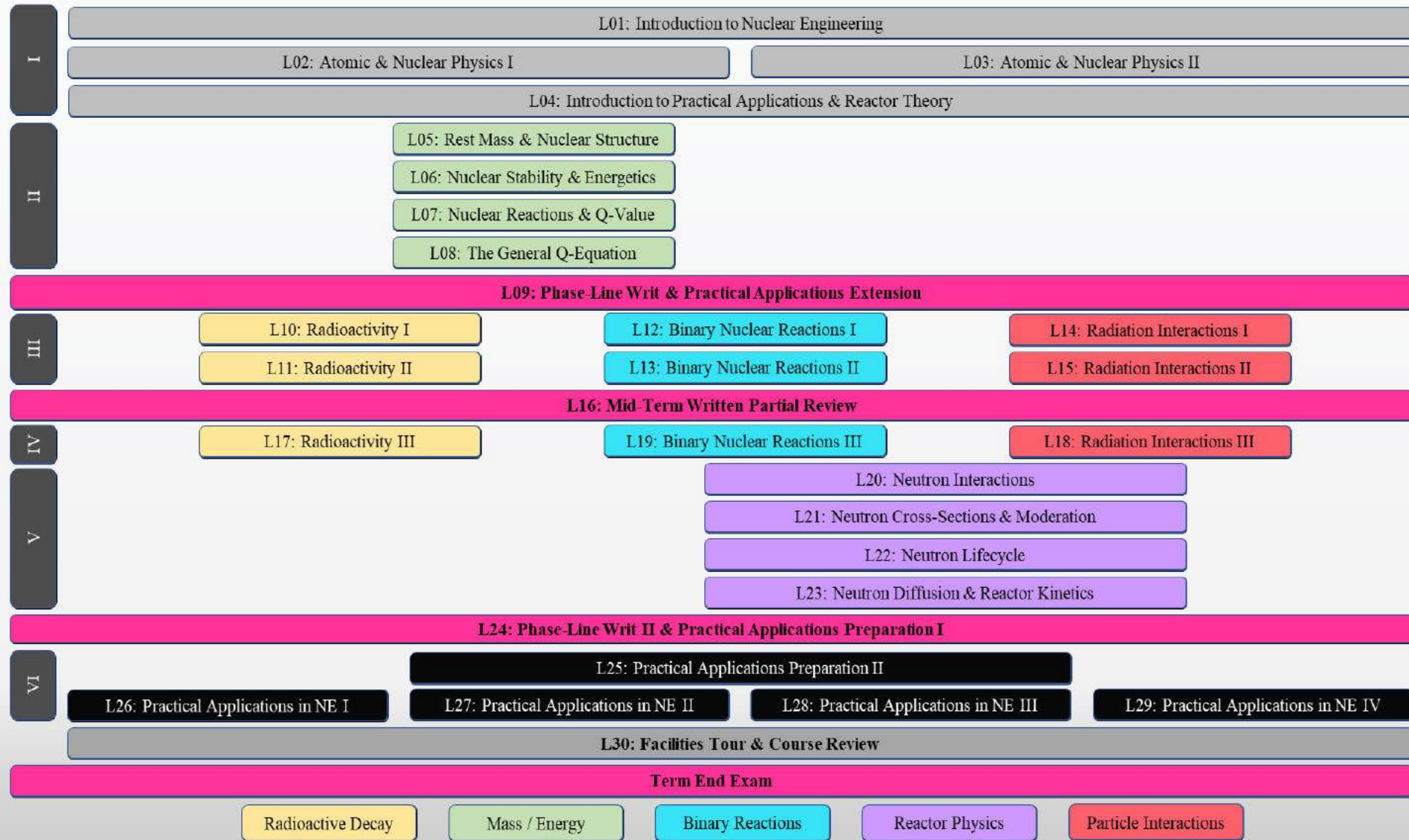
- ✓ NE Outcomes assessment is: 1, 3, 7 *Ongoing*
- ✓ Implement CAT recommendations *Ongoing*
- ✓ Be prepared for ABET accreditation visit in Nov *Complete*
- ✓ Prepare for in-person, remote or hybrid classes for AY21 *Complete*
- ✓ Assess use of remote technology to enhance course even when taught in-person such as asynchronous video lessons for “flipped classrooms” or video AI for cadets who are on quarters or on trips. *Complete*
- ✓ Be NE proponent for review and maintenance of FNEK Cards and NERDP *Complete*

NE Student Outcomes

Cadets possess:

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. an ability to communicate effectively with a range of audiences
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

NE300 Concept Map AY21-2



Embedded Indicators

Embedded Indicators (Tools)	NE Student Outcome						
	1	2	3	4	5	6	7
Term End Exam	X						
Problem Sets	X						X
Phase-Line Writs	X						X
Written Partial Review	X						
FNEK Writ			X				X
Group Project			X				X

4-Step Evaluation & Assessment Process

1. Identify the connections between:
 - Course block objectives and Student Outcomes
 - Lesson learning objectives and course block objectives by developing a concept map
2. Select multiple embedded indicators to measure against established rubrics that will:
 - Assess the cadets' retention of previously learned concepts and procedures (reinforcement-level instruments)
 - Assess the cadets' ability to synthesize multiple concepts and procedures (extension-level instruments)
3. Assess the level of attainment of Student Outcomes based on the population distribution of cadet performance on the embedded indicators as:

<u>Descriptor</u>	<u>Color Code</u>	<u>Likert Rating</u>	<u>Criteria</u>
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Good	Green	4	most frequent grade is A and B, some D or F
Acceptable	Yellow	3	most frequent grade is C or better, with < 20% D or F
Marginally Acceptable	Orange	2	most frequent grade is C or better, with at least 20% D or F but < 40% D or F
Unacceptable	Red	1	most frequent grade is D or F

4. Analyze results and propose recommendations for change if necessary or needed

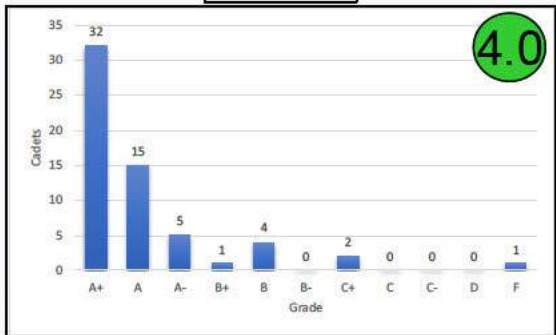
Best Assessment Instrument

- SO 1 – TEE Problem 1: Neutron Lifecycle (85 marks)
- SO 3 – Practical Application Presentation – Individual Component (50 marks)
- SO 7 – Practical Application Presentation – Group Component (150 marks)

Outcome 1

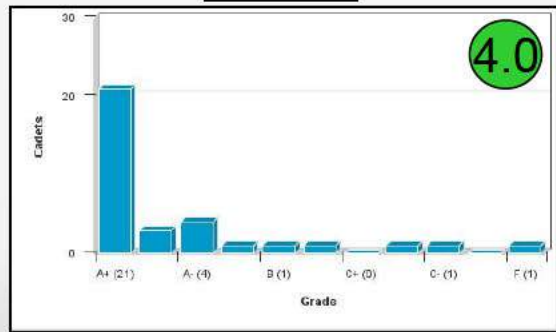
TEE Q1 (85)

AY 21-1



Course Director: 4.0
CDT Survey: 4.2

AY 21-2

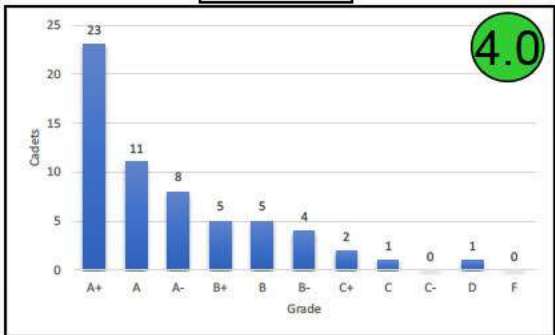


Course Director: 4.0
CDT Survey: 3.8

Outcome 3

Presentation - Individual (50)

AY 21-1



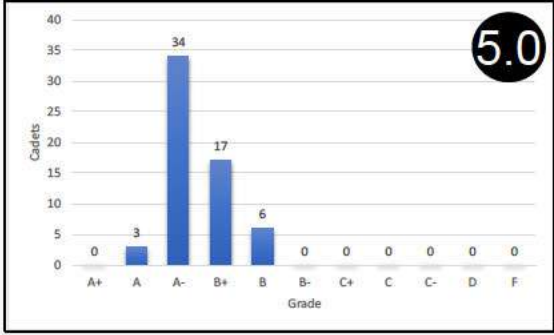
Course Director: 4.0
CDT Survey: 4.2

Course Director: 4.0
CDT Survey: 3.9

Outcome 7

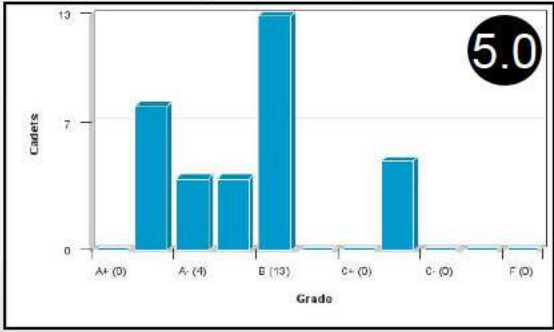
Presentation - Group (150)

AY 21-1



Course Director: 4.5
CDT Survey: 4.2

AY 21-2



Course Director: 4.5
CDT Survey: 4.0

NE Outcomes Assessment

NE Outcome	CDT Survey (Fall / Spring)	Embedded Indicator	CD	Course Director Comments
1	4.2 / 3.8	4.0	4.0	Cadets are generally able to identify and formulate solutions to nuclear engineering problems with only minor errors. The strong link between Problem Set material and exam calculations helps students develop and reinforce these skills.
3	4.2 / 3.9	4.0	4.0	Cadets effectively communicate technical information to their peers and instructors in a clear, concise manner using appropriate visual aids.
7	4.2 / 4.0	5.0	4.5	Cadets each identified practical applications of NE and formed groups by similar interest. In these groups they demonstrated the ability to learn independently by locating scholarly sources, researching a topic, and then collectively presenting new material.

KEY – Cadet Survey

- 5 – Strongly Agree
- 4 – Agree
- 3 – Neutral
- 2 – Disagree
- 1 – Strongly Disagree

KEY – Embedded Indicators and CD

- 5  -- Excellent
- 4  -- Good
- 3  -- Acceptable
- 2  -- Marginally Acceptable
- 1  -- Unacceptable

AY21 Recommended Changes (For AY22)

- **Sustain brevity and organization of the RDC to keep it nested with key concepts as presented on course map**
- **Sustain order and type of major graded events (MGEs): PLW, WPR, PLW, TEE**
- **Sustain exam question development methods with a deliberate link to key learning objectives and problem sets**
- **Sustain course structure organized by key concepts and built with multiple touch points on each**
- **Sustain material density across the course and within each lesson to maintain proper balance across a broad range of students**
- **Sustain minimization of duplicated effort in NE350 / NE452 while keeping the course appropriately nested with each**
- **Sustain the Practical Applications block at the end of the semester to allow Cadets the flexibility to pursue topics of interest**
- **Maintain FNEK cards and NERD-P to reflect content across all courses**

AY21 Recommended Changes (For AY22)

- **Continue to expand content and deepen Cadet ownership of Practical Applications block; Develop new ways to scaffold critical concepts by helping them link the Physics to prior lesson concepts and learning objectives**
- **Improve long-term retention of FNEK materials and concepts across the course and program**
- **Review the Radiation Practical exercise with NE350 Course Director to build in hands-on exercises to reinforce key concepts**
- **Update Nuclear Reactor Supplement and Standards for Writing Technical Reports**
- **Maintain the Chart of Nuclides as an optional text, and expand the use of digital database resources in class and problem sets to allow Cadets to gain familiarity with the resources available**
- **Maintain the use of the new Lesson Supplements and continue to expand or improve the content and organization to maximize their use as a study resource**
- **Re-evaluate the inclusion of individual presentation topics, extension problems, and technical reading assignments to determine where they may add value to course concepts and the practical applications block**

Resource Requirements

- **Acquisitions this AY**
 - **N/A**
- **Requirements for next AY**
 - NERDP ~ 150 copies on hand in 470

Questions?



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SECTION II

EXAMINATIONS



UNITED STATES MILITARY ACADEMY
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TAB G

WPR #1

Instructor Solution

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE300, Fundamentals of Nuclear Engineering
Written Partial Review I

1. Instructions:

- a. Bring only your TI-30XIIS or TI-36XPRO calculator, a straight-edge, and pencil(s) into the examination room. **No phones, cameras, or smartwatches are allowed in the examination room.**
- b. Check your exam for 6 problems. Check for the Reference Data Card and NERD-P provided with the examination.
- c. For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question.
- d. For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- e. Take up to 75 minutes to complete the examination.
- f. An instructor is in the hall or in the classroom for any questions.
- g. **When you turn in your exam check your name is written on every page.**

2. Exam Security:

- a. I will not discuss this exam with anyone until it is released from academic security on **16 October 2025 at 1700.**
- b. Cadet Signature: _____
- c. Cadet Name (print): _____

Problem	Weight	Score
1	60	
2	50	
3	40	
4	30	
5	40	
6	30	
Total:	250	

A: 27 (11%)
B: 93 (37%)
D: 130 (52%)
Total: 250

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Wt. No.

Cadet: _____

- 60 1. A new type of nuclear fuel is being considered for use in advanced reactors. The fuel consists of uranium nitride (UN), where the uranium is enriched to 12.927 w/o ^{235}U and the remainder is ^{238}U . The nitrogen is naturally occurring. The density of the UN fuel is 11.597 g/cm³.

Calculate the number densities (in atoms/cm³) of ^{235}U , ^{238}U , and N in the UN fuel. Report your answers to five significant figures.

$$\mathcal{A}_x = \left[\sum \frac{\omega_i}{\mathcal{A}_i} \right]^{-1} \quad \text{8 Equation 3}$$

8 Equation 6

$$N_i = \frac{\omega_i \rho_x N_a}{\mathcal{A}_i}$$

$$\mathcal{A}_U = \left[\frac{\omega_{235}}{\mathcal{A}_{235}} + \frac{\omega_{238}}{\mathcal{A}_{238}} \right]^{-1}$$

$$N_{235} = \frac{\omega_{235} \rho_U N_a}{\mathcal{A}_{235}}$$

4 Correct w/o (2 per)

$$\mathcal{A}_U = \left[\frac{0.12927}{235.0439231} + \frac{0.87073}{238.0507826} \right]^{-1}$$

$$\mathcal{A}_U = 237.6577633 \text{ u}$$

$$N_{235} = \frac{0.12927(10.95155448)N_a}{235.0439231}$$

$$N_{235} = 3.6272 \times 10^{21} \frac{\text{atoms}}{\text{cm}^3}$$

4 Substitutes mass frac and density (2 per)

$$\omega_x = \frac{n\mathcal{A}_x}{n\mathcal{A}_x + m\mathcal{A}_y}$$

8 Equation 2

$$\omega_U = \frac{\mathcal{A}_U}{\mathcal{A}_U + \mathcal{A}_N}$$

$$N_{238} = \frac{0.87073(10.95155448)N_a}{238.0507826}$$

$$N_{238} = 2.4124 \times 10^{22} \frac{\text{atoms}}{\text{cm}^3}$$

2 Substitutes 238 mass frac and mass

$$\omega_U = \frac{237.6577633}{237.6577633 + 14.0067}$$

$$\omega_U = 0.9443437511$$

$$\omega_N = \frac{14.0067}{237.6577633 + 14.0067}$$

$$\omega_N = 0.05565624887$$

4 Calculates new mass frac for N

$$\rho_x = \omega_x \rho$$

8 Equation 5

$$\rho_U = \omega_U \rho_{UN}$$

$$\rho_U = 0.9443437511(11.597)$$

4 Correct density used

$$\rho_U = 10.95155448 \frac{\text{g}}{\text{cm}^3}$$

$$N_N = \frac{0.05565624887(11.597)N_a}{14.0067}$$

$$N_N = 2.7751 \times 10^{22} \frac{\text{atoms}}{\text{cm}^3}$$

2 Substitutes N mass frac and mass

^{235}U

$$3.6272 \times 10^{21} \frac{\text{atoms}}{\text{cm}^3}$$

Correct Answer
Report your answer in this box.

^{238}U

$$2.4124 \times 10^{22} \frac{\text{atoms}}{\text{cm}^3}$$

Correct Answer
Report your answer in this box.

N

$$2.7751 \times 10^{22} \frac{\text{atoms}}{\text{cm}^3}$$

Correct Answer
Report your answer in this box.

A: 6 (10%)

B: 22 (37%)

D: 32 (53%)

Total: 60

Wt. **No.**

Cadet: _____

50

2.

During the Manhattan Project, polonium-210 was used as a neutron initiator in early atomic bomb designs. For a scheduled test 42 days from now, the initiator must contain enough polonium-210 to ensure an activity of exactly 1.5967 Curies at the moment of detonation.

Calculate the mass, in grams, of polonium-210 that must be initially included in the bomb to achieve this required activity at the time of the test. Report your answer to five significant figures.

$$T_{\frac{1}{2}} = \frac{\ln 2}{\lambda}$$

7 Equation 22

$$A = \lambda N$$

7 Equation 21

$$\lambda = \frac{\ln 2}{T_{\frac{1}{2}}}$$

$$N = \frac{A_0}{\lambda}$$

4 Converts to Bq

$$N = \frac{7.291114644 \times 10^{10}}{5.797636015 \times 10^{-8}}$$

$$\lambda = \frac{0.693147181}{138.376 \text{ days}} \quad 4 \text{ Correct half life (day or sec)}$$

$$N = 1.257601309 \times 10^{18} \text{ atoms}$$

$$\lambda = 5.009157517 \times 10^{-3} \text{ day}^{-1}$$

$$N = \frac{m N_a}{\mathcal{A}} \quad 7 \text{ Equation 20}$$

$$\lambda = 5.797636015 \times 10^{-8} \text{ sec}^{-1}$$

$$m = \frac{N \mathcal{A}}{N_a}$$

$$A(t) = A_0 e^{-\lambda t} \quad 7 \text{ Equation 25}$$

$$1.5967 = A_0 e^{-5.009157517 \times 10^{-3} (42)}$$

$$A_0 = 1.970571525 \text{ Ci} \quad 4 \text{ Substitutes } \lambda \text{ and } A$$

$$A_0 = 7.291114644 \times 10^{10} \text{ Bq}$$

4 Uses matched time units

$$m = \frac{1.257601309 \times 10^{18} (209.982857)}{N_a}$$

4 Substitutes N and $\mathcal{A}$

$$m = 4.3851 \times 10^{-4} \text{ g}$$

2 Correct Answer

$$4.3851 \times 10^{-4} \text{ g}$$

Report your answer in this box.

A: 6 (12%)
B: 16 (32%)
D: 28 (56%)
Total: 50

Wt. **No.**

Cadet: _____

- 40 3. Uranium enrichment for nuclear fuel production generates a small amount of uranium-234 as a byproduct. While ^{234}U is not fissile with thermal neutrons, it can be used in atomic weapons as it is fissionable by fast neutrons.

Calculate the minimum kinetic energy, in MeV, that a neutron must have to induce fission in ^{234}U . Report your answer to five significant figures. Note: the value for S_n in the NERD-P is incorrectly rounded and thus cannot be used.

$$S_n[^{234}\text{U}] = [M[^{234}\text{U}] + m_n - M[^{235}\text{U}]] c^2 \quad \boxed{10} \text{ Equation 15}$$

$$S_n[^{234}\text{U}] = [234.0409456 + 1.0086649233 - 235.0439231] c^2$$

$$S_n[^{238}\text{U}] = 0.0056874233 * 931.494013 \quad \textcircled{6} \text{ Correct masses}$$

$$S_n[^{238}\text{U}] = 5.297800753 \text{ MeV} \quad \textcircled{4} \text{ Converts to MeV}$$

$$E^* = S_n + T_n \quad \boxed{10} \text{ Equation 28}$$

$$T_n = E^* - S_n$$

$$T_n = 5.75 - 5.297800753 \quad \textcircled{4} \text{ Looks up any } E_{\text{crit}}$$

$$T_n = 0.45220 \text{ MeV} \quad \textcircled{4} \text{ Correctly substitutes } E_{\text{crit}}$$

0.45220 MeV $\textcircled{2}$ Correct Answer

Report your answer in this box.

A: 6 (15%)
B: 14 (35%)
D: 20 (50%)
Total: 40

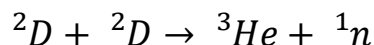
Wt. **No.**

Cadet: _____

30

4.

In boosted nuclear weapons, a small amount of deuterium-tritium (D-T) gas is placed at the core to enhance the weapon's yield through fusion. At times, a D-D fusion will occur instead. The fusion reaction is:



Calculate the threshold energy, in MeV, required for the D-D fusion reaction to occur. Report your answer to five significant figures. Note: the Q value in the NERD-P does not have enough significant figures and thus cannot be used.

$$E^K = -Q \left(1 + \frac{m_1}{m_2} \right) \quad \boxed{5} \text{ Equation 26}$$

$$Q = [M[{}^2D] + M[{}^2D] - M[{}^3He] - m_n]c^2 \quad \boxed{5} \text{ Equation 17}$$

$$Q = [2.0141017780 + 2.0141017780 - 3.0160293097 - 1.0086649233]c^2$$

$$Q = 0.003509323 * 931.494013 \quad \textcircled{4} \text{ Correct masses}$$

$$Q = 3.268913364 \text{ MeV}$$

Positive Q indicates $E^k = 0 \text{ MeV}$

$$E^C = 1.2 \frac{Z_1 Z_2}{A_1^{\frac{1}{3}} + A_2^{\frac{1}{3}}} \quad \boxed{5} \text{ Equation 27}$$

$$E^C = 1.2 \frac{1(1)}{2^{\frac{1}{3}} + 2^{\frac{1}{3}}} \quad \textcircled{4} \text{ Correct A and Z}$$

$$E^C = 0.4762203156 \text{ MeV}$$

$$E^{th} = \max(E^K, E^C)$$

$$E^{th} = 0.47622 \text{ MeV}$$

$\textcircled{5}$ Selects greater value

0.47622 MeV $\textcircled{2}$ Correct Answer

Report your answer in this box.

A: 2 (7%)
B: 13 (43%)
D: 15 (50%)
Total: 30

Wt. No.

Cadet: _____

30 5a. Uranium-233 is a potential isotope for both nuclear reactors and weapons.

(a) Predict the binding energy, in MeV, of ^{233}U using the liquid drop model. Report your answer to five significant figures.

(b) Calculate the binding energy, in MeV, of ^{233}U using the atomic mass data. Report your answer to five significant figures.

④ Coefficients correct (1 per)

④ A and Z Correct (2 per)

$$E_B = a_v A - a_s A^{\frac{2}{3}} - \frac{a_c Z^2}{A^{\frac{1}{3}}} - \frac{a_a (A - 2Z)^2}{A} - \frac{a_p}{\sqrt{A}} \quad \boxed{5} \text{ Equation 13}$$

① a_p correct

$$E_B = 15.835(233) - 18.33(233^{\frac{2}{3}}) - \frac{0.714(92^2)}{233^{\frac{1}{3}}} - \frac{23.20(233 - 2(92))^2}{233} - \frac{0}{\sqrt{233}}$$

$$E_B = 3689.555 - 694.0643625 - 982.0989034 - 239.0695279 - 0$$

$$E_B = 1774.322206 \text{ MeV}$$

$$BE = [ZM(^1_1\text{H}) + (A - Z)m_n - M(^A_Z\text{X})]c^2 \quad \boxed{5} \text{ Equation 14}$$

$$BE = [92(1.0078250321) + 141(1.0086649233) - 233.039628]c^2$$

④ A and Z correct (2 per)

$$BE = [92.71990295 + 142.2217542 - 233.039628]c^2$$

③ Masses correct

$$BE = 1.90202915 * 931.494013$$

$$BE = 1771.728766 \text{ MeV}$$

Liquid Drop Model Binding Energy

1774.3 MeV ② Correct Answer

Report your answer in this box.

Calculated Binding Energy

1771.7 MeV ② Correct Answer

Report your answer in this box.

- 10 5b. Which of the following best describes the differences between the liquid drop model and the direct calculation of nuclear binding energy?
- ☐ The liquid drop model incorporates shell effects and magic numbers, while direct binding energy calculation relies solely on semi-empirical parameters.
 - ☐ The liquid drop model is applicable only to fission processes, while direct binding energy calculation is used for fusion reactions.
 - ☐ Direct binding energy calculation accounts for pairing and asymmetry terms, while the liquid drop model focuses on Coulomb repulsion.

⑩

● Direct binding energy calculation uses experimental atomic masses whereas the liquid drop model predicts using a semi-empirical formula

A: 5 (12%)
B: 15 (38%)
D: 20 (50%)
Total: 40

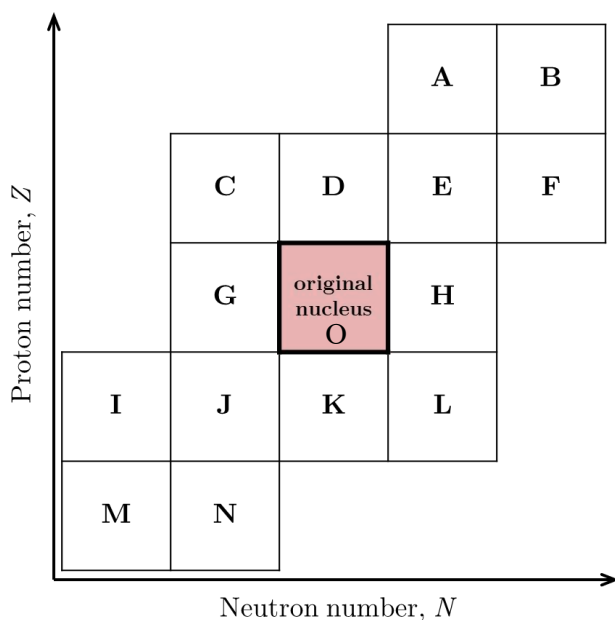
Wt. **No.**

Cadet: _____

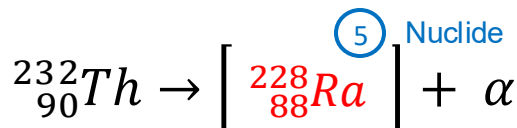
30

6.

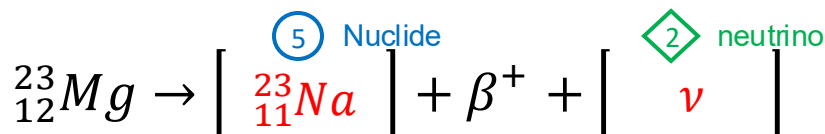
Refer to the diagram from the Chart of the Nuclides below. For each decay process shown, fill in the boxes to complete the decay equation, and then write the letter that corresponds to the relative position of the decay daughter in the blank provided.



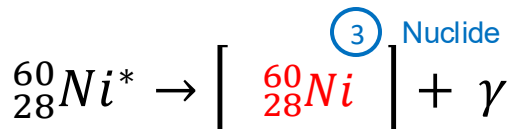
- (10) a. Thorium-232 decays via alpha emission: M 5 Position



- (10) b. Magnesium-23 decays via beta-plus emission: L 5 Position



- (10) c. Nickel-60 decays via gamma emission: O 5 Position



Outstanding Student Work

WPR1

WPR1 V2

● Graded

Student

[REDACTED]

[REDACTED]

223 / 250 pts

Question 1

Number Density

60 / 60 pts

1.1 — 235U

48 / 48 pts

✓ - 0 pts Correct (3.6272×10^{21} atoms/cm³)

U Atomic Mass

- 8 pts Doesn't use Eq 3
- 2 pts Doesn't substitute correct 235 w/o
- 2 pts Doesn't substitute correct 238 w/o

U Mass Fraction

- 8 pts Doesn't use Eq 2
- 2 pts Doesn't use correct N atomic mass

U Density

- 8 pts Doesn't use Eq 5
- 4 pts Doesn't use correct fuel density

U Number Density

- 8 pts Doesn't use equation 6
- 2 pts Doesn't substitute 235 mass fraction
- 2 pts Doesn't substitute U density
- 1 pt 235 Wrong number or sig figs
- 1 pt 235 Wrong or missing units

1.2 — 238U

4 / 4 pts

✓ - 0 pts Correct (2.8981×10^{22} atoms/cm³)

- 2 pts Doesn't substitute 238 mass fraction and/or atomic mass
- 1 pt Wrong answer and/or sig figs
- 1 pt Wrong or missing units

✓ - 0 pts Correct (3.4276×10^{22} atoms/cm³)

- 4 pts Doesn't calculate new mass fraction for N
- 2 pts Doesn't substitute N mass fraction and/or N mass
- 1 pt Wrong answer and/or sig figs
- 1 pt Wrong or missing units

– 0 pts Correct (4.3851×10^{-4} grams)

Decay Constant

– 7 pts Doesn't use Eq 22

– 4 pts Wrong half-life (138.376 day or 11,955,686.4 sec)

Activity

– 7 pts Doesn't use Eq 25

– 4 pts Doesn't substitute decay constant and/or activity

– 4 pts Doesn't keep time units constant

Number of Atoms

– 7 pts Doesn't use Eq 21

– 4 pts Doesn't convert to Bq

Mass

– 7 pts Doesn't use Eq 20

– 4 pts Doesn't substitute number of atoms and/or atomic mass

✓ – 1 pt Wrong answer and/or sig figs

– 1 pt Wrong or missing units

– 0 pts Correct (1.3345×10^{-4} grams)

Decay Constant

– 7 pts Doesn't use Eq 22

– 4 pts Wrong half-life (138.376 day or 11,955,686.4 sec)

Activity

– 7 pts Doesn't use Eq 25

– 4 pts Doesn't substitute decay constant and/or activity

– 4 pts Doesn't keep time units constant

Number of Atoms

– 7 pts Doesn't use Eq 21

– 4 pts Doesn't convert to Bq

Mass

- 7 pts Doesn't use Eq 20
- 4 pts Doesn't substitute number of atoms and/or atomic mass

- 1 pt Wrong answer and/or sig figs
- 1 pt Wrong or missing units

💬 Calculation error

Question 3

Fission

15 / 40 pts

- 0 pts Correct (0.69375 MeV)

- 10 pts Doesn't use Eq 15

✓ - 6 pts Doesn't use correct masses

- 4 pts Doesn't convert to MeV

✓ - 10 pts Doesn't use Eq 28

✓ - 4 pts Doesn't attempt to look up any E-crit

✓ - 4 pts Doesn't use correct E-crit (5.75)

✓ - 1 pt Wrong answer or sig figs

- 1 pt Wrong or missing units

- 0 pts Correct (0.69375 MeV)

- 10 pts Doesn't use Eq 15

- 6 pts Doesn't use correct masses

- 4 pts Doesn't convert to MeV

- 10 pts Doesn't use Eq 28

- 4 pts Doesn't attempt to look up any E-crit

- 4 pts Doesn't use correct E-crit (5.5)

- 1 pt Wrong answer or sig figs

- 1 pt Wrong or missing units

Question 4

Threshold Energy

29 / 30 pts

– 0 pts Correct (0.44409 MeV)

– 5 pts Doesn't use Eq 26

– 5 pts Doesn't use Eq 17

– 4 pts Doesn't use correct masses

– 5 pts Doesn't use Eq 27

– 4 pts Doesn't use correct A and/or Z

– 5 pts Doesn't select greater E value

✓ – 1 pt Wrong answer or sig figs

– 1 pt Wrong or missing units

– 0 pts Correct (0.44409 MeV)

– 5 pts Doesn't use Eq 26 or indicate $E^K = 0$

– 5 pts Doesn't use Eq 17

– 4 pts Doesn't use correct masses

– 5 pts Doesn't use Eq 27

– 2 pts Incorrect Z

– 2 pts Incorrect A

– 5 pts Doesn't select greater E value

– 1 pt Wrong answer or sig figs

– 1 pt Wrong or missing units

– 15 pts Missing coulombic barrier

Question 5

Binding Energy

40 / 40 pts

5.1 LDM

16 / 16 pts

✓ - 0 pts Correct (1774.3MeV)

- 5 pts Doesn't use Eq 13

Coefficients

- 1 pt Wrong a v (15.835)

- 1 pt Wrong a s (18.33)

- 1 pt Wrong a c (0.714)

- 1 pt Wrong a a (23.20)

- 1 pt Wrong a p (0)

- 2 pts Incorrect A

- 2 pts Incorrect Z

- 1 pt LDM wrong answer or sig figs

- 1 pt LDM wrong or missing units

5.2 BE Calc

14 / 14 pts

✓ - 0 pts Correct (1771.7MeV)

- 5 pts Doesn't use Eq 14

- 2 pts Incorrect A

- 2 pts Incorrect Z

- 3 pts Incorrect masses (any)

- 1 pt BE wrong answer or sig figs

- 1 pt BE wrong or missing units

5.3 Concept

10 / 10 pts

- 10 pts A

- 10 pts B

- 10 pts C

✓ - 0 pts D

Question 6

Decay Types

30 / 30 pts

6.1 Alpha Position

5 / 5 pts

✓ - 0 pts Correct (M)

- 5 pts Wrong position

6.2 Alpha Daughter

5 / 5 pts

✓ - 0 pts Correct (237 Np)

- 5 pts Wrong nuclide

6.3 Beta Position

5 / 5 pts

✓ - 0 pts Correct (L

- 5 pts Wrong position

6.4 Beta Daughter

5 / 5 pts

✓ - 0 pts Correct (14 N)

- 5 pts Wrong nuclide

6.5 Beta Subatomic

2 / 2 pts

✓ - 0 pts Correct (antineutrino)

- 2 pts Wrong particle

6.6 Gamma Position

5 / 5 pts

✓ - 0 pts Correct (O)

- 5 pts Wrong position

6.7 Gamma Daughter

3 / 3 pts

✓ - 0 pts Correct (131 Xe)

- 3 pts Wrong nuclide

Time of completion _____
6/7 Oct 2025

Cadet _____

(First)

(Last)

Section _____

H2

Version 2

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE300, Fundamentals of Nuclear Engineering
Written Partial Review I

1. Instructions:

- Bring only your TI-30XIIS or TI-36XPRO calculator, a straight-edge, and pencil(s) into the examination room. **No phones, cameras, or smartwatches are allowed in the examination room.**
- Check your exam for 6 problems. Check for the Reference Data Card and NERD-P provided with the examination.
- For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question.
- For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- Take up to 75 minutes to complete the examination.
- An instructor is in the hall or in the classroom for any questions.
- When you turn in your exam check your name is written on every page.**

2. Exam Security:

- I will not discuss this exam with anyone until it is released from academic security on **16 October 2025 at 1700.**
- Cadet Signature: _____
- Cadet Name (print): _____

Problem	Weight	Score
1	60	
2	50	
3	40	
4	30	
5	40	
6	30	
Total:	250	

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Wt. No.

60

1.

Cadet:

A new type of nuclear fuel is being considered for use in advanced reactors. The fuel consists of uranium nitride (UN), where the uranium is enriched to 12.927 w/o ^{235}U and the remainder is ^{238}U . The nitrogen is naturally occurring. The density of the UN fuel is 11.597 g/cm³.

Calculate the number densities (in atoms/cm³) of ^{235}U , ^{238}U , and N in the UN fuel. Report your answers to five significant figures.

$$\rho_{\text{UN}} = 11.597 \text{ g/cm}^3$$

$$w_{^{235}\text{U}} = 0.12927$$

$$w_{^{238}\text{U}} = 0.87073$$

$$A_{\text{N}} = 14.0067$$

$$A_{^{235}\text{U}} = 235.0439231$$

$$A_{^{238}\text{U}} = 238.0507826$$

$$N_{^{238}\text{U}} = \frac{w_{^{238}\text{U}} \rho_{\text{UN}} N_{\text{A}}}{A_{^{238}\text{U}}}$$

$$N_{^{238}\text{U}} = \frac{(0.87073)(11.597)(6.022 \times 10^{23})}{238.0507826}$$

$$N_{^{238}\text{U}} = 2.4124 \times 10^{22} \text{ atoms/cm}^3$$

$$N_{\text{N}} = \frac{w_{\text{N}} \rho_{\text{UN}} N_{\text{A}}}{A_{\text{N}}}$$

$$w_{\text{N}} = \frac{14.0067}{14.0067 + 237.6577633}$$

$$w_{\text{N}} = 0.055656249$$

$$N_{\text{N}} = \frac{(0.055656249)(11.597)(6.022 \times 10^{23})}{14.0067}$$

$$N_{\text{N}} = 2.7751 \times 10^{22} \text{ atoms/cm}^3$$

$$N_{^{235}\text{U}} = \frac{(0.12927)(11.597)(6.022 \times 10^{23})}{235.0439231}$$

$$N_{^{235}\text{U}} = 3.6272 \times 10^{21} \text{ atoms/cm}^3$$

$$N_{^{235}\text{U}} = \frac{w_{^{235}\text{U}} \rho_{\text{UN}} N_{\text{A}}}{A_{^{235}\text{U}}}$$

$$\rho_{\text{U}} = w_{\text{U}} \rho_{\text{UN}}$$

$$w_{\text{U}} = \frac{A_{\text{U}}}{A_{\text{U}} + A_{\text{N}}}$$

$$A_{\text{U}} = \left[\frac{w_{^{235}\text{U}}}{A_{^{235}\text{U}}} + \frac{w_{^{238}\text{U}}}{A_{^{238}\text{U}}} \right]^{-1}$$

$$A_{\text{U}} = \left[\frac{0.12927}{235.0439231} + \frac{0.87073}{238.0507826} \right]^{-1}$$

$$A_{\text{U}} = 237.6577633$$

$$w_{\text{U}} = \frac{237.6577633}{237.6577633 + 14.0067}$$

$$w_{\text{U}} = 0.944343751$$

$$\rho_{\text{U}} = 0.944343751(11.597)$$

$$\rho_{\text{U}} = 10.95155448$$

$$N_{^{235}\text{U}} = \frac{(0.12927)(10.95155448)(6.022 \times 10^{23})}{235.0439231}$$

$$N_{^{235}\text{U}} = 3.62723168 \times 10^{21} \text{ atoms/cm}^3$$

$$N_{^{238}\text{U}} = \frac{(0.87073)(10.95155448)(6.022 \times 10^{23})}{238.0507826}$$

$$N_{^{238}\text{U}} = 2.4124 \times 10^{22} \text{ atoms/cm}^3$$

$$N_{^{235}\text{U}} = 3.6272 \times 10^{21} \text{ atoms/cm}^3$$

$$N_{\text{N}} = 2.7751 \times 10^{22} \text{ atoms/cm}^3$$

Wt. No.

50

2.

Cadet: [REDACTED]

During the Manhattan Project, polonium-210 was used as a neutron initiator in early atomic bomb designs. For a scheduled test 42 days from now, the initiator must contain enough polonium-210 to ensure an activity of exactly 1.5967 Curies at the moment of detonation.

Calculate the mass, in grams, of polonium-210 that must be initially included in the bomb to achieve this required activity at the time of the test. Report your answer to five significant figures.

$$^{210}_{84}\text{Po} \\ A_{210\text{Po}} = 209.982857 \\ A = 1.5967 \text{ Ci} \\ \text{at detonation}$$

$$? \text{ mass (g) } ^{210}_{84}\text{Po} \\ \text{initially} \\ t = 42 \text{ days}$$

$$^{210}_{84}\text{Po}$$

$$T_{1/2} = 138.376 \text{ days}$$

$$A(t) = A_0 e^{-\lambda t} \\ T_{1/2} = \frac{\ln 2}{\lambda}$$

$$\lambda = \frac{\ln 2}{T_{1/2}}$$

$$\lambda = \frac{\ln 2}{11955486.4}$$

$$\lambda = 5.8 \cdot 10^{-8} \frac{\text{dec}}{\text{sec}}$$

$$A(t) = A_0 e^{-\lambda t}$$

$$\frac{A(t)}{e^{-\lambda t}} = A_0$$

$$\frac{5.90779 \cdot 10^{10}}{e^{(-5.8 \cdot 10^{-8} (3628800))}} = A_0$$

$$A_0 = 6.033449231 \cdot 10^{10} \text{ Bq}$$

$$A = \lambda N$$

$$N = \frac{A}{\lambda}$$

$$138.376 \text{ days} \cdot \frac{24 \text{ hr}}{1 \text{ day}} \cdot \frac{60 \text{ min}}{1 \text{ hr}} \cdot \frac{60 \text{ sec}}{1 \text{ min}} = 11955486.4 \text{ sec}$$

$$42 \text{ days} \cdot \frac{24 \text{ hr}}{1 \text{ day}} \cdot \frac{60 \text{ min}}{1 \text{ hr}} \cdot \frac{60 \text{ sec}}{1 \text{ min}} = 3628800 \text{ sec}$$

$$1.5967 \text{ Ci} \cdot \frac{3.7 \cdot 10^{10} \text{ Bq}}{1 \text{ Ci}} = 5.90779 \cdot 10^{10} \text{ Bq}$$

$$N = \frac{6.033449231 \cdot 10^{10}}{5.8 \cdot 10^{-8}}$$

$$N = 1.040249867 \cdot 10^{18}$$

$$N = \frac{m N_A}{A}$$

$$N A = m N_A$$

$$m = \frac{N A}{N_A} = \frac{(1.040249867 \cdot 10^{18})(209.982857)}{6.022 \cdot 10^{23}}$$

$$0.06036272 \text{ g}$$

Report your answer in this box.

$$m = 0.000362719 \text{ g}$$

Wt. No.

Cadet: XXXXXXXXXX

40

3.

Uranium enrichment for nuclear fuel production generates a small amount of uranium-234 as a byproduct. While ^{234}U is not fissile with thermal neutrons, it can be used in atomic weapons as it is fissionable by fast neutrons.

Calculate the minimum kinetic energy, in MeV, that a neutron must have to induce fission in ^{234}U . Report your answer to five significant figures. Note: the value for S_n in the NERD-P is incorrectly rounded and thus cannot be used.



T of neutron

$$Q = T_3 \left(\frac{m_3}{m_4} + 1 \right)$$

$$S_n \left({}^A_Z\text{X} \right) = [M({}^{A-1}_Z\text{X}) + m_n - M({}^A_Z\text{X})] c^2$$

$$S_n({}^{234}_{92}\text{U}) = [{}^{233}_{92}\text{U} + m_n - M({}^{234}_{92}\text{U})] c^2$$

$$S_n({}^{234}_{92}\text{U}) = [233.039628 + 1.008664913 - 234.0409456] c^2$$

$$S_n = 0.007347823 \text{ a.u.}$$

$$S_n = 6.843987666 \text{ MeV}$$

$$Q = T_3 \left(\frac{m_3}{m_4} + 1 \right)$$

$$m_3 = \text{neutron}$$

$$m_4 = {}^{234}\text{U}$$

$$\frac{Q}{\frac{m_3}{m_4} + 1} = T_3 \quad \frac{6.843987666}{\frac{1.008664913}{234.0409456} + 1} = T_3$$

$$T_3 = 6.814618163 \text{ MeV}$$

6.8146 MeV

Report your answer in this box.

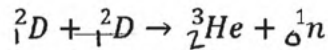
Wt. No.

Cadet: XXXXXXXXXX

30

4.

In boosted nuclear weapons, a small amount of deuterium-tritium (D-T) gas is placed at the core to enhance the weapon's yield through fusion. At times, a D-D fusion will occur instead. The fusion reaction is:



Calculate the threshold energy, in MeV, required for the D-D fusion reaction to occur. Report your answer to five significant figures. Note: the Q value in the NERD-P does not have enough significant figures and thus cannot be used.

$$Q = [\sum m_{\text{oi}} - \sum m_{\text{of}}] \quad Q = [M({}^2_1\text{H}) + M({}^2_1\text{H}) - M({}^3_2\text{He}) - m_{\text{n}}]c^2$$

$$Q = [2.0141017780 + 2.0141017780 - 3.0160293097 - 1.0086649123]c^2$$

$$Q = 0.003509323 \cdot 931.494013$$

$$Q = 3.268913364 \text{ MeV}$$

$$E^c \approx 1.2 \frac{Z_1 Z_2}{A_1^{1/3} + A_2^{1/3}}$$

$$E^c \approx 1.2 \frac{(1)(1)}{2^{1/3} + 2^{1/3}} = 0.476220316 \text{ MeV}$$

Threshold energy $E = Q + E^c$

$$E = 3.268913364 + 0.476220316$$

$$E = 3.74513368 \text{ MeV}$$

3.7451 MeV

Report your answer in this box.

Wt. No.

Cadet: XXXXXXXXXX

30 5a. Uranium-233 is a potential isotope for both nuclear reactors and weapons.

 ${}^{233}_{92}\text{U}$ (a) Predict the binding energy, in MeV, of ${}^{233}\text{U}$ using the liquid drop model. Report your answer to five significant figures.(b) Calculate the binding energy, in MeV, of ${}^{233}\text{U}$ using the atomic mass data. Report your answer to five significant figures.

liquid drop

$$E_B = a_v A - a_s A^{2/3} - a_c \frac{Z^2}{A^{1/3}} - a_a \frac{(A-2Z)^2}{A} - \frac{\pm a_p}{\sqrt{A}}$$

$$E_B = (15.835)(233) - 18.33(233)^{2/3} - 0.714 \left(\frac{92^2}{233^{1/3}} \right) - 23.20 \left(\frac{(233-2(92))^2}{233} \right) - \cancel{0}$$

$$E_B = 1774.322204$$

$$BE({}^{233}_{92}\text{U}) = [Zm({}^1_1\text{H}) + (A-Z)m_n - m({}^A_Z\text{X})]c^2$$

$$BE({}^{233}_{92}\text{U}) = [92(1.007825032) + (233-92)(1.008664923) - (233.039028)]c^2$$

$$BE = 1.902029139 \cdot 931.494013$$

$$BE = 1771.728765 \text{ MeV}$$

Liquid Drop Model Binding Energy

1774.3 MeV

Report your answer in this box.

Calculated Binding Energy

1771.7 MeV

Report your answer in this box.

- 10 5b. Which of the following best describes the differences between the liquid drop model and the direct calculation of nuclear binding energy?
- ☐ The liquid drop model incorporates shell effects and magic numbers, while direct binding energy calculation relies solely on semi-empirical parameters.
 - ☐ The liquid drop model is applicable only to fission processes, while direct binding energy calculation is used for fusion reactions.
 - ☐ Direct binding energy calculation accounts for pairing and asymmetry terms, while the liquid drop model focuses on Coulomb repulsion.
 - ☒ Direct binding energy calculation uses experimental atomic mass data, whereas the liquid drop model predicts using a semi-empirical formula.

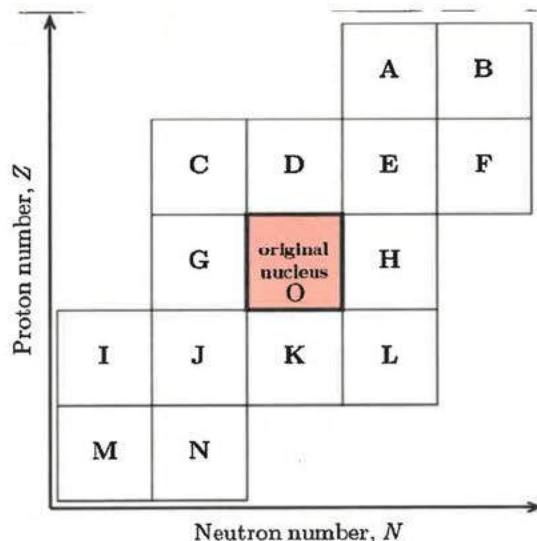
Wt. No.

30

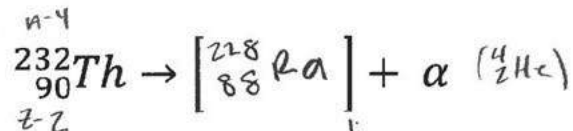
6.

Cadet: XXXXXXXXXX

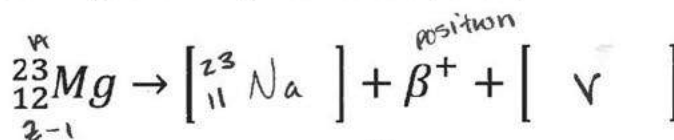
Refer to the diagram from the Chart of the Nuclides below. For each decay process shown, fill in the boxes to complete the decay equation, and then write the letter that corresponds to the relative position of the decay daughter in the blank provided.



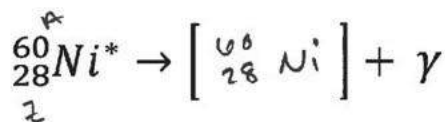
- (10) a. Thorium-232 decays via alpha emission: M



- (10) b. Magnesium-23 decays via beta-plus emission: L



- (10) c. Nickel-60 decays via gamma emission: O



Poor Student Work

Student



Total Points
187 / 250 pts

Question 1

Number Density

60 / 60 pts

1.1 — 235U

48 / 48 pts

✓ - 0 pts Correct (5.2958×10^{21} atoms/cm³)

1.2 — 238U

4 / 4 pts

✓ - 0 pts Correct (2.8981×10^{22} atoms/cm³)

1.3 — N

8 / 8 pts

✓ - 0 pts Correct (3.4276×10^{22} atoms/cm³)

Question 2

Decay

34 / 50 pts

Activity

✓ - 7 pts Doesn't use Eq 25

✓ - 4 pts Doesn't substitute decay constant and/or activity

✓ - 4 pts Doesn't keep time units constant

✓ - 1 pt Wrong answer and/or sig figs

Question 3

Fission

1 / 40 pts

- ✓ - 10 pts Doesn't use Eq 15
- ✓ - 6 pts Doesn't use correct masses
- ✓ - 4 pts Doesn't convert to MeV
- ✓ - 10 pts Doesn't use Eq 28
- ✓ - 4 pts Doesn't attempt to look up any E-crit
- ✓ - 4 pts Doesn't use correct E-crit (5.5)
- ✓ - 1 pt Wrong answer or sig figs

Question 4

Threshold Energy

30 / 30 pts

- ✓ - 0 pts Correct (0.44409 MeV)

Question 5

Binding Energy

34 / 40 pts

5.1 LDM

10 / 16 pts

Coefficients

- ✓ - 5 pts All Coefficients wrong/missing

- ✓ - 1 pt LDM wrong answer or sig figs

5.2 BE Calc

14 / 14 pts

- ✓ - 0 pts Correct (1806.9 MeV)

5.3 Concept

10 / 10 pts

- ✓ - 0 pts B

Question 6

Decay Types

28 / 30 pts

6.1 — Alpha Position 5 / 5 pts

✓ - 0 pts Correct (M)

6.2 — Alpha Daughter 5 / 5 pts

✓ - 0 pts Correct (237 Np)

6.3 — Beta Position 5 / 5 pts

✓ - 0 pts Correct (C)

6.4 — Beta Daughter 5 / 5 pts

✓ - 0 pts Correct (14 N)

6.5 — Beta Subatomic 0 / 2 pts

✓ - 2 pts Wrong particle

6.6 — Gamma Position 5 / 5 pts

✓ - 0 pts Correct (O)

6.7 — Gamma Daughter 3 / 3 pts

✓ - 0 pts Correct (131 Xe)

Time of completion 1525
6/7 Oct 2025

Cadet

(First)

(Last)

Section

J1

Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE300, Fundamentals of Nuclear Engineering
Written Partial Review I

1. Instructions:

- Bring only your TI-30XIIS or TI-36XPRO calculator, a straight-edge, and pencil(s) into the examination room. **No phones, cameras, or smartwatches are allowed in the examination room.**
- Check your exam for 6 problems. Check for the Reference Data Card and NERD-P provided with the examination.
- For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question.
- For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- Take up to 75 minutes to complete the examination.
- An instructor is in the hall or in the classroom for any questions.
- When you turn in your exam check your name is written on every page.**

2. Exam Security:

- I will not discuss this exam with anyone until it is released from academic security on **16 October 2025 at 1700.**
- Cadet Signature: _____
- Cadet Name (print): _____

Problem	Weight	Score
1	60	
2	50	
3	40	
4	30	
5	40	
6	30	
Total:	250	

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Wt. No.

Cadet: _____

60

1.

A new type of nuclear fuel is being considered for use in advanced reactors. The fuel consists of uranium nitride (UN), where the uranium is enriched to 15.285 w/o ^{235}U and the remainder is ^{238}U . The nitrogen is naturally occurring. The density of the UN fuel is 14.320 g/cm^3 .

Calculate the number densities (in atoms/cm³) of ^{235}U , ^{238}U , and N in the UN fuel. Report your answers to five significant figures.

$$w/o \text{ } ^{235}\text{U} = 0.15285 \quad \rho_{\text{UN}} = 14.320 \text{ g/cm}^3$$

$$w/o \text{ } ^{238}\text{U} = 0.84715$$

$$N_{\text{U}235} = \frac{w_{\text{U}235} \rho_{\text{U}} N_{\text{A}}}{A_{\text{U}235}} \quad A_{\text{U}} = \left(\frac{w_{\text{U}235}}{A_{\text{U}235}} + \frac{w_{\text{U}238}}{A_{\text{U}238}} \right)^{-1} = 237.5862130$$

$$\rho_{\text{U}} = w_{\text{U}} \rho_{\text{UN}} = 13.52277586$$

$$N_{\text{U}235} = 5.2958 \times 10^{21} \text{ atoms/cm}^3$$

$$w_{\text{U}} = \frac{1 \cdot A_{\text{U}}}{1 \cdot A_{\text{U}} + 1 \cdot A_{\text{N}}} = 0.944327923$$

$$N_{\text{U}238} = \frac{w_{\text{U}238} \rho_{\text{U}} N_{\text{A}}}{A_{\text{U}238}} = 2.8981 \times 10^{22} \text{ atoms/cm}^3$$

$$N_{\text{N}} = \frac{w_{\text{N}} \rho_{\text{UN}} N_{\text{A}}}{A_{\text{N}}} = 3.4276 \times 10^{22} \text{ atoms/cm}^3$$

$$w_{\text{N}} = \frac{A_{\text{N}}}{A_{\text{N}} + A_{\text{U}}} = 0.055672077$$

$$^{235}\text{U} \quad 5.2958 \times 10^{21} \text{ atoms/cm}^3$$

Report your answer in this box.

$$^{238}\text{U} \quad 2.8981 \times 10^{22} \text{ atoms/cm}^3$$

Report your answer in this box.

$$\text{N} \quad 3.4276 \times 10^{22} \text{ atoms/cm}^3$$

Report your answer in this box.

Wt. No.

50

2.

Cadet: _____

During the Manhattan Project, polonium-210 was used as a neutron initiator in early atomic bomb designs. For a scheduled test 42 days from now, the initiator must contain enough polonium-210 to ensure an activity of exactly 0.48592 Curies at the moment of detonation.

Calculate the mass, in grams, of polonium-210 that must be initially included in the bomb to achieve this required activity at the time of the test. Report your answer to five significant figures.

$$t = 42 \text{ d}$$

$$A = 0.48592 \text{ Ci}$$

$$m = ?$$

$$N = \frac{m N_a}{A} \rightarrow \frac{N A}{N_a} = m$$

$$A = \lambda N \rightarrow \frac{A}{\lambda} = N$$

~~scribble~~

$$\lambda = \frac{\ln 2}{T_{1/2}}$$

~~scribble~~

$$\frac{\left(\frac{A}{\lambda} \right) A}{N_a} = m$$

$$\frac{\left(\frac{A}{\frac{\ln 2}{T_{1/2}}} \right) A_{\text{Po210}}}{N_a} = m$$

$$m = 7.5648 \times 10^{-19} \text{ g}$$

$$T_{1/2 \text{ Po210}} = 138.376 \text{ d} \quad \begin{array}{|c|c|c|c|} \hline 24 \text{ hrs} & 60 \text{ min} & 60 \text{ s} & \\ \hline 1 \text{ d} & 1 \text{ hr} & 1 \text{ min} & \\ \hline \end{array}$$

$$T_{1/2 \text{ Po210}} = 11955686.4 \text{ s}$$

$$\frac{0.48592 \text{ Ci}}{1 \text{ Ci}} \times \frac{3.7 \times 10^{10} \text{ Bq}}{1 \text{ Ci}} = 1.797904 \times 10^{10} \text{ Bq} = A$$

$$7.5648 \times 10^{-19} \text{ g}$$

Report your answer in this box.

Wt. No.

40

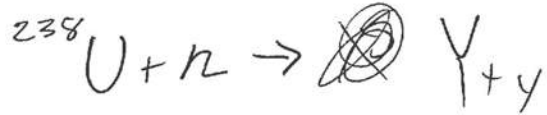
3.

Cadet: _____

Uranium enrichment for nuclear fuel production generates large quantities of uranium-238 as a byproduct. While ^{238}U is not fissile with thermal neutrons, it can be used in atomic weapons as a tamper, and as an added benefit, it is fissionable by fast neutrons.

Calculate the minimum kinetic energy, in MeV, that a neutron must have to induce fission in ^{238}U . Report your answer to five significant figures. Note: the value for S_n in the NERD-P is incorrectly rounded and thus cannot be used.

$$E_i^{\text{th}} = \max(E_i^{\text{K}}, E_i^{\text{C}})$$



$$E_i^{\text{C}} = 0 \text{ since its neutral}$$

$$Q = T_f - T_i$$

$$T_i = T_f - Q \quad T_i = E^{\text{th}} - Q$$

$$E'_{\min} = \alpha E \quad ^{238}\text{U}$$

$$\alpha = \left(\frac{238-1}{238+1} \right)^2$$

$$E(^{238}_{92}\text{U}) = [92M(^1_1\text{H}) + (238-92)m_n - M(^{238}_{92}\text{U})]c^2$$
$$= 1.934199155 \text{ u c}^2, 931.494013 \text{ MeV/c}^2$$

$$E = 1801.694933 \text{ MeV}$$

$$E'_{\min} = 1771.66721 \text{ MeV}$$

1771, 7 MeV

Report your answer in this box.

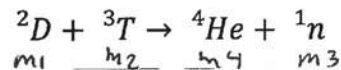
Wt. No.

30

4.

Cadet: _____

In boosted nuclear weapons, a small amount of deuterium-tritium (D-T) gas is placed at the core to enhance the weapon's yield through fusion. The fusion reaction is:



Calculate the threshold energy, in MeV, required for the D-T fusion reaction to occur. Report your answer to five significant figures. Note: the Q value in the NERD-P does not have enough significant figures and thus cannot be used.

$$Q = [M({}^2_1\text{H}) + M({}^3_1\text{H}) - M({}^4_2\text{He}) - m_n] c^2$$

$$= 0.018882873 \text{ u} - 0.001008665 \text{ u} = 0.017874208 \text{ u}$$

$$= 17.58928268 \text{ MeV}$$

$$E_1^K \approx -\left(1 + \frac{m_{2p}}{m_{3T}}\right) Q$$

$$\approx -29.3353128 \text{ MeV}$$

$$E_1^C \approx 1.2 \frac{1.1}{2^{\frac{1}{3}} + 3^{\frac{1}{3}}} \approx 0.444087428 \text{ MeV}$$

$$E_1^{th} = \max(E_1^K, E_1^C)$$

~~0.44409~~ 0.44409 MeV

Report your answer in this box.

Wt. No.

30

5a.

Cadet: _____

Plutonium-239 is a key isotope in both nuclear reactors and weapons.

(a) Predict the binding energy, in MeV, of ^{239}Pu using the liquid drop model. Report your answer to five significant figures.

(b) Calculate the binding energy, in MeV, of ^{239}Pu using the atomic mass data. Report your answer to five significant figures.

$N=145$ $^{239}_{94}\text{Pu} \rightarrow e/o = 0$

a) ~~$E_B = a_v (239)^{2/3} - a_s \frac{94^2}{239^{1/3}} - a_c \frac{(239-2(94))^2}{239} - \frac{0}{\sqrt{239}}$~~

$$E_B = a_v 239 - a_s (239)^{2/3} - a_c \frac{94^2}{239^{1/3}} - a_a \frac{(239-2 \cdot 94)^2}{239} - \frac{0}{\sqrt{239}}$$

$$= 2401.719713 \text{ MeV}$$

$$BE(^{239}_{94}\text{Pu}) = [94 \cdot M(^1_1\text{H}) + (239-94)m_n - M(^{239}_{94}\text{Pu})]c^2$$

$$= 1.939810396 \text{ u} \cdot c^2 \cdot 931.494013 \text{ MeV/u}$$

$$= 1806.92177 \text{ MeV}$$

Liquid Drop Model Binding Energy

2401.7 MeV

Report your answer in this box.

Calculated Binding Energy

1806.9 MeV

Report your answer in this box.

- 10 5b. Which of the following best describes the differences between the liquid drop model and the direct calculation of nuclear binding energy?
- ☐ The liquid drop model incorporates shell effects and magic numbers, while direct binding energy calculation relies solely on semi-empirical parameters.
 - ☒ Direct binding energy calculation uses experimental atomic mass data, whereas the liquid drop model predicts using a semi-empirical formula.
 - ☐ The liquid drop model is applicable only to fission processes, while direct binding energy calculation is used for fusion reactions.
 - ☐ Direct binding energy calculation accounts for pairing and asymmetry terms, while the liquid drop model focuses on Coulomb repulsion.

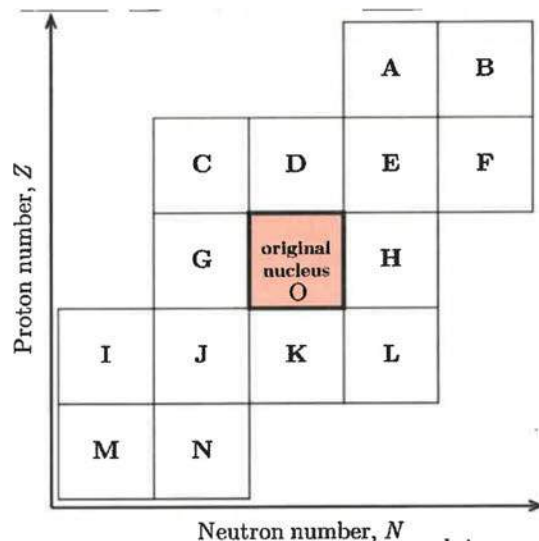
Wt. No.

30

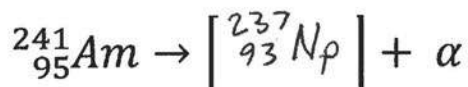
6.

Cadet: _____

Refer to the diagram from the Chart of the Nuclides below. For each decay process shown, fill in the boxes to complete the decay equation, and then write the letter that corresponds to the relative position of the decay daughter in the blank provided.



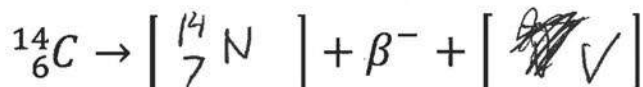
- (10) a. Americium-241 decays via alpha emission: M



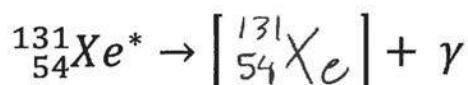
- (10) b. Carbon-14 decays via beta-minus emission: C

β^-

$Z+1$



- (10) c. Xenon-131 decays via gamma emission: O





UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB H

WPR #2

Instructor Solution

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE300, Fundamentals of Nuclear Engineering
Written Partial Review II

1. Instructions:

- a. Bring only your TI-30XIIS or TI-36XPRO calculator, a straight-edge, and pencil(s) into the examination room. **No phones, cameras, or smartwatches are allowed in the examination room.**
- b. Check your exam for 6 problems. Check for the Reference Data Card and NERD-P provided with the examination.
- c. For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question.
- d. For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- e. Take up to 75 minutes to complete the examination.
- f. An instructor is in the hall or in the classroom for any questions.
- g. When you turn in your exam check your name is written on every page.**

2. Exam Security:

- a. I will not discuss this exam with anyone until it is released from academic security on **28 November 2025 at 1700.**
- b. Cadet Signature: _____
- c. Cadet Name (print): _____

A: 23 (9%)

B: 87 (35%)

D: 140 (56%)

Total: 250

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Wt. **No.**

Cadet: _____

60 1 Sheldon, a clumsy cat, leaps onto MAJ Pert's desk in search of scratches. In doing so, he knocks a 6.70 mCi source of 0.200 MeV photons into a spherical fishbowl filled with heavy water. The diameter of the fishbowl is 13.4 cm and the source is suspended in the center.

(50) a. Calculate the intensity of uncollided photons MAJ Pert is exposed to 60.3 cm away from the edge of the bowl. You can neglect attenuation from the bowl itself and air. Note: light and heavy water have the same mass attenuation coefficients. Report your answer to **three** significant figures.

12 Geometric component

12 Decay component

$$\phi = \frac{S_p}{4\pi r^2} e^{-\mu_t x}$$

4 Converts to Bq

2 Includes any density

2 Subs activity

4 Correct density

$$\phi = \frac{2.479 \times 10^8 \text{ s}^{-1}}{4\pi (67 \text{ cm})^2} e^{-0.136 \frac{\text{cm}^2}{\text{g}} * 1.106 \frac{\text{g}}{\text{cm}^3} * 6.7 \text{ cm}}$$

4 Correct MAC

4 Correct x value

4 Correct r value

$$\phi = \frac{2.479 \times 10^8 \text{ s}^{-1}}{56410 \text{ cm}^2} e^{-1.0077872}$$

$$\phi = 1.605596 \times 10^3 \text{ cm}^{-2} \text{ s}^{-1}$$

$$1.61 \times 10^3 \frac{1}{\text{cm}^2 \text{ s}}$$

Report your answer in this box.

2 Correct Answer/Units

(10) b. Suppose the water in the fishbowl is replaced with a material that has a higher linear attenuation coefficient, but the mass attenuation coefficient remains the same. The source and geometry are unchanged. Which of the following best explains what happens to the fraction of uncollided photons reaching MAJ Pert, and why?

1

3

6

- ☒ The fraction decreases, because a higher linear attenuation coefficient means photons are more likely to interact per unit length.
- ☐ The fraction increases, because a higher linear attenuation coefficient means photons are less likely to interact per unit length.
- ☐ The fraction remains unchanged, because the mass attenuation coefficient is the same.
- ☐ The fraction could increase or decrease, depending on the density of the new material.

A: 7 (12%)
B: 23 (38%)
D: 30 (50%)
Total: 60

Wt. **No.**

60

2

Cadet: _____

Pam, a small cat, slips into MAJ Pert's gardening shed, where an AmBe source is producing collimated thermal neutrons for illicit study. The neutron beam is incident on a sample of aluminum oxide (Al_2O_3) body armor with a density of 3.99 g/cm^3 and Pam lays down on the other side.

Calculate the thickness of the tank armor required to protect Pam from 67% of the uncollided neutrons. Use natural compositions of elements. Report your answer to **three** significant figures.

6 Equation 2

$$\omega_{Al} = \frac{4\mathcal{A}_{Al}}{4\mathcal{A}_{Al} + \mathcal{A}_O}$$

2 Correct coefficients

2 Correct masses

$$\omega_{Al} = \frac{2 \left(26.9815 \frac{\text{g}}{\text{mol}} \right)}{2 \left(26.9815 \frac{\text{g}}{\text{mol}} \right) + 3 \left(15.9994 \frac{\text{g}}{\text{mol}} \right)}$$

$$\omega_{Al} = 0.52925$$

$$\omega_O = 1 - \omega_{Al} = 0.47075$$

6 Equation 6

$$N_{Al} = \frac{\omega_{Al} \rho_{\text{Al}_2\text{O}_3} N_a}{\mathcal{A}_{Al}}$$

2 Subs w/o

2 Correct density

$$N_{Al} = \frac{0.52925 \left(3.99 \frac{\text{g}}{\text{cm}^3} \right) N_a}{26.9815 \frac{\text{g}}{\text{mol}}}$$

2 Correct mass

$$N_{Al} = 4.71323 \times 10^{22} \frac{\text{atoms}}{\text{cm}^3}$$

$$N_O = \frac{3N_{Al}}{2} = 7.06985 \times 10^{22} \frac{\text{atoms}}{\text{cm}^3}$$

4 Finds other N

6 Equation 35

$$\Sigma_t = \Sigma_s + \Sigma_a$$

6 Equation 37

$$\Sigma_t = N_{Al} \sigma_{s,Al} + N_O \sigma_{s,O} + N_{Al} \sigma_{a,Al} + N_O \sigma_{a,O}$$

$$\Sigma_t = N_{Al} 1.49 + N_O 3.76 + N_{Al} 0.230 + N_O 0.00027$$

$$\Sigma_t = 0.07023 + 0.26583 + 0.01084 + 1.9089 \times 10^{-5}$$

$$\Sigma_t = 0.33606 + 0.01086$$

2 Absorption includes both elements

$$\Sigma_t = 0.34691 \text{ cm}^{-1}$$

2 Scatter includes both elements

6 Equation 31 $\phi = \phi_0 e^{-\Sigma_t x}$

$$\frac{\phi}{\phi_0} = e^{-\Sigma_t x}$$

4 Microscopic XS correct (1 per)

$$\ln\left(\frac{\phi}{\phi_0}\right) = -\Sigma_t x$$

2 Takes natural log

$$\frac{\ln(0.33)}{-0.34691 \text{ cm}^{-1}} = x$$

4 Correct ratio

$$3.1957938 \text{ cm} = x$$

2 Correct Answer/Units

3.20 cm

Report your answer in this box.

A: 6 (10%)
B: 24 (40%)
D: 30 (50%)
Total: 60

Wt. No.

Cadet: _____

70 3 In 1958 the first criticality accident in a uranium processing facility occurred at the Y-12 enrichment plant. A 1:750 mix of uranyl nitrate ($^{235}\text{UO}_8\text{N}_2$) and light water was accidentally allowed to drain into a cylindrical tank at 20°C until criticality was achieved. Assume $\epsilon p = 1.0670$. Report all answers to **five** significant figures.

- (30) a. Calculate the thermal utilization factor for this solution.
 (10) b. If 2.067 fast neutrons are produced per thermal neutron absorption in fuel, determine the multiplication factor for an infinitely large tank.
 (30) c. If the diameter of the tank is 67.000 cm, what was the height of the mixture in the tank? Assume negligible thermal neutron leakage.

8 Equation 51

$$f = \frac{\Sigma_a^F}{\Sigma_a^F + \Sigma_a^{NF}}$$

$k_\infty = \epsilon p f \eta$ **6** Equation 48

$$k_\infty = 1.067(0.572506)(2.067)$$

6 Equation 35

2 Identifies correct fuel

$k_\infty = 1.262655$ **2** Subs eta

$$f = \frac{N_{235}\sigma_{a235}g_{a235}}{N_{235}\sigma_{a235}g_{a235} + N_O\sigma_{aO} + N_N\sigma_{aN} + N_{H_2O}\sigma_{H_2O}}$$

2 Identifies correct non fuels

$$f = \frac{\gamma_{235}\sigma_{a235}g_{a235}}{\gamma_{235}\sigma_{a235}g_{a235} + \gamma_O\sigma_{aO} + \gamma_N\sigma_{aN} + \gamma_{H_2O}\sigma_{H_2O}}$$

4 Divides by Uranium number density

$$f = \frac{1(687)(0.0978)}{1(687)(0.0978) + 8(0.00027)(1) + 2(1.85)(1) + 750(0.664)(1)}$$

2 Correct coefficients

$1(687)(0.0978)$ **2** Correct microscopic XS

$$f = 0.572506$$

2 Correct Wescott

8 Equation 47

$$k_{eff} = k_\infty P_{NL}^f P_{NL}^{th}$$

6 Equation 49

$$1 = k_\infty e^{-B_C^2 \tau_T} (1)$$

2 Thermal NL = 1

$$\ln\left(\frac{1}{k_\infty}\right) = -B_C^2 \tau_T$$

2 Takes natural log

$$\sqrt{\frac{\pi}{\tau_T} - \left(\frac{2.405}{R}\right)^2} = H$$

$$\sqrt{\frac{\pi}{\tau_T} - \left(\frac{2.405}{R}\right)^2} = H$$

$$53.2266559 \text{ cm} = H$$

$$\frac{\ln\left(\frac{1}{k_\infty}\right)}{\tau_T} = -\left(\frac{2.405}{R}\right)^2 - \left(\frac{\pi}{H}\right)^2$$

4 Correct buckling eq

2 Correct Fermi age

2 Subs k_∞

2 Correct radius

2 Correct Answer/Units
0.57251

Report your answer in this box.

2 Correct Answer/Units
1.2627

Report your answer in this box.

2 Correct Answer/Units
53.227

Report your answer in this box.

A: 6 (9%)
B: 30 (43%)
D: 34 (49%)
Total: 70

Wt. **No.**

Cadet: _____

20 4 For the particles and media outlined below, select the most likely interaction mechanism.

(5) a. Thermal neutron in krypton (Kr)

☐ Photoelectric Effect

☒ 2 ☐ Scattering

☐ Compton Scattering

☐ Coulombic

☐ Pair Production

☐ Bremsstrahlung Radiation

☒ 1 ☐ 1 ☒ 3 ☒ Absorption

☐ Cherenkov Radiation

(5) b. 1 keV gamma particle in lead

☒ 1 ☐ 1 ☒ 3 ☒ Photoelectric Effect

☐ Scattering

☒ 2 ☐ Compton Scattering

☐ Coulombic

☒ 2 ☐ Pair Production

☐ Bremsstrahlung Radiation

☐ Absorption

☐ Cherenkov Radiation

(5) c. 5 MeV alpha particle in tissue

☐ Photoelectric Effect

☐ Scattering

☐ Compton Scattering

☒ 1 ☐ 1 ☒ 3 ☒ Coulombic

☐ Pair Production

☒ 2 ☐ Bremsstrahlung Radiation

☐ Absorption

☒ 2 ☐ Cherenkov Radiation

(5) d. 2 keV beta particle in silver (Ag)

☐ Photoelectric Effect

☐ Scattering

☐ Compton Scattering

☒ 1 ☐ 1 ☒ 3 ☒ Coulombic

☐ Pair Production

☒ Bremsstrahlung Radiation

☐ Absorption

☒ 2 ☐ Cherenkov Radiation

A: 4 (20%)
B: 4 (20%)
D: 12 (60%)
Total: 20

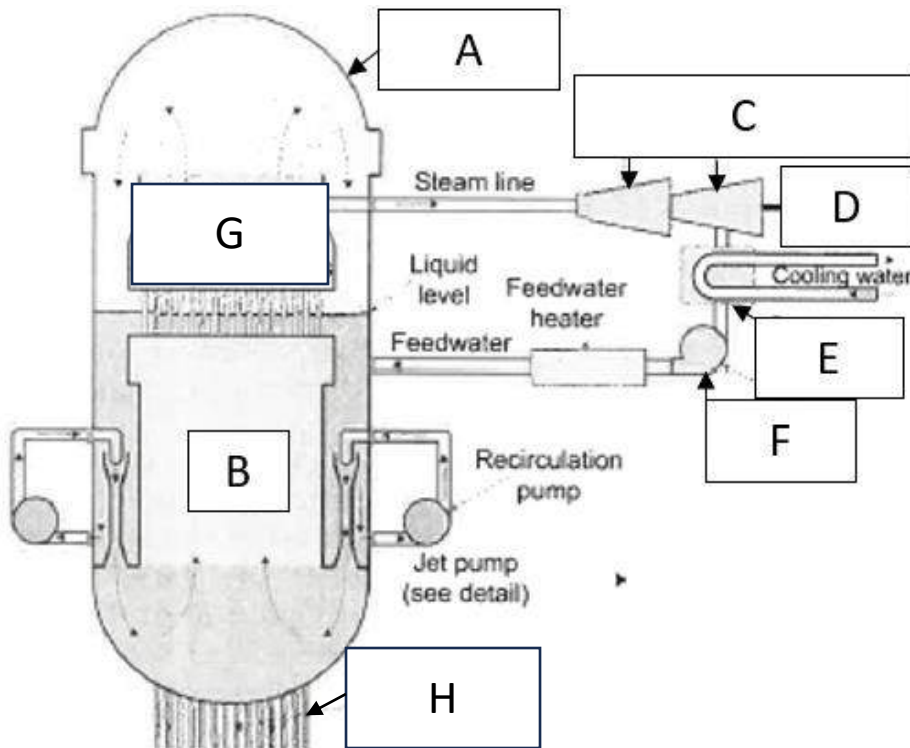
Wt. **No.**

20 5

Cadet: _____

Refer to the diagram of the reactor system below. Match the lettered components on the diagram to the most proper reactor system component listed below and answer the follow-on questions at the bottom of the page.

(16) a.



Component	Letter
Condenser	E
Control Rods	H
Core	B
Generator	D
Pressure Vessel	A
Reactor Coolant Pump	F
Steam Dryer	G
Turbine	C

2

2

2

2

2

2

2

2

(4) b. The reactor depicted above is a:

☐ Pressurized Water Reactor

☒ Boiling Water Reactor

☐ Heavy Water Reactor

☐ Gas Cooled Reactor

☐ Liquid Metal Fast Breeder Reactor

A: 0 (0%)

B: 0 (0%)

D: 20 (100%)

Total: 20

Wt. No.

Cadet: _____

10

6

Complete the missing sections of the once-through fuel cycle flow chart below
(Process steps are boarded by a solid rectangle, materials by a dashed rectangle).

2

Mining

Uranium Ore

Milling

U_3O_8

2

2 Conversion

UF_6

Enrichment

UF_6

2

Fuel Fabrication

UO_2

2

10 7 For the various uses or conditions outlined below, select the most appropriate detector.

(5) a. Rapid radiation surveys of large geographical areas

☐ Semiconductor Detector

☐ Scintillation Detector

5 ☒ Gas-filled Detector

(5) b. Nuclear forensics of weapon fallout for attribution to responsible actors

5 ☒ Semiconductor Detector

☐ Scintillation Detector

☐ Gas-filled Detector

Outstanding Student Work

Student

Total Points
224 / 250 pts

Question 1

Photon Attenuation

58 / 60 pts

1.1 — Calculation

48 / 50 pts

✓ - 1 pt Incorrect answer

✓ - 1 pt Incorrect Units

💬 think through units

1.2 — Concept

10 / 10 pts

✓ - 0 pts A (Correct)

Question 2

Neutron Attenuation

60 / 60 pts

✓ - 0 pts Correct - 3.20 cm

Question 3

6 Factor Formula

55 / 70 pts

3.1 Thermal Utilization

23 / 30 pts

✓ - 2 pts Does not identify correct non-fuels (O, N, H₂O)

✓ - 2 pts Incorrect coefficients

✓ - 2 pts Incorrect micro XS (any)

✓ - 1 pt Incorrect Answer

3.2 k infinite

7 / 10 pts

✓ - 2 pts Does not substitute correct eta (2.067)

✓ - 1 pt Incorrect Answer

3.3 Buckling

25 / 30 pts

✓ - 2 pts Does not take natural log

✓ - 2 pts Does not use correct fermi age (27)

✓ - 1 pt Incorrect answer

Question 4

Interactions

15 / 20 pts

4.1 Neutron

0 / 5 pts

✓ - 5 pts B

4.2 Photon

5 / 5 pts

✓ - 0 pts A (Correct)

4.3 Alpha

5 / 5 pts

✓ - 0 pts F (Correct)

4.4 Beta

5 / 5 pts

✓ - 0 pts G (Correct)

Question 5

Reactors

20 / 20 pts

5.1 Components

16 / 16 pts

✓ - 0 pts Correct

5.2 Type

4 / 4 pts

✓ - 0 pts B (Correct)

Question 6

Fuel Cycle

6 / 10 pts

✓ - 2 pts Incorrect Purification/Conversion

✓ - 2 pts Incorrect UF6/greensalt

Question 7

Detectors

10 / 10 pts

7.1 Gas Filled

5 / 5 pts

✓ - 0 pts C (Correct)

7.2 Semiconductor

5 / 5 pts

✓ - 0 pts A (Correct)

Time of completion 0849
19/20 Nov 2025

Cadet

(First)

(Last)

Section

62
Version 2

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE300, Fundamentals of Nuclear Engineering
Written Partial Review II

1. Instructions:

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- Take up to 75 minutes to complete the examination.
- An instructor is in the hall or in the classroom for any questions.
- When you turn in your exam check your name is written on every page.**

2. Exam Security:

- I will not discuss this exam with anyone until it is released from academic security on **28 November 2025 at 1700.**
- Cadet Signature: _____
- Cadet Name (print): _____

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Wt. No.

60

1

Cadet: _____

Sheldon, a clumsy cat, leaps onto MAJ Pert's desk in search of scratches. In doing so, he knocks a 6.70 mCi source of 0.200 MeV photons into a spherical fishbowl filled with heavy water. The diameter of the fishbowl is 13.4 cm and the source is suspended in the center.

$$r = 73.7 \text{ cm}$$

(50)

- a. Calculate the intensity of uncollided photons MAJ Pert is exposed to 60.3 cm away from the edge of the bowl. You can neglect attenuation from the bowl itself and air. Note: light and heavy water have the same mass attenuation coefficients. Report your answer to **three** significant figures.

$$I(x) = \frac{S_p}{4\pi r^2} e^{-\mu_t x}$$

$$\frac{6.70 \text{ mCi}}{1} \times \frac{1 \text{ Ci}}{1000 \text{ mCi}} \times \frac{3.7 \times 10^{10} \text{ Bq}}{1 \text{ Ci}} = 2.479 \times 10^8 \text{ Bq}$$

$$S_p = 2.479 \times 10^8 \text{ Bq}$$

$$\mu_t = \left(\frac{\mu_t}{\rho} \right) \rho \quad \mu_t = .136 (1.105)$$

$$I(1) = \frac{2.479 \times 10^8}{4\pi (67)^2} e^{-.15028 (6.7)}$$

$$I = 1605.596 \text{ cm}^{-1}$$

$$1.61 \text{ cm}^{-1}$$

Report your answer in this box.

(10)

- b. Suppose the water in the fishbowl is replaced with a material that has a higher linear attenuation coefficient, but the mass attenuation coefficient remains the same. The source and geometry are unchanged. Which of the following best explains what happens to the fraction of uncollided photons reaching MAJ Pert, and why?
- ☒ The fraction decreases, because a higher linear attenuation coefficient means photons are more likely to interact per unit length.
 - ☐ The fraction increases, because a higher linear attenuation coefficient means photons are less likely to interact per unit length.
 - ☐ The fraction remains unchanged, because the mass attenuation coefficient is the same.
 - ☐ The fraction could increase or decrease, depending on the density of the new material.

Wt. No.

60

2

Cadet: _____

Pam, a small cat, slips into MAJ Pert's gardening shed, where an AmBe source is producing collimated thermal neutrons for illicit study. The neutron beam is incident on a sample of aluminum oxide (Al_2O_3) body armor with a density of 3.99 g/cm^3 and Pam lays down on the other side.

Calculate the thickness of the tank armor required to protect Pam from 67% of the uncollided neutrons. Use natural compositions of elements. Report your answer to three significant figures.

$$\rho_{\text{Al}_2\text{O}_3} = 3.99 \text{ g/cm}^3 \quad \Sigma_i = \frac{N_{\sigma_i}(E)g_i(T)}{x}$$

$$I = I_0 e^{-\Sigma_t x} \quad N_{\text{Al}} = \frac{w_{\text{Al}} \rho_{\text{Al}_2\text{O}_3} N_A}{A_{\text{Al}}}$$

$$I_0 = 1$$

$$I = .33$$

$$x = ?$$

$$w_{\text{Al}} = \frac{n A_{\text{Al}}}{n A_{\text{Al}} + m A_{\text{O}}}$$

$$w_{\text{Al}} = \frac{2(26.9815)}{2(26.9815) + 3(15.9994)}$$

$$w_{\text{O}} = 0.470749658$$

$$w_{\text{Al}} = .529250342$$

$$N_{\text{Al}} = \frac{.529250342(3.99)N_A}{26.9815}$$

$$N_{\text{Al}} = 4.713233371 \times 10^{22}$$

$$N_{\text{O}} = \frac{.470749658(3.99)N_A}{15.9994}$$

$$N_{\text{O}} = 7.069850069 \times 10^{22}$$

$$I = I_0 e^{-\Sigma_t x}$$

$$\ln \frac{I}{I_0} = -\Sigma_t x$$

$$-\ln \left(\frac{I}{I_0} \right) = \Sigma_t x$$

$$\Sigma_{\text{Al},a} = 4.713233371 \times 10^{22} (.230 \times 10^{-24})(1)$$

$$\Sigma_{\text{Al},a} = .010840437$$

$$\Sigma_{\text{Al},s} = 4.713233371 \times 10^{22} (1.49 \times 10^{-24})(1)$$

$$\Sigma_{\text{Al},s} = .070227177$$

$$\Sigma_{\text{O},a} = 7.069850069 \times 10^{22} (.00027 \times 10^{-24})(1)$$

$$\Sigma_{\text{O},a} = .000019089$$

$$\Sigma_{\text{O},s} = 7.069850069 \times 10^{22} (3.76 \times 10^{-24})(1)$$

$$\Sigma_{\text{O},s} = 0.265826363$$

$$\Sigma_t = \Sigma_{\text{Al},a} + \Sigma_{\text{Al},s} + \Sigma_{\text{O},a} + \Sigma_{\text{O},s}$$

$$\Sigma_t = 0.346913066$$

$$x = \frac{-\ln(.33)}{0.346913066}$$

$$x = 3.20 \text{ cm}$$

3.20cm

Report your answer in this box.

Wt. No.

Cadet: _____

70

3

In 1958 the first criticality accident in a uranium processing facility occurred at the Y-12 enrichment plant. A 1:750 mix of uranyl nitrate ($^{235}\text{UO}_2\text{N}_2$) and light water was accidentally allowed to drain into a cylindrical tank at 20°C until criticality was achieved. Assume $\epsilon_p = 1.0670$. Report all answers to five significant figures.

- (30) a. Calculate the thermal utilization factor for this solution.
 (10) b. If 2.067 fast neutrons are produced per thermal neutron absorption in fuel, determine the multiplication factor for an infinitely large tank.
 (30) c. If the diameter of the tank is 67.000 cm, what was the height of the mixture in the tank? Assume negligible thermal neutron leakage.

$$\epsilon_p = 1.0670 \quad N_{^{235}\text{UO}_2\text{N}_2} = 1 \quad N_{\text{H}_2\text{O}} = 750 \quad 20^\circ\text{C} \quad N_{^{235}} =$$

a)
$$f = \frac{\sum_a^F}{\sum_a^F + \sum_a^{NF}}$$

$$\sum_i = N_i \sigma_i$$

$$f = \frac{(1.687 \times 10^{-24})(.978)}{(1.687 \times 10^{-24})(.978) + (750)(.664 \times 10^{-24})(1)}$$

$$f = 0.574317498$$

b)
$$K_{\infty} = \epsilon_p f \eta$$

$$\nu = 2.067$$

$$\eta = \nu \frac{\sum_f^F}{\sum_a^F}$$

$$\eta = 2.067 \frac{(1.687 \times 10^{-24})(.9759)}{(1.687 \times 10^{-24})(.978)}$$

$$K_{\infty} = 1.067(0.574317498)(1.762334341)$$

$$K_{\infty} = 1.079952792$$

c) $r = 33.5 \text{ cm}$ $B_c^2 = ?$ $\eta = 1.762334341$

$$P_{NL}^{th} = \frac{1}{1 + L_T^2 B_c^2}$$

$$K_{eff} = \epsilon_p R F_{NL} P_{NL}^F \eta$$

$$1 = \frac{1 + L_T^2 B_c^2}{1 + L_T^2 B_c^2}$$

$$B_c^2 = .009870715$$

$$P_{NL}^{th} = 1$$

$$P_{NL}^F = e^{-B_c^2 D_T}$$

$$B_c^2 = \left(\frac{2.405}{R}\right)^2 + \left(\frac{\pi}{H}\right)^2$$

$$1.079952792 = 1 + 18.1 B_c^2$$

$$H = \frac{\pi}{\sqrt{.009870715 - \left(\frac{2.405}{33.5}\right)^2}}$$

$$H = 45.743 \text{ cm}$$

a.

0.57432

Report your answer in this box.

b.

1.0800

Report your answer in this box.

c.

45.743 cm

Report your answer in this box.

Wt. **No.**

Cadet: _____

20

4

For the particles and media outlined below, select the most likely interaction mechanism.

- (5) a. Thermal neutron in krypton (Kr)
- | | |
|---|--|
| ☐ Photoelectric Effect | ☐ Scattering |
| ☒ Compton Scattering | ☐ Coulombic |
| ☐ Pair Production | ☐ Bremsstrahlung Radiation |
| ☐ Absorption | ☐ Cherenkov Radiation |
- (5) b. 1 keV gamma particle in lead
- | | |
|---|--|
| ☒ Photoelectric Effect | ☐ Scattering |
| ☐ Compton Scattering | ☐ Coulombic |
| ☐ Pair Production | ☐ Bremsstrahlung Radiation |
| ☐ Absorption | ☐ Cherenkov Radiation |
- (5) c. 5 MeV alpha particle in tissue
- | | |
|--|--|
| ☐ Photoelectric Effect | ☐ Scattering |
| ☐ Compton Scattering | ☒ Coulombic |
| ☐ Pair Production | ☐ Bremsstrahlung Radiation |
| ☐ Absorption | ☐ Cherenkov Radiation |
- (5) d. 2 keV beta particle in silver (Ag)
- | | |
|--|---|
| ☐ Photoelectric Effect | ☐ Scattering |
| ☐ Compton Scattering | ☐ Coulombic |
| ☐ Pair Production | ☒ Bremsstrahlung Radiation |
| ☐ Absorption | ☐ Cherenkov Radiation |

Wt. No.

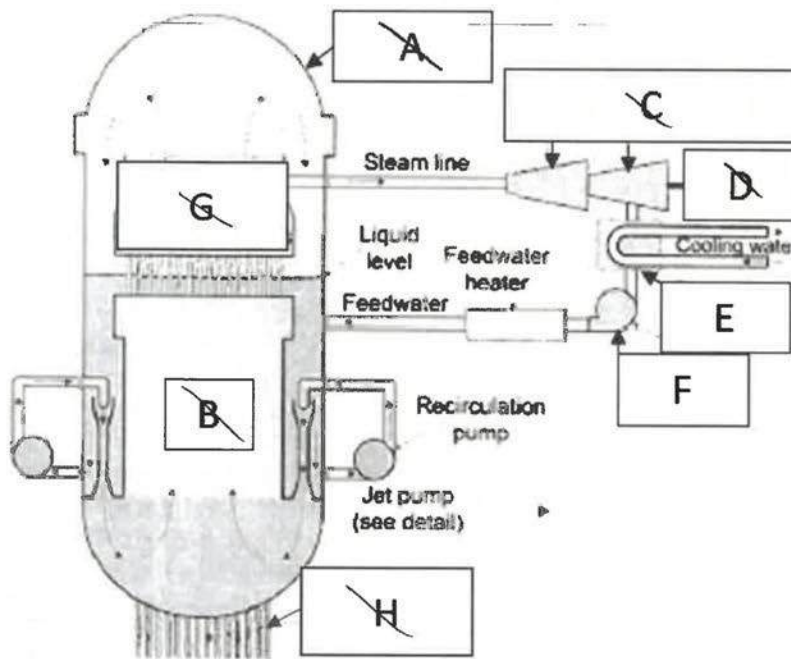
20

5

Cadet: _____

Refer to the diagram of the reactor system below. Match the lettered components on the diagram to the most proper reactor system component listed below and answer the follow-on questions at the bottom of the page.

(16) a.



Component	Letter
Condenser	E
Control Rods	H
Core	B
Generator	P
Pressure Vessel	A
Reactor Coolant Pump	F
Steam Dryer	G
Turbine	C

(4) b. The reactor depicted above is a:

- ☐ Pressurized Water Reactor
- ☒ Boiling Water Reactor
- ☐ Heavy Water Reactor
- ☐ Gas Cooled Reactor
- ☐ Liquid Metal Fast Breeder Reactor

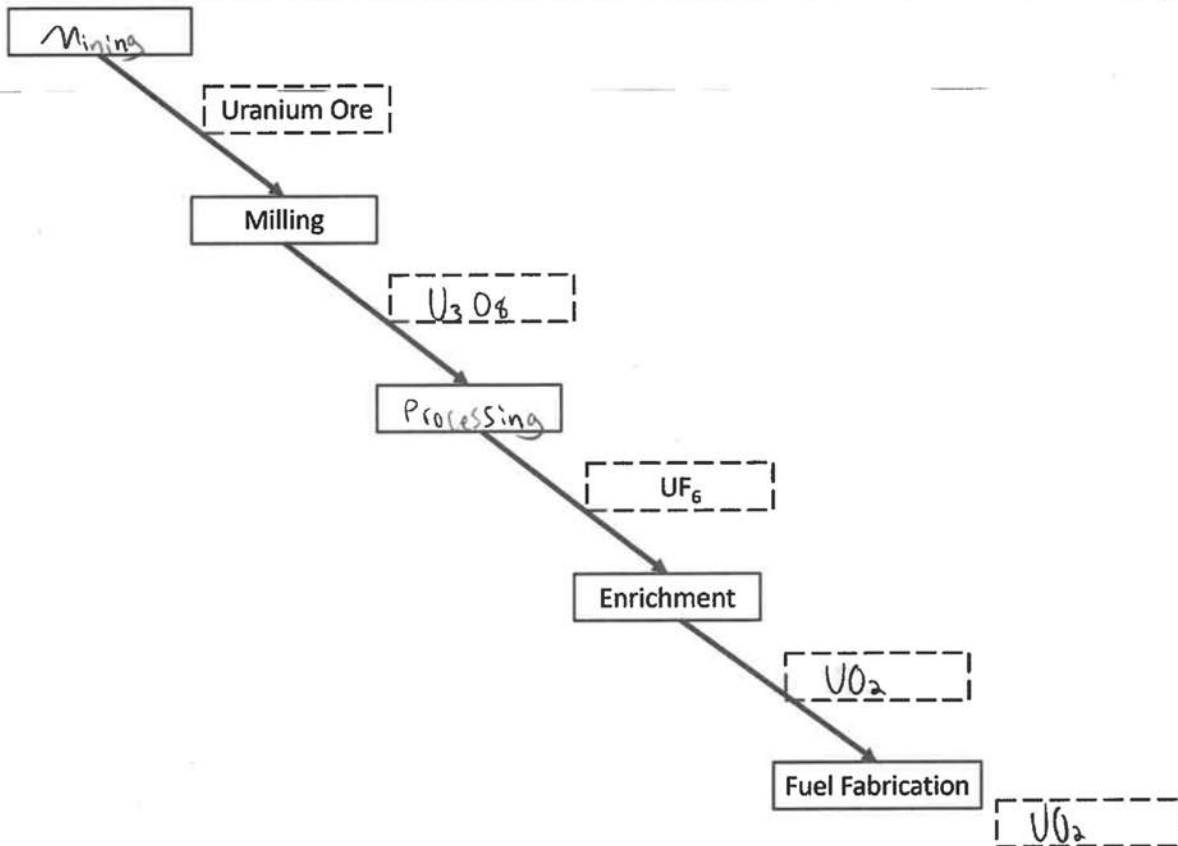
Wt. No.

10

6

Cadet: _____

Complete the missing sections of the once-through fuel cycle flow chart below
(Process steps are boarded by a solid rectangle, materials by a dashed rectangle).



10 7 For the various uses or conditions outlined below, select the most appropriate detector.

(5) a. Rapid radiation surveys of large geographical areas

- ☐ Semiconductor Detector
- ☐ Scintillation Detector
- ☒ Gas-filled Detector

(5) b. Nuclear forensics of weapon fallout for attribution to responsible actors

- ☒ Semiconductor Detector
- ☐ Scintillation Detector
- ☐ Gas-filled Detector

Poor Student Work

Student

Total Points
127 / 250 pts

Question 1

Photon Attenuation

26 / 60 pts

1.1 Calculation

16 / 50 pts

✓ - 12 pts Does not use geometric component of equation 39

✓ - 4 pts Does not convert to Bq

✓ - 2 pts Does not substitute any activity

✓ - 4 pts Incorrect r value (67 cm)

✓ - 2 pts Incorrect MAC (0.136)

✓ - 4 pts Incorrect density (1.105 g/cm³)

✓ - 4 pts Incorrect x value (6.7 cm)

✓ - 1 pt Incorrect answer

✓ - 1 pt Incorrect Units

💬 Need to include geometric attenuation and convert activity to Bq. Wrong x value (helps to draw a picture)

1.2 Concept

10 / 10 pts

✓ - 0 pts A (Correct)

Equation 2 Weight Percent

✓ - 6 pts Does not use equation 2

✓ - 2 pts Incorrect coefficients

✓ - 2 pts Incorrect masses

Equation 6 - Number Density

✓ - 6 pts Does not use equation 6

✓ - 2 pts Does not substitute weight percent

✓ - 2 pts Does not use correct mass

✓ - 4 pts Does not find other element's number density

Equation 35 Macroscopic Cross Section

✓ - 6 pts Does not use equation 35

✓ - 6 pts Does not use equation 37

✓ - 2 pts Does not find XS absorption for both elements

✓ - 2 pts Does not find XS scatter for both elements

✓ - 1 pt Incorrect micro XS scatter Aluminum (1.49)

✓ - 1 pt Incorrect micro XS scatter Oxygen (3.76)

✓ - 1 pt Incorrect micro XS absorption Aluminum (0.230)

✓ - 1 pt Incorrect micro XS absorption Oxygen (0.00027)

Equation 31 - Attenuation

✓ - 4 pts Does not substitute correct ratio (0.33)

✓ - 1 pt Incorrect Answer

✓ - 1 pt Incorrect Units

Question 3

6 Factor Formula

54 / 70 pts

3.1 Thermal Utilization

23 / 30 pts

✓ - 2 pts Does not identify correct fuel (U235)

✓ - 2 pts Does not identify correct non-fuels (O, N, H₂O)

✓ - 2 pts Incorrect coefficients

✓ - 1 pt Incorrect Answer

3.2 k infinite

7 / 10 pts

✓ - 2 pts Does not substitute correct eta (2.067)

✓ - 1 pt Incorrect Answer

3.3 Buckling

24 / 30 pts

✓ - 2 pts Does not use correct fermi age (27)

✓ - 2 pts Does not use correct radius (33.5 cm)

✓ - 1 pt Incorrect answer

✓ - 1 pt Incorrect units

Question 4

Interactions

10 / 20 pts

4.1 Neutron

5 / 5 pts

✓ - 0 pts D (Correct)

4.2 Photon

5 / 5 pts

✓ - 0 pts A (Correct)

4.3 Alpha

0 / 5 pts

✓ - 5 pts C

4.4 Beta

0 / 5 pts

✓ - 5 pts B

Question 5

Reactors

16 / 20 pts

5.1 Components

12 / 16 pts

✓ - 2 pts Incorrect Condenser (E)

✓ - 2 pts Incorrect Coolant Pump (F)

5.2 Type

4 / 4 pts

✓ - 0 pts B (Correct)

Question 6

Fuel Cycle

6 / 10 pts

✓ - 2 pts Incorrect Recovery/Mining

✓ - 2 pts Incorrect UF6/greensalt

Question 7

Detectors

5 / 10 pts

7.1 Gas Filled

5 / 5 pts

✓ - 0 pts C (Correct)

7.2 Semiconductor

0 / 5 pts

✓ - 5 pts B

Time of completion _____
19/20 Nov 2025

Cadet _____

(First)

(Last)

Section _____

Version 2

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NE300, Fundamentals of Nuclear Engineering
Written Partial Review II

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- c. Cadet Name (print): _____

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Wt. No.

Cadet: _____

60 1 Sheldon, a clumsy cat, leaps onto MAJ Pert's desk in search of scratches. In doing so, he knocks a 6.70 mCi source of 0.200 MeV photons into a spherical fishbowl filled with heavy water. The diameter of the fishbowl is 13.4 cm and the source is suspended in the center.

- (50) a. Calculate the intensity of uncollided photons MAJ Pert is exposed to 60.3 cm away from the edge of the bowl. You can neglect attenuation from the bowl itself and air. Note: light and heavy water have the same mass attenuation coefficients. Report your answer to three significant figures.

Equations

$$I(x) = I_0 e^{-\mu x}$$

$$\mu = \left(\frac{\mu}{\rho} \right) \rho$$

$$\left(\frac{\mu}{\rho} \right) = \sum \left(\frac{\mu_i}{\rho} \right)$$

6.7 mCi

0.200 MeV photons
Fish Bowl 13.4 cm
heavy water
 $13.4 \cdot \pi r^2$

H₂O & D₂O have same $\frac{\mu}{\rho}$ so $\rho_{20} = \frac{\mu}{\rho}$ at 0.2 MeV = 0.136

$$\frac{\mu}{\rho}$$

mCi source 0.200 MeV

$$I(x) = I_0 e^{-\mu x} \quad 60.3$$

looking for



Report your answer in this box.

- (10) b. Suppose the water in the fishbowl is replaced with a material that has a higher linear attenuation coefficient, but the mass attenuation coefficient remains the same. The source and geometry are unchanged. Which of the following best explains what happens to the fraction of uncollided photons reaching MAJ Pert, and why?
- ☒ The fraction decreases, because a higher linear attenuation coefficient means photons are more likely to interact per unit length.
 - ☐ The fraction increases, because a higher linear attenuation coefficient means photons are less likely to interact per unit length.
 - ☐ The fraction remains unchanged, because the mass attenuation coefficient is the same.
 - ☐ The fraction could increase or decrease, depending on the density of the new material.

Wt. No.

60

2

Cadet: _____

Pam, a small cat, slips into MAJ Pert's gardening shed, where an AmBe source is producing collimated thermal neutrons for illicit study. The neutron beam is incident on a sample of aluminum oxide (Al_2O_3) body armor with a density of 3.99 g/cm^3 and Pam lays down on the other side.

Calculate the thickness of the tank armor required to protect Pam from 67% of the uncollided neutrons. Use natural compositions of elements. Report your answer to three significant figures.

Equations: Looking for x (thickness)

$$I(x) = I_0 e^{-\mu t x}$$

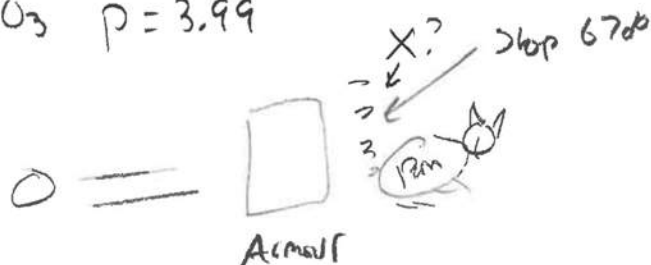
$$\phi(r) = \frac{SP}{4\pi r^2} e^{-\mu t x}$$

AmBe source

producing thermal neutrons

The neutron beam is incident on a sample of (Al_2O_3) with a density of 3.99 g/cm^3

$\text{Al}_2\text{O}_3 \quad \rho = 3.99$



67%

$$\mu t = \left(\frac{\mu}{\rho} \right) \rho x$$

$$I(x) = I_0 e^{-\mu t x}$$

.67 I_0

$$.67 I_0 = I_0 e^{-\mu t x}$$

$$\ln(.67) = \ln(e^{-\mu t x})$$

$$\ln(.67) = -\mu t x$$

$-\mu t$

$$-\frac{\ln(.67)}{\mu t} = x$$

$$-\frac{\ln(.67)}{\mu t} = x$$

Report your answer in this box.

Cadet:

70

3

In 1958 the first criticality accident in a uranium processing facility occurred at the Y-12 enrichment plant. A 1:750 mix of uranyl nitrate ($^{235}\text{UO}_8\text{N}_2$) and light water was accidentally allowed to drain into a cylindrical tank at 20°C until criticality was achieved. Assume $\epsilon_p = 1.0670$. Report all answers to **five** significant figures.

- (30) a. Calculate the thermal utilization factor for this solution.
- (10) b. If 2.067 fast neutrons are produced per thermal neutron absorption in fuel, determine the multiplication factor for an infinitely large tank.
- (30) c. If the diameter of the tank is 67.000 cm, what was the height of the mixture in the tank? Assume negligible thermal neutron leakage.

Equations

a) Thermal Utilization factor
— F

$$f = \frac{\bar{\Sigma}_a^F}{\bar{\Sigma}_a^F + \bar{\Sigma}_a^{NF}}, \quad \Sigma_i = N \sigma_i(E) g_i(T)$$

Giver

11,750

$$(235008 N_2) \text{ 8 kg}$$

Cylindrical tank at 20°C

 $K, H = 1$ (critically damped)
$$\varepsilon_p = 1.0670$$

$$\begin{aligned} \text{m.i.x} &\rightarrow \frac{1}{750} (\cancel{\overset{N_{\text{O}_2}}{2350}} \cdot \sigma_{\text{O}_2,a} \cdot g_{\text{O}_2,a}) + \cancel{(8) \cdot \sigma_{\text{O}_2,a} \cdot g_{\text{O}_2,a}} + \cancel{(2) \cdot \sigma_{\text{H}_2\text{O},a} \cdot g_{\text{H}_2\text{O},a}} \\ &= \frac{1}{750} [\underset{\uparrow}{\overset{N_{\text{H}_2\text{O}}}{2350}} \cdot \sigma_{\text{H}_2\text{O},a} \cdot g_{\text{H}_2\text{O},a}] + \dots \end{aligned}$$

And hexoff
factors for
Oxygen, light water
SLN

$$b) K_{eff} = \sum p f n$$

$$k_{eff\infty}$$

don't now

$$K_{eff} \infty = 1.6670 \cdot 0.93117 \cdot k$$

$$\eta = \frac{v \sum f}{\sum a} = \frac{2.067}{0.93117} = 2.2301 \times 10^4$$

c) $K_{eff} = \epsilon P_{NL} P_{NL} f_n$ lock. may be $\textcircled{A}$

get rid of pH_{NL} (Hemolysis)

$$\frac{1}{\Sigma P \cdot f \cdot h} = P_{NL}^F e^{-B_c^2 \pi}$$

$$f \cdot h = e^{\left(\frac{(2.405)^2}{R} + \left(\frac{\pi}{H} \right)^2 \right)} \quad \text{Buckling}$$

$$= e^{\left(\frac{(2.405)^2}{R} + \left(\frac{\pi}{H} \right)^2 \right)}$$

0.93117

Report your answer in this box.

$$5.32881 \times 10^{49}$$

Report your answer in this box

$$\sqrt{\frac{H}{\Sigma P \cdot f_n}} = (H)$$

Report your answer in this box.

Wt. No.

Cadet: _____

20 4 For the particles and media outlined below, select the most likely interaction mechanism.

(5) a. Thermal neutron in krypton (Kr)

☐ Photoelectric Effect

☐ Scattering

☐ Compton Scattering

☐ Coulombic

☐ Pair Production

☐ Bremsstrahlung Radiation

☒ Absorption

☐ Cherenkov Radiation

(5) b. 1 keV gamma particle in lead

☒ Photoelectric Effect

☐ Scattering

☐ Compton Scattering

☐ Coulombic

☐ Pair Production

☐ Bremsstrahlung Radiation

☐ Absorption

☐ Cherenkov Radiation

(5) c. 5 MeV alpha particle in tissue

☐ Photoelectric Effect

☐ Scattering

☐ Compton Scattering

☐ Coulombic

☒ Pair Production

☐ Bremsstrahlung Radiation

☐ Absorption

☐ Cherenkov Radiation

(5) d. 2 keV beta particle in silver (Ag)

☐ Photoelectric Effect

☐ Scattering

☒ Compton Scattering

☐ Coulombic

☐ Pair Production

☐ Bremsstrahlung Radiation

☐ Absorption

☐ Cherenkov Radiation

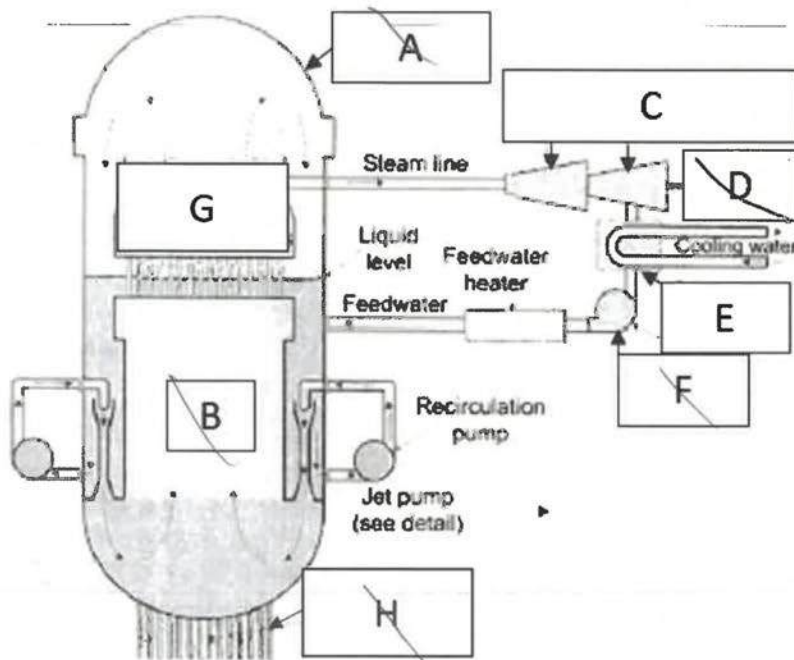
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20 5

Cadet: _____

Refer to the diagram of the reactor system below. Match the lettered components on the diagram to the most proper reactor system component listed below and answer the follow-on questions at the bottom of the page.

(16) a.



Component	Letter
Condenser	F
Control Rods	H
Core	B
Generator	D
Pressure Vessel	A
Reactor Coolant Pump	E
Steam Dryer	G
Turbine	C

(4) b. The reactor depicted above is a:

- ☐ Pressurized Water Reactor
- ☒ Boiling Water Reactor
- ☐ Heavy Water Reactor
- ☐ Gas Cooled Reactor
- ☐ Liquid Metal Fast Breeder Reactor

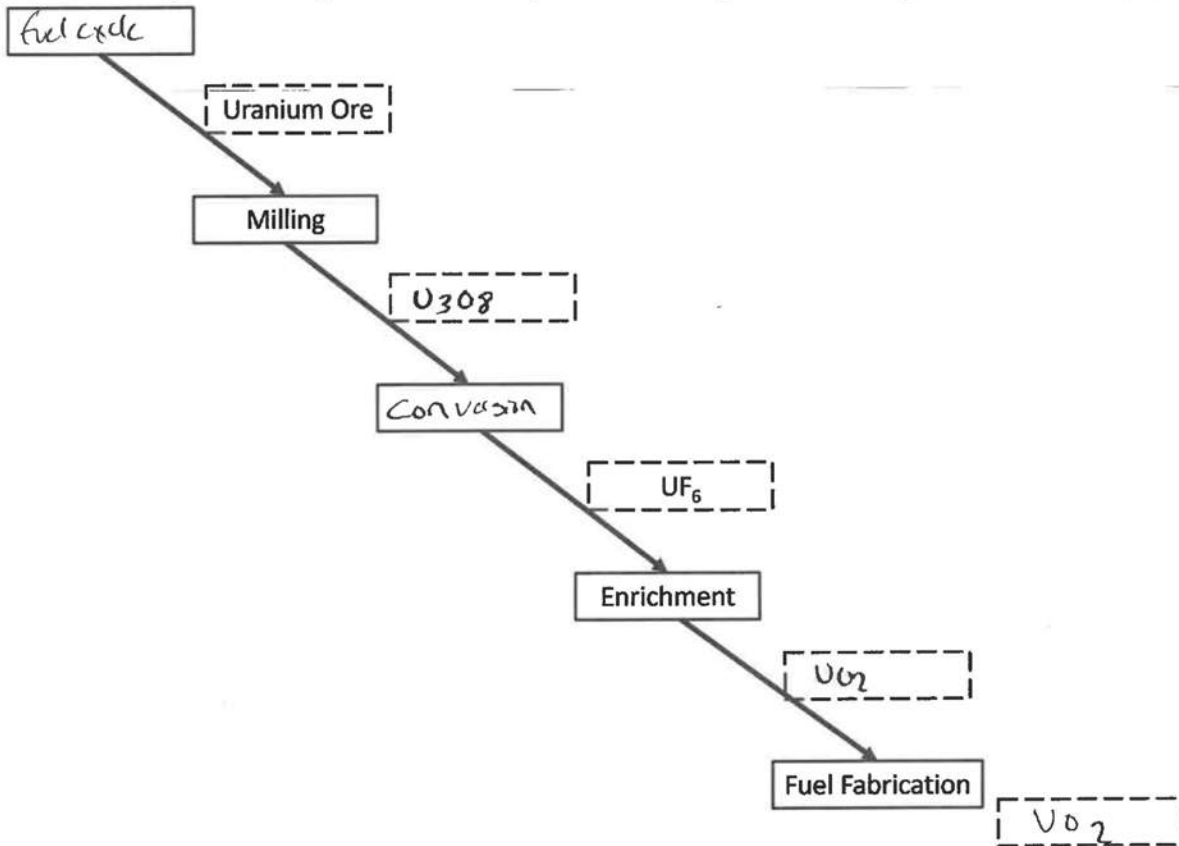
Wt. No.

10

6

Cadet: _____

Complete the missing sections of the once-through fuel cycle flow chart below
(Process steps are boarded by a solid rectangle, materials by a dashed rectangle).



10 7 For the various uses or conditions outlined below, select the most appropriate detector.

(5) a. Rapid radiation surveys of large geographical areas

- ☐ Semiconductor Detector
- ☐ Scintillation Detector
- ☒ Gas-filled Detector

(5) b. Nuclear forensics of weapon fallout for attribution to responsible actors

- ☐ Semiconductor Detector
- ☒ Scintillation Detector
- ☐ Gas-filled Detector



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB I

TERM END EXAMINATION

Instructor Solution

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE300, Fundamentals of Nuclear Engineering
Term End Exam

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- g. When you turn in your exam check your name is written on every page.**

2. Exam Security:

- a. I will not discuss this exam with anyone until it is released from academic security on **22 December 2025 at 1700.**
- b. Cadet Signature: _____
- c. Cadet Name (print): _____

A: 47 (9%)

B: 205 (41%)

D: 248 (50%)

Total: 500

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Wt. No.

Cadet: _____

- 90 1. Consider a PWR which operates at a constant temperature of 400°C and is fueled with UO₂ enriched to 3.82387 w/o ²³⁵U and a mass density of 11.1538 g/cm³. The remainder of the uranium is ²³⁸U, and all the oxygen is of natural assay. Fast fissions add 2.863% of neutrons relative to thermal fissions, and thermal fissions regenerate 1.86300 neutrons per absorption in the fuel. During a given lifecycle, 1 in 8 fast neutrons leak from the core and 1 in 12 neutrons are absorbed in resonance peaks as they slow. Exactly 78% of all thermal neutrons go on to be absorbed in the core. The fuel is defined to be all of the uranium.

- (40) a. Calculate the number density for ²³⁵U, ²³⁸U, and O. Report your answers to **four** significant figures.

5 Eq 3
$$\mathcal{A}_x = \left[\sum \frac{\omega_i}{\mathcal{A}_i} \right]^{-1}$$

5 Eq 6
$$N_i = \frac{\omega_i \rho_x N_a}{\mathcal{A}_i}$$

2 Correct w/o and mass
$$\mathcal{A}_U = \left[\frac{\omega_{235}}{\mathcal{A}_{235}} + \frac{\omega_{238}}{\mathcal{A}_{238}} \right]^{-1}$$

$$N_{235} = \frac{\omega_{235} \rho_U N_a}{\mathcal{A}_{235}}$$
 2 Subs w_235 and rho_U

$$\mathcal{A}_U = \left[\frac{0.0382387}{235.0439231} + \frac{0.9617613}{238.0507826} \right]^{-1}$$

$$\mathcal{A}_U = 237.934390 \text{ u}$$

$$N_{235} = \frac{0.0382387(9.8135905)N_a}{235.0439231}$$

$$N_{235} = 9.6322748 \times 10^{20} \frac{\text{atoms}}{\text{cm}^3}$$

5 Eq 2
$$\omega_x = \frac{n\mathcal{A}_x}{n\mathcal{A}_x + m\mathcal{A}_y}$$

$$N_{238} = \frac{0.9617613(9.8135905)N_a}{238.0507826}$$

$$\omega_U = \frac{\mathcal{A}_U}{\mathcal{A}_U + (2)\mathcal{A}_O}$$

$$N_{238} = 2.3920621 \times 10^{22} \frac{\text{atoms}}{\text{cm}^3}$$

2 Correct sub and coefficients
$$\omega_U = \frac{237.934390}{237.934390 + (2)15.9994}$$

$$\omega_O = \frac{(2)15.9994}{237.934390 + (2)15.9994}$$

$$\omega_U = 0.88145659$$

$$\omega_O = 0.1185434$$
 3 Attempts N_O, any method

5 Eq 5
$$\rho_x = \omega_x \rho$$

$$\rho_U = \omega_U \rho_{UO_2}$$

$$\rho_U = 0.88145659(11.1538)$$

$$N_O = \frac{0.1185434(11.1538)N_a}{15.9994}$$

2 Subs w_U and rho_UO2

$$\rho_U = 9.8135905 \frac{\text{g}}{\text{cm}^3}$$

$$N_O = 4.97676979 \times 10^{22} \frac{\text{atoms}}{\text{cm}^3}$$

²³⁵U

$$9.632 \times 10^{20} \frac{\text{atoms}}{\text{cm}^3}$$

Report your answer in this box.

²³⁸U

$$2.392 \times 10^{22} \frac{\text{atoms}}{\text{cm}^3}$$

Report your answer in this box.

O

$$4.977 \times 10^{22} \frac{\text{atoms}}{\text{cm}^3}$$

Report your answer in this box.

A: 6 (15%)
 B: 14 (35%)
 D: 20 (50%)
 Total: 40



Wt. **No.**

Cadet: _____

- 90 1. Consider a PWR which operates at a constant temperature of 400°C and is fueled with UO_2 enriched to 3.82387 w/o ^{235}U and a mass density of 11.1538 g/cm³. The remainder of the uranium is ^{238}U , and all the oxygen is of natural assay. Fast fissions add 2.863% of neutrons relative to thermal fissions, and thermal fissions regenerate 1.86300 neutrons per absorption in the fuel. During a given lifecycle, 1 in 8 fast neutrons leak from the core and 1 in 12 neutrons are absorbed in resonance peaks as they slow. Exactly 78% of all thermal neutrons go on to be absorbed in the core. The fuel is defined to be all of the uranium.
- (30) b. Calculate the reactor's thermal utilization factor. Assume the listed number density in NERD-P for the moderator is valid. Report your answer to **six** significant figures.

10 Eq 51

$$f = \frac{\Sigma_a^F}{\Sigma_a^F + \Sigma_a^{NF}}$$

5 Eq 37

$$f = \frac{N_{235}\sigma_{a235}g_{a235} + N_{238}\sigma_{a238}g_{a238}}{N_{235}\sigma_{a235}g_{a235} + N_{238}\sigma_{a238}g_{a238} + N_O\sigma_{aO} + N_{H_2O}\sigma_{H_2O}}$$

3 Correctly ID's Fuel

3 Correctly ID's Non-fuel

$$f = \frac{9.632 \times 10^{20} (687 \times 10^{-24}) (0.9294) + 2.392 \times 10^{22} (2.73 \times 10^{-24}) (1.0085)}{\text{numerator} + 4.977 \times 10^{22} (0.00027 \times 10^{-24}) + 3.343 \times 10^{22} (0.664 \times 10^{-24})}$$

5 Correct Subs

2 ID's LW as moderator

$$f = \frac{0.61501863 + 0.06585837}{0.61501863 + 0.06585837 + 0.000013437278 + 0.0222}$$

$$f = \frac{6.80877}{7.03090}$$

$$f = 0.968406$$

2 Correct Ans

0.968406

Report your answer in this box.

A: 2 (7%)
B: 13 (43%)
D: 15 (50%)
Total: 30

Wt. No.

Cadet: _____

- 90 1. Consider a PWR which operates at a constant temperature of 400°C and is fueled with UO₂ enriched to 3.82387 w/o ²³⁵U and a mass density of 11.1538 g/cm³. The remainder of the uranium is ²³⁸U, and all the oxygen is of natural assay. Fast fissions add 2.863% of neutrons relative to thermal fissions, and thermal fissions regenerate 1.86300 neutrons per absorption in the fuel. During a given lifecycle, 1 in 8 fast neutrons leak from the core and 1 in 12 neutrons are absorbed in resonance peaks as they slow. Exactly 78% of all thermal neutrons go on to be absorbed in the core. The fuel is defined to be all of the uranium.
- (20) c. Report the following reactor kinetics terms in the chart below. Then calculate the reactor's neutron multiplication factor to **six** significant figures and select its criticality:

4 Eq 47 $k_{eff} = \epsilon p f \eta P_{NL}^f P_{NL}^{th}$

$$k_{eff} = (1.02863)(0.916667)(0.968406)(1.863)(0.875)(0.78)$$

$$k_{eff} = 1.16103$$

Term	Value
ϵ	1.02863
P_{NL}^f	0.875000
p	0.916667
P_{NL}^{th}	0.780000
f	0.968406
η	1.86300
k_{eff}	1.16103

② Fast fission factor

② Fast non-leak

② Resonance escape

② Thermal non-leak

② Thermal utilization/follows

② Thermal reproduction

② k

By calculation, the reactor above is:

☐ Subcritical

☐ Critical

☒ Supercritical

② Supercritical/follows

Wt. **No.**

Cadet: _____

40

2.

The relative atomic masses and critical energies of fission for several isotopes of a hypothetical element X are listed below.

Determine by appropriate calculations if ^{251}X is fissile or fissionable. If it is fissionable, calculate what neutron energy would be needed to result in fission.

Report your answer to **three** significant digits if needed.

Isotope	Atomic Weight (u)	Critical Energy of Fission (MeV)
^{249}X	249.074854	5.08
^{250}X	250.076406	4.17
^{251}X	251.079587	5.02
^{252}X	252.081630	6.43

3 Attempts compound nucleus

3 Correct compound nucleus

6 Eq 15

$$S_n[^{252}\text{X}] = [M[^{251}\text{X}] + m_n - M[^{252}\text{X}]] c^2$$

6 Correct masses/follows

$$S_n[^{252}\text{X}] = [251.079587 + 1.0086649233 - 252.08163] c^2$$

$$S_n[^{252}\text{X}] = 0.0066219233 * 931.494013$$

$$S_n[^{252}\text{X}] = 6.1682819 \text{ MeV}$$

3 Correct S_n/follows

6 Eq 28

$$E^* = S_n + T_n$$

$$T_n = E^* - S_n$$

5 Correct E_crit/follows

$$T_n = 6.43 - 6.1682819$$

$$T_n = 0.261718 \text{ MeV}$$

4 Correct Ans/units

4 Fissionable/follows

☐ Fissile

☒ Fissionable

0.262 MeV

Report your answer in this box if needed.

A: 4 (10%)
B: 15 (38%)
D: 21 (52%)
Total: 40

Wt. **No.**

40 3.

Cadet: _____

(20) a. Calculate the exact binding energy of ^{112}Cd to **six** significant figures.

10 Eq 14

$$BE = [ZM({}_1^1\text{H}) + (A - Z)m_n - M({}_Z^AX)]c^2$$

$$BE = [48(1.0078250321) + 64(1.0086649233) - 111.9027572]c^2$$

4 Correct A/Z $BE = [48.37560154 + 64.55455509 - 111.9027572]c^2$

$$BE = 1.0273994 * 931.494013$$
 4 Correct masses

$$BE = 957.016 \text{ MeV}$$

2 Correct Ans/Units

957.016 MeV

Report your answer in this box.

(20) b. Utilize the liquid drop model to calculate the approximate binding energy of a theoretical isotope, $^{311}_{121}\text{Un}$, to **six** significant figures.

10 Eq 13

$$E_B = a_v A - a_s A^{\frac{2}{3}} - \frac{a_c Z^2}{A^{\frac{1}{3}}} - \frac{a_a (A - 2Z)^2}{A} - \frac{a_p}{\sqrt{A}}$$

2 Correct A/Z

$$E_B = 15.835(311) - 18.33(311)^{\frac{2}{3}} - \frac{0.714(121^2)}{311^{\frac{1}{3}}} - \frac{23.20(311 - 2(121))^2}{311} - \frac{0}{\sqrt{311}}$$

4 Correct subs $E_B = 4924.685 - 841.4004 - 1542.939 - 355.1614 - 0$ **2** Correct a_p

$$E_B = 2185.18393 \text{ MeV}$$

2 Correct Ans/Units

2185.18 MeV

Report your answer in this box.

A: 4 (10%)
B: 16 (40%)
D: 20 (50%)
Total: 40

Wt. No.

Cadet: _____

- 40 4. Spent fuel is removed from a reactor and placed into a dry cask for storage. The cask of spent fuel contains 385.000 grams of the fission byproduct ^{131}I which has a half-life of 8.02520 days.
- (30) a. Calculate the activity in curies of the ^{131}I which remains after the cask is stored for exactly 365.000 days to **six** significant figures.

4 Eq 22

$$T_{\frac{1}{2}} = \frac{\ln 2}{\lambda}$$

4 Eq 20

$$N = \frac{mN_a}{\mathcal{A}}$$

2 Subs masses

$$N = \frac{385 N_a}{130.9061242}$$

$$N = 1.771135123 \times 10^{24} \text{ atoms}$$

$$\lambda = \frac{\ln 2}{T_{\frac{1}{2}}}$$

4 Eq 21

$$A = \lambda N$$

$$A = 9.996681468 \times 10^{-7} (1.771135369 \times 10^{24})$$

2 Converts A to per sec

$$A = 1.770547366 \times 10^{18} \text{ Bq}$$

2 Correct HL

$$\lambda = \frac{0.693147181}{8.0252 \text{ days}}$$

$$\lambda = 0.086371328 \text{ day}^{-1}$$

$$\lambda = 9.996681468 \times 10^{-7} \text{ sec}^{-1}$$

4 Eq 24 or 25

$$A(t) = A_0 e^{-\lambda t}$$

3 Matched exp units

$$A = 1.770547612 \times 10^{18} e^{-0.086371328(365)}$$

$$A = 3.603643325 \times 10^4 \text{ Bq}$$

$$A = 9.73958 \times 10^{-7} \text{ Ci}$$

2 Converts to Ci

3 Correct Ans/Units

$$9.73958 \times 10^{-7} \text{ Ci}$$

Report your answer in this box.

- (10) b. Consider a 1.0-Ci source of ^{137}Cs produced in the year 1995. Select the closest present-day activity of the source.

☐ 1.3x10⁵ Bq

☐ 3.7x10⁹ Bq

☒ 1.8x10¹⁰ Bq

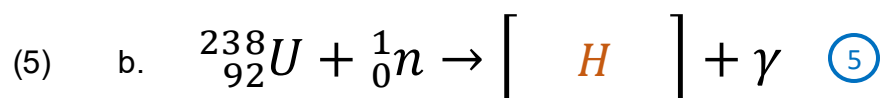
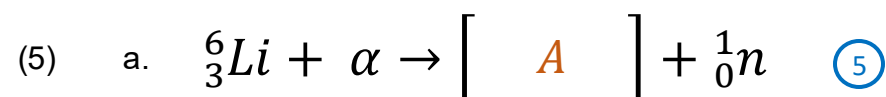
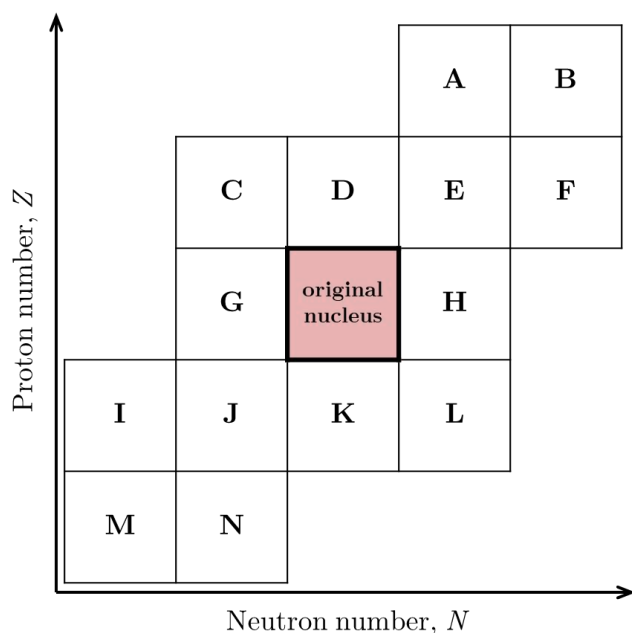
6

3

1

☐ 2.9x10¹⁰ Bq

- 35 5. Consider the figure below. For each decay or reaction shown, **write the letter** corresponding to the position of the missing nuclide in the box. Note: Some of the standard nomenclature may be omitted



- (10) f. In any given material, the range of the particle emitted in part c. is relatively _____ compared to those of other radiations.

☐ long

☐ the same

☒ short (6) (3) (1)

☐ cannot be determined

A: 1 (3%)
B: 13 (37%)
D: 21 (60%)
Total: 35

Wt. **No.**

Cadet: _____

- 30 6. A 0.62202 MeV neutron enters a target of ^{14}C . Calculate the average number of collisions between the neutron and the carbon nuclei required to slow the neutron to thermal energy (room temperature).

5 Eq 43

$$\alpha = \left(\frac{A - 1}{A + 1} \right)^2$$

2 Correct A

$$\alpha = \left(\frac{14 - 1}{14 + 1} \right)^2$$

$$\alpha = 0.751111111$$

5 Eq 44

$$\xi = 1 + \frac{\alpha}{1 - \alpha} \ln(\alpha)$$

2 Subs A

$$\xi = 1 + \frac{0.751111111}{1 - 0.751111111} \ln(0.751111111)$$

$$\xi = 0.136284194$$

5 Eq 45

$$n = \frac{1}{\xi} \ln \left(\frac{E_1}{E_2} \right)$$

$$n = \frac{1}{0.136284194} \ln \left(\frac{0.62202 \times 10^6}{0.0253} \right)$$

2 Matched units

2 Subs xi

$$n = 124.869$$

$$n = 125$$

4 Correct thermal energy

3 Correct Ans/rounded

125

Report your answer in this box.

A: 3 (10%)
B: 12 (40%)
D: 15 (50%)
Total: 30

Wt. No.

Cadet: _____

- 55 7. An experiment produces a beam of 0.200 MeV photons with an intensity of $8.83157 \times 10^{12} \text{ cm}^{-2} \text{ s}^{-1}$ and there is a need to evaluate the feasibility of using 1.62973 cm of copper, Cu, as shielding material. The copper shield has a density of 8.95852 g/cm³. Any photon that does not collide in the shield will enter 5.28375 cm of tissue afterwards. Tissue has a density of 1.03468 g/cm³.

- (30) a. Calculate the uncollided intensity that results after passing through both the copper and the tissue to **six** significant figures.

$$I(x) = I_0 e^{-\mu_t x} \quad \text{Eq 30}$$
$$I_{Cu} = 8.83157 \times 10^{12} e^{-0.147(8.95852)(1.62973)} \quad \text{Eq 34}$$
$$I_{Cu} = 1.032659371 \times 10^{12} \text{ cm}^{-2} \text{ s}^{-1}$$
$$I_{tissue} = I_{Cu} e^{-\mu_t x}$$
$$I_{tissue} = 1.032659371 \times 10^{12} e^{-0.132(1.03468)(5.28375)}$$
$$I_{tissue} = 5.01824 \times 10^{11} \text{ cm}^{-2} \text{ s}^{-1}$$

3 Correct Ans/units

$5.01824 \times 10^{11} \frac{\text{photons}}{\text{cm}^2 \text{ s}}$

Report your answer in this box.

- (25) b. Calculate the minimum thickness of copper required for exactly 95% of the photon beam to be attenuated before reaching the tissue to **six** significant figures.

$$I(x) = I_0 e^{-\mu_t x} \quad \text{Eq 30}$$
$$\ln\left(\frac{I}{I_0}\right) = -\mu_t x$$
$$x = \frac{\ln(0.05)}{-0.147 \frac{\text{cm}^2}{\text{g}} \left(8.95852 \frac{\text{g}}{\text{cm}^3}\right)}$$
$$x = 2.27483 \text{ cm}$$

3 Correct Ans/units

2.27483 cm

Report your answer in this box.

A: 6 (11%)
B: 24 (44%)
D: 25 (45%)
Total: 55

Wt. No.

Cadet: _____

50 8. A $^{230}_{90}\text{Th}$ nucleus undergoes alpha decay.

(10) a. Write down the complete reaction equation for this decay event using standard form.

2 Th is a reactant 3 Alpha/He is a product



1 All A/Z correct

4 Ra is a product

(40) b. Calculate the kinetic energy of the alpha particle produced in this reaction to **six** significant figures.

10 Eq 17

$$Q = (M[^{230}_{90}\text{Th}] - M[^{226}_{88}\text{Ra}] - M[^4_2\text{He}]) c^2$$

$$Q = (230.0331266 - 226.0254026 - 4.0026032497) 931.494013$$

6 Correct masses

$$Q = 4.76995 \text{ MeV}$$

10 Eq 19 $Q = T_1 \left(\frac{m_1}{m_4} - 1 \right) - \frac{2}{m_4} \sqrt{m_1 m_3 T_1 T_3} \cos(\theta) + T_3 \left(\frac{m_3}{m_4} + 1 \right)$

4 Simplification $Q = T_3 \left(\frac{m_3}{m_4} + 1 \right)$

$$T_3 = \frac{Q}{\left(\frac{m_3}{m_4} + 1 \right)}$$

2 Subs masses

4 Correct m3/m4

$$T_3 = \frac{4.76995}{\left(\frac{4.00260325}{226.0254026} + 1 \right)}$$

$$T_3 = 4.68695 \text{ MeV}$$

4 Correct Ans/units

$$4.68695 \text{ MeV}$$

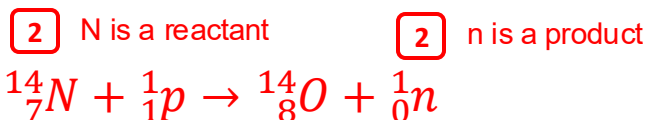
Report your answer in this box.

Wt. No.

Cadet: _____

70 9. A 13.57945 MeV proton is absorbed by nitrogen 14, resulting in a daughter isotope and a neutron.

(10) a. Write down the complete nuclear equation in standard form.



1 All A/Z correct 1 p is a reactant 4 O is a product

(50) b. Calculate the threshold energy for this equation to **six** significant figures.

8 Eq 17 $Q = [M[{}^{14}_7\text{N}] + M[{}^1_1\text{H}] - M[{}^{14}_8\text{O}] - m_n]c^2$

$$Q = [14.0030740052 + 1.0078250321 - 14.00859529 - 1.0086649233]c^2$$

4 Subs masses from part a $Q = -0.006361176 * 931.494013$

$$Q = -5.925393596 \text{ MeV}$$

8 Eq 26 $E^K = -Q \left(1 + \frac{m_1}{m_2} \right)$ 8 Eq 27 $E^C = 1.2 \frac{Z_1 Z_2}{A_1^{\frac{1}{3}} + A_2^{\frac{1}{3}}}$

$$E^K = -(-5.925393596) \left(1 + \frac{1.00727646688}{14.0030740052} \right)$$

4 Correct Z/A

2 Subs Q $E^K = 6.35162615 \text{ MeV}$ 2 Subs masses 4 Correct m1/m2

$$E^C = 1.2 \frac{1 (7)}{1^{\frac{1}{3}} + 14^{\frac{1}{3}}}$$

5 Takes max value/follows

$$E^{th} = \max(E^K, E^C)$$

$$E^C = 2.463240343 \text{ MeV}$$

$$E^{th} = 6.35163 \text{ MeV}$$

5 Correct Ans/units

$$6.35163 \text{ MeV}$$

Report your answer in this box.

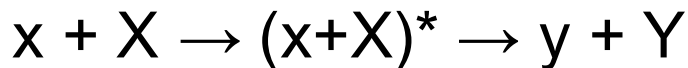
(10) c. Based on your calculations, this reaction

☒ can occur 6 3 1

☐ cannot occur

☐ not enough information given

- 15 10. Consider a generic nuclear reaction with the form and conditions shown below:



Conditions:

1. In this reaction, $Q = -1.19 \text{ MeV}$
2. Reactant 'x' possesses a net positive charge
3. Reactant 'X' and product 'Y' are different elements

- (5) a. The nuclear reaction indicated is an example of a(n) _____ reaction. (Choose all that apply)

3

☒ Binary

-1

☐ Radioactive Decay

2

☒ Endoergic

-1

☐ Exoergic

-1

☐ Fission

- (5) b. Indicate all reaction threshold(s) which exist for the reaction above. (Choose all that apply).

2

☒ Coulombic Threshold

-1

☐ Critical Energy of Fission Threshold

3

☒ Kinematic Threshold

-1

☐ Radioactive Decay Threshold

-1

☐ There are not thresholds for this reaction

-1

☐ Not enough information to determine

- (5) c. Which of the following statements are true about the reaction above? (Choose all that apply)

3

☒ Total of (mass + energy) and charge are all conserved in the reaction.

2

☒ The reactants have less collective rest mass than the products.

-1

☐ The total kinetic energy of the products must be greater than the reactants.

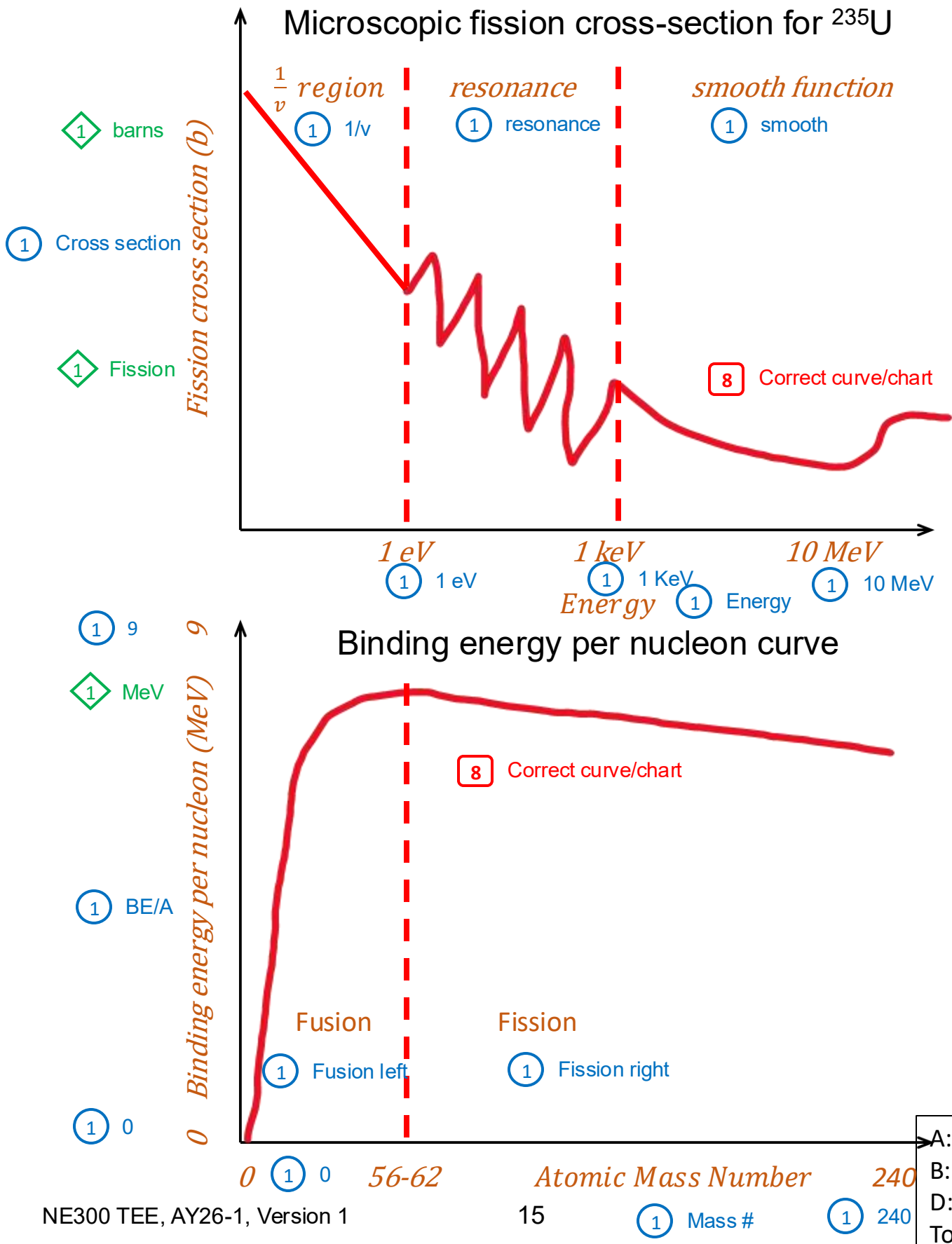
Wt. No.

Cadet: _____

35

11.

Sketch the microscopic fission cross section for ^{235}U and binding energy per nucleon curve using the axes provided below. For full credit, label all axes appropriately, including units as applicable and indicate any critical values. Then, indicate significant regions of the plot.



Outstanding Student Work

Student

Total Points
473 / 500 pts

Question 1

Reactor Physics 88 / 90 pts

1.1 — **Number Density 235U** 30 / 30 pts

The rubric is hidden for this question.

1.2 — **Number Density 238U** 5 / 5 pts

The rubric is hidden for this question.

1.3 — **Number Density O** 5 / 5 pts

The rubric is hidden for this question.

1.4 — **Thermal Utilization Factor** 30 / 30 pts

The rubric is hidden for this question.

1.5 — **Six Factor Formula** 18 / 20 pts

The rubric is hidden for this question.

Question 2

Fission 40 / 40 pts

2.1 — **Fission** 36 / 36 pts

The rubric is hidden for this question.

2.2 — **Fissionable** 4 / 4 pts

The rubric is hidden for this question.

Question 3

Binding Energy 37 / 40 pts

3.1 — **Binding Energy Formula** 20 / 20 pts

The rubric is hidden for this question.

3.2 — **Liquid Drop Model** 17 / 20 pts

The rubric is hidden for this question.

Question 4

Radioactive Decay

36 / 40 pts

4.1 Activity Decay

26 / 30 pts

The rubric is hidden for this question.

4.2 Half Life

10 / 10 pts

The rubric is hidden for this question.

Question 5

Decay Equations

30 / 35 pts

5.1 Chart of Nuclides

20 / 25 pts

The rubric is hidden for this question.

5.2 Particle Penetration

10 / 10 pts

The rubric is hidden for this question.

Question 6

Thermalization

29 / 30 pts

The rubric is hidden for this question.

Question 7

Shielding

55 / 55 pts

7.1 Intensity Calculation

30 / 30 pts

The rubric is hidden for this question.

7.2 Thickness Calculation

25 / 25 pts

The rubric is hidden for this question.

Question 8

Alpha Decay

50 / 50 pts

8.1 Decay Equation

10 / 10 pts

The rubric is hidden for this question.

8.2 Kinetic Energy

40 / 40 pts

The rubric is hidden for this question.

Question 9

Binary Reaction

62 / 70 pts

9.1 Equation

9 / 10 pts

The rubric is hidden for this question.

9.2 Threshold Energy

43 / 50 pts

The rubric is hidden for this question.

9.3 Threshold Interpretation

10 / 10 pts

The rubric is hidden for this question.

Question 10

Reaction Concepts

15 / 15 pts

10.1 Reaction Type

5 / 5 pts

The rubric is hidden for this question.

10.2 Reaction Thresholds

5 / 5 pts

The rubric is hidden for this question.

10.3 Reaction Mass and Energy

5 / 5 pts

The rubric is hidden for this question.

Question 11

FNEK Curves

31 / 35 pts

11.1 Microscopic Fission XS

17 / 18 pts

The rubric is hidden for this question.

11.2 BE Per Nucleon

14 / 17 pts

The rubric is hidden for this question.

Time of completion _____
17/19 Dec 2025

Cadet _____

(First)

(Last)

Section HZ
Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE300, Fundamentals of Nuclear Engineering
Term End Exam

1. Instructions:

- a. Bring only your TI-30XIIS or TI-36XPRO calculator, pencil(s), and a hydration source into the examination room. **No phones, cameras, or smartwatches are allowed in the examination room.**
- b. Check your exam for 11 problems worth 500 marks. Check for the instructor provided Reference Data Card and NERD-P with the examination.
- c. For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question, unless otherwise stated.
- d. For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- e. Take up to 210 minutes (3.5 hours) to complete the examination.
- f. An instructor is in the hall or in the classroom for any questions.
- g. **When you turn in your exam check your name is written on every page.**

2. Exam Security:

- a. I will not discuss this exam with anyone until it is released from academic security on **22 December 2025 at 1700.**
- b. Cadet Signature: _____
- c. Cadet Name (print): _____

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Wt. No.

Cadet: [REDACTED]

90

1.

Consider a PWR which operates at a constant temperature of 400°C and is fueled with UO_2 enriched to 3.82387 w/o ^{235}U and a mass density of 11.1538 g/cm^3 . The remainder of the uranium is ^{238}U , and all the oxygen is of natural assay. Fast fissions add 2.863% of neutrons relative to thermal fissions, and thermal fissions regenerate 1.86300 neutrons per absorption in the fuel. During a given lifecycle, 1 in 8 fast neutrons leak from the core and 1 in 12 neutrons are absorbed in resonance peaks as they slow. Exactly 78% of all thermal neutrons go on to be absorbed in the core. The fuel is defined to be all of the uranium.

$$w_{235} = 0.0382387$$

$$w_{238} = 0.9617613$$

$$\lambda_{235} = 235.0439231$$

$$\lambda_{238} = 238.0507826$$

$$A_0 = 15.9994$$

- (40) a. Calculate the number density for ^{235}U , ^{238}U , and O. Report your answers to four significant figures.

$$\rho = 11.1538$$

$$N_{235} = \frac{w_{235} \rho N_A}{\lambda_{235}} \rightarrow N_{235} = 9.632 \times 10^{20}$$

$$N_{238} = \frac{w_{238} \rho N_A}{\lambda_{238}}$$

$$N_{238} = 2.392 \times 10^{22}$$

$$\rho_U = w_U \rho \quad \rho_U = 9.831590548 \text{ g/cm}^3$$

$$w_U = \frac{A_U}{A_U + 2A_0} \quad w_U = 0.881456593$$

$$A_U = \left[\frac{w_{235}}{\lambda_{235}} + \frac{w_{238}}{\lambda_{238}} \right]^{-1}$$

$$A_U = 237.9343402$$

$$w_O = \frac{w_O \rho N_A}{\lambda_O}$$

$$N_O = \lambda_O$$

$$w_O = \frac{2A_0}{2A_0 + A_U}$$

$$w_O = 0.118583407$$

$$N_O = 4.977 \times 10^{22} \text{ atoms/cm}^3$$

235U

$$9.632 \times 10^{20} \frac{\text{atoms } ^{235}\text{U}}{\text{cm}^3}$$

Report your answer in this box.

238U

$$2.392 \times 10^{22} \frac{\text{atoms } ^{238}\text{U}}{\text{cm}^3}$$

Report your answer in this box.

O

$$4.977 \times 10^{22} \frac{\text{atoms O}}{\text{cm}^3}$$

Report your answer in this box.

90

1.

Consider a PWR which operates at a constant temperature of 400°C and is fueled with UO_2 enriched to 3.82387 w/o ^{235}U and a mass density of 11.1538 g/cm³. The remainder of the uranium is ^{238}U , and all the oxygen is of natural assay. Fast fissions add 2.863% of neutrons relative to thermal fissions, and thermal fissions regenerate 1.86300 neutrons per absorption in the fuel. During a given lifecycle, 1 in 8 fast neutrons leak from the core and 1 in 12 neutrons are absorbed in resonance peaks as they slow. Exactly 78% of all thermal neutrons go on to be absorbed in the core. The fuel is defined to be all of the uranium.

fuel: $^{235}\text{U}, ^{238}\text{U}$
 non fuel: $\text{O}, \text{H}_2\text{O}$
 moderator

- (30) b. Calculate the reactor's thermal utilization factor. Assume the listed number density in NERD-P for the moderator is valid. Report your answer to six significant figures.

$$f = \frac{\Sigma_a^F}{\Sigma_a^F + \Sigma_a^{NF}}$$

$$\Sigma_a^F = N_{235} \sigma_{a,235} g_{a,235} + N_{238} \sigma_{a,238} g_{a,238} \quad \sigma_{a,235} = 687.06$$

$$\sigma_{a,238} = 2.736$$

$$\Sigma_a^{NF} = N_O \sigma_{a,O} + N_{H_2O} \sigma_{a,H_2O}$$

$$\sigma_{a,O} = 0.000276$$

$$\Sigma_a^F = (4.632 \times 10^{20}) [687.06 (0.9294) + (2.392 \times 10^{22}) (2.736) (1.0085)] \quad \sigma_{a,H_2O} = 0.6646$$

$$\Sigma_a^F = 6.868577445 \times 10^{23} \text{ b}$$

$$g_{a,235} = 0.9294$$

$$g_{a,238} = 1.0085$$

$$N_{H_2O} = 0.03343 \times 10^{24}$$

$$\Sigma_a^{NF} = (4.977 \times 10^{20}) (0.000276) + (0.03343 \times 10^{24}) (0.6646)$$

$$\Sigma_a^{NF} = 2.22109579 \times 10^{22} \text{ b}$$

$$f = 0.968409$$

$$f = 0.968409$$

Report your answer in this box.

- 90 1. Consider a PWR which operates at a constant temperature of 400°C and is fueled with UO_2 enriched to 3.82387 w/o ^{235}U and a mass density of 11.1538 g/cm³. The remainder of the uranium is ^{238}U , and all the oxygen is of natural assay. Fast fissions add 2.863% of neutrons relative to thermal fissions, and thermal fissions regenerate 1.86300 neutrons per absorption in the fuel. During a given lifecycle, 1 in 8 fast neutrons leak from the core and 1 in 12 neutrons are absorbed in resonance peaks as they slow. Exactly 78% of all thermal neutrons go on to be absorbed in the core. The fuel is defined to be all of the uranium.

- (20) c. Report the following reactor kinetics terms in the chart below. Then calculate the reactor's neutron multiplication factor to six significant figures and select its criticality:

$\epsilon = 1 + 0.2863$
 $\eta = 1.86300$
 $p_{NL}^f = 1 - \frac{1}{8}$
 $p = 1 - \frac{1}{12}$
 $p_{NL}^{th} = 0.78$
 $f = 0.968409$

$\epsilon = \text{fast fissions}$
 $p = \text{resonance escape prob}$
 $p_{NL}^f = \text{fast non-leakage prob}$
 $p_{NL}^{th} = \text{thermal non-leakage prob}$
 $f = \text{thermal utilization factor}$
 $\eta = \text{thermal reproduction factor}$

Term	Value
ϵ	1.2863
p_{NL}^f	0.875
p	0.917
p_{NL}^{th}	0.78
f	0.968409
η	1.86300
k_{eff}	1.45187

By calculation, the reactor above is:

- ☐ Subcritical
☐ Critical
☒ Supercritical

$k_{eff} > 1$

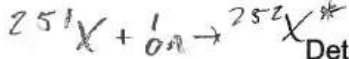
Wt. No.

Cadet: [REDACTED]

40

2.

The relative atomic masses and critical energies of fission for several isotopes of a hypothetical element X are listed below.



Determine by appropriate calculations if ^{251}X is fissile or fissionable. If it is fissionable, calculate what neutron energy would be needed to result in fission. Report your answer to three significant digits if needed.

Isotope	Atomic Weight (u)	Critical Energy of Fission (MeV)
^{249}X	249.074854	5.08
^{250}X	250.076406	4.17
^{251}X	251.079587	5.02
^{252}X	252.081630	$E^* = 6.43$

^{251}X is fissile if $S_n(^{250}\text{X}) < 5.02 \text{ MeV}$

$$S_n(^{250}\text{X}) = [m(^{249}\text{X}) + m_n - m(^{250}\text{X})]c^2 = [249.074854 + 1.0086649233 - 250.076406]c^2$$

$$S_n(^{250}\text{X}) = 6.625 \text{ MeV}$$

6.625 MeV is not less than 5.02 MeV, so ^{251}X is fissionable.

$T_n = ?$

$$T_n = E_{252}^* - S_n(^{252}\text{X})$$

$$S_n(^{252}\text{X}) = [m(^{251}\text{X}) + m_n - m(^{252}\text{X})]c^2 = [251.079587 + 1.0086649233 - 252.081630]c^2 = 931.494013 \text{ MeV}$$

$$S_n(^{252}\text{X}) = 6.168281908 \text{ MeV}$$

$$T_n = 6.43 - 6.168281908 \text{ MeV} \rightarrow T_n = 0.262$$

☐ Fissile

☒ Fissionable

0.262 MeV

Report your answer in this box if needed.

Wt. No.
40 3.

Cadet: [REDACTED]

- (20) a. Calculate the exact binding energy of ^{112}Cd to **six** significant figures.

$$BE(^{112}_{48}\text{Cd}) = [(48)(1.0078250321) + (112 - 48)(1.0086649233) - 111.9027572]c^2$$

$$= 931.494013 \text{ MeV}$$

$$BE(^{112}_{48}\text{Cd}) = 957.016 \text{ MeV}$$

957.016 MeV

Report your answer in this box.

- (20) b. Utilize the liquid drop model to calculate the approximate binding energy of a theoretical isotope, $^{311}_{121}\text{Un}$, to **six** significant figures. $A = 311$ $Z = 121$ 0.6×11.2

$$E_B = (15.835)(311) - (18.33)(311)^{2/3} - [0.714](121)^2 / (311)^{1/3} - (23.20)(311 - 2(121))^2 / 311$$

$$- (11.2) / \sqrt{311} \quad E_B = 2184.55 \text{ MeV}$$

2184.55 MeV

Report your answer in this box.

Wt. No.

40 4.

Cadet: [REDACTED]

1 day = 86400 s

Spent fuel is removed from a reactor and placed into a dry cask for storage. The cask of spent fuel contains 385.000 grams of the fission byproduct ^{131}I which has a half-life of 8.02520 days.

- (30) a. Calculate the activity in curies of the ^{131}I which remains after the cask is stored for exactly 365.000 days to six significant figures.

$$N = \frac{(385.000\text{g})(6.0221419 \times 10^{23})}{130.9061242}$$

$$N = 1.771135369 \times 10^{24}$$

$$\lambda = 0.086371328 \frac{1}{\text{days}}$$

$$A(t) = A_0 e^{(-\lambda t)}$$

$$A_0 = \lambda N$$

$$N = \frac{m N_A}{A_{^{131}\text{I}}}$$

$$A_0 = ?$$

$$A(t) = ?$$

$$\lambda = ?$$

$$t = 365.000 \text{ days}$$

$$T_{1/2} = 8.02520 \text{ days}$$

$$m = 385.000\text{g}$$

$$A_0 = (0.086371328)(1.771135369 \times 10^{24})$$

$$A_0 = 1.529753139 \times 10^{23}$$

$$A(t) = A_0 e^{(-\lambda t)} \cdot \frac{1\text{Ci}}{3.7 \times 10^{10} \text{Bq}} \rightarrow 0.0841500 \text{ Ci}$$

0.0841500 Ci

Report your answer in this box.

- (10) b. Consider a 1.0-Ci source of ^{137}Cs produced in the year 1995. Select the closest present-day activity of the source.

☐ 1.3x10⁵ Bq

☐ 3.7x10⁹ Bq

☒ 1.8x10¹⁰ Bq

☐ 2.9x10¹⁰ Bq

$$A(t) = (1.0) e^{-(0.023043457)(30)}$$

$$T_{1/2} = 30.08 \text{ y}$$

$$A(t) = 0.500922585 \text{ Ci}$$

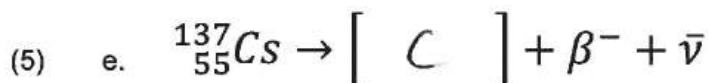
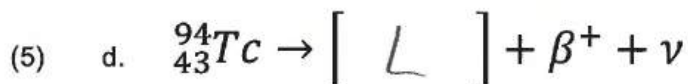
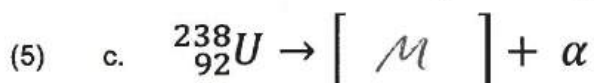
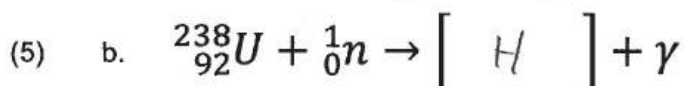
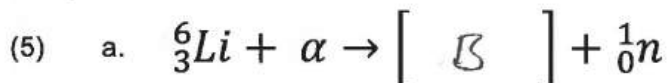
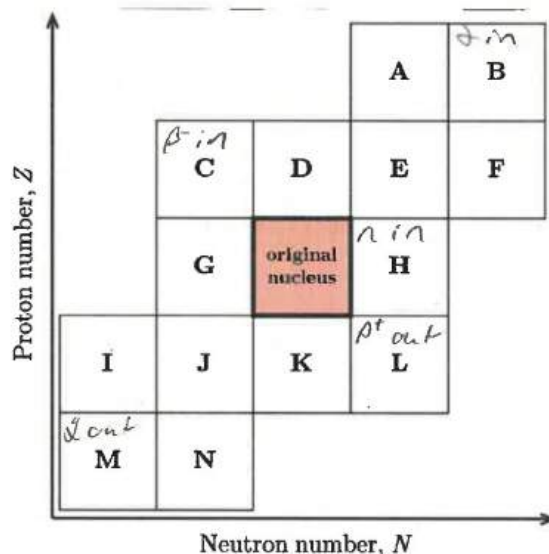
$$\lambda = \frac{\ln(2)}{30.08} \rightarrow \lambda = 0.023043457 \frac{1}{\text{years}}$$

$$= \frac{3.7 \times 10^{10} \text{Bq}}{1\text{Ci}}$$

35

5.

Consider the figure below. For each decay or reaction shown, write the letter corresponding to the position of the missing nuclide in the box. Note: Some of the standard nomenclature may be omitted



- (10) f. In any given material, the range of the particle emitted in part c. is relatively _____ compared to those of other radiations.

☐ long

☐ the same

☒ short ✓

☐ cannot be determined

Wt. No.

30

6.

A 0.62202 MeV neutron enters a target of ^{14}C . Calculate the average number of collisions between the neutron and the carbon nuclei required to slow the neutron to thermal energy (room temperature).

Cadet: [REDACTED]

$$E_{\text{th}} = 0.0253 \text{ eV} \rightarrow E_2 = 0.0253 \times 10^{-6} \text{ MeV} \quad A = 14$$

$$E_1 = 0.62202 \text{ MeV}$$

$$n = \frac{1}{\xi} \ln \left(\frac{E_1}{E_2} \right) \quad \xi = 1 + \frac{\alpha}{1-\alpha} \ln \alpha \quad \alpha = \left(\frac{A-1}{A+1} \right)^2$$

$$\alpha = \left(\frac{14-1}{14+1} \right)^2$$

$$\xi = 0.136284194$$

$$\alpha = 0.751111111$$

$$n = \frac{1}{0.136284194} \ln \left(\frac{0.62202}{0.0253 \times 10^{-6}} \right) \quad n = 124.869 \text{ collisions}$$

124.869 collisions

Report your answer in this box.

55

7.

An experiment produces a beam of 0.200 MeV photons with an intensity of $8.83157 \times 10^{12} \text{ cm}^{-2} \text{ s}^{-1}$ and there is a need to evaluate the feasibility of using 1.62973 cm of copper, Cu, as shielding material. The copper shield has a density of 8.95852 g/cm³. Any photon that does not collide in the shield will enter 5.28375 cm of tissue afterwards. Tissue has a density of 1.03468 g/cm³.

$$I_0 = 8.83157 \times 10^{12}$$

$$x_1 = 1.62973 \text{ cm}$$

$$x_2 = 5.28375 \text{ cm}$$

$$\rho_{\text{Cu}} = 8.95852$$

$$\rho_t = 1.03468$$

- (30) a. Calculate the uncollided intensity that results after passing through both the copper and the tissue to six significant figures.

$$I(x) = I_0 e^{-\mu_{\text{Cu}} x_1} e^{-\mu_t x_2} \rightarrow I(x) = I_0 e^{-(\mu_{\text{Cu}} x_1 + \mu_t x_2)}$$

$$\mu_{\text{Cu}} = \left(\frac{\mu}{\rho}\right)_{\text{Cu}} \cdot \rho_{\text{Cu}} \quad \left(\frac{\mu}{\rho}\right)_{\text{Cu}} = 0.147 \quad \mu_{\text{Cu}} = (0.147)(8.95852) \rightarrow \mu_{\text{Cu}} = 1.31690244$$

$$\mu_t = \left(\frac{\mu}{\rho}\right)_t \cdot \rho_t \quad \left(\frac{\mu}{\rho}\right)_t = 0.132 \quad \mu_t = (0.132)(1.03468) \rightarrow \mu_t = 0.13657776$$

$$I(x) = (8.83157 \times 10^{12}) e^{-(1.31690244 \cdot 1.62973 + 0.13657776 \cdot 5.28375)}$$

$$I(x) = 5.01824 \times 10^{11} \frac{1}{\text{cm}^2 \cdot \text{s}}$$

$$5.01824 \times 10^{11} \frac{1}{\text{cm}^2 \cdot \text{s}}$$

Report your answer in this box.

- (25) b. Calculate the minimum thickness of copper required for exactly 95% of the photon beam to be attenuated before reaching the tissue to six significant figures.

$$\frac{I(x)}{I_0} = 0.05 \quad \frac{I(x)}{I_0} = e^{-\mu_{\text{Cu}} x} \rightarrow \ln\left(\frac{I(x)}{I_0}\right) = -\mu_{\text{Cu}} x \rightarrow x = \frac{\ln(0.05)}{-\mu_{\text{Cu}}}$$

$$\mu_{\text{Cu}} = 1.31690244$$

$$x = \frac{\ln(0.05)}{-1.31690244} \rightarrow x = 2.27483 \text{ cm}$$

$$2.27483 \text{ cm}$$

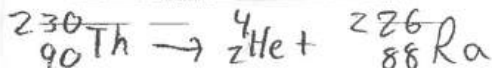
Report your answer in this box.

Wt. No.

Cadet: [REDACTED]

50 8. A $^{230}_{90}\text{Th}$ nucleus undergoes alpha decay.

(10) a. Write down the complete reaction equation for this decay event using standard form.



(40) b. Calculate the kinetic energy of the alpha particle produced in this reaction to **six** significant figures.

$$Q_{\text{tot}} = [M(^{230}_{90}\text{Th}) - M(^4_2\text{He}) - M(^{226}_{88}\text{Ra})]c^2$$

$$Q_{\text{tot}} = [230.0331266 - 226.0254026 - 4.002603249] \text{ u} \cdot 931.494013 \text{ MeV/u}$$

$$Q_{\text{tot}} = 4.769948247 \text{ MeV}$$

$$Q_{\alpha} = \frac{M(^{226}_{88}\text{Ra})}{M(^{226}_{88}\text{Ra}) + M(^4_2\text{He})} \cdot Q_{\text{tot}} \rightarrow \left(\frac{226.0254026}{226.0254026 + 4.002603249} \right) \cdot Q_{\text{tot}}$$

$$Q_{\alpha} = 4.68695 \text{ MeV}$$

4.68695 MeV

Report your answer in this box.

Wt. No.

Cadet: [REDACTED]

- 70 9. A 13.57945 MeV proton is absorbed by nitrogen 14, resulting in a daughter isotope and a neutron.

- (10) a. Write down the complete nuclear equation in standard form.



- (50) b. Calculate the threshold energy for this equation to six significant figures.

$$E^{\text{th}} = \max(E^{\text{K}}, E^{\text{C}}) \quad E^{\text{K}} = -\left(1 + \frac{m_1}{m_2}\right) Q \quad E^{\text{C}} = 1.2 \frac{Z_1 Z_2}{A_1^{1/3} + A_2^{1/3}}$$

$$Q = [m({}^{14}_7\text{N}) + m({}^1_1\text{H}) - m({}^{14}_8\text{O}) - m_{\text{n}}] c^2$$

$$Q = [14.0030740052 + 1.0078250321 - 14.00859529 - 1.0086649233] c^2$$

$$Q = -5.92539763 \text{ MeV}$$

$$-931.494013 \text{ MeV}$$

$$E^{\text{K}} = -\left(1 + \frac{1.0078250321}{14.0030740052}\right) (-5.92539763)$$

$$E^{\text{K}} = 6.35186 \text{ MeV}$$

$$E^{\text{C}} = (1.2) \left(\frac{7 \cdot 1}{7^{1/3} + 1^{1/3}} \right)$$

$$E^{\text{C}} = 2.88369 \text{ MeV}$$

$$E^{\text{th}} = E^{\text{K}} = 6.35186 \text{ MeV}$$

6.35186 MeV

Report your answer in this box.

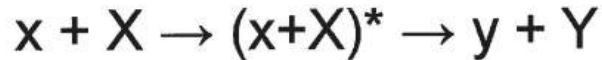
- (10) c. Based on your calculations, this reaction

☒ can occur

☐ cannot occur

☐ not enough information given

15 10. Consider a generic nuclear reaction with the form and conditions shown below:

Conditions:

1. In this reaction, $Q = -1.19 \text{ MeV}$
2. Reactant 'x' possesses a net positive charge
3. Reactant 'X' and product 'Y' are different elements

(5) a. The nuclear reaction indicated is an example of a(n) _____ reaction. (Choose all that apply)

- ☒ Binary
- ☐ Radioactive Decay
- ☒ Endoergic
- ☐ Exoergic
- ☐ Fission

(5) b. Indicate all reaction threshold(s) which exist for the reaction above. (Choose all that apply).

- ☒ Coulombic Threshold
- ☐ Critical Energy of Fission Threshold
- ☒ Kinematic Threshold
- ☐ Radioactive Decay Threshold
- ☐ There are not thresholds for this reaction
- ☐ Not enough information to determine

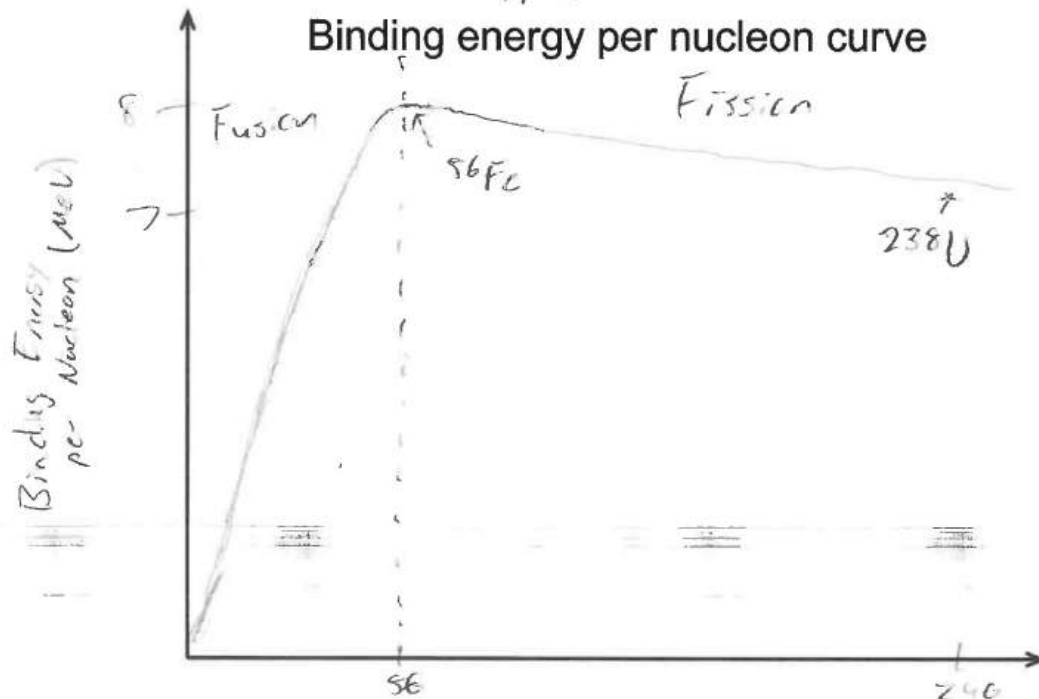
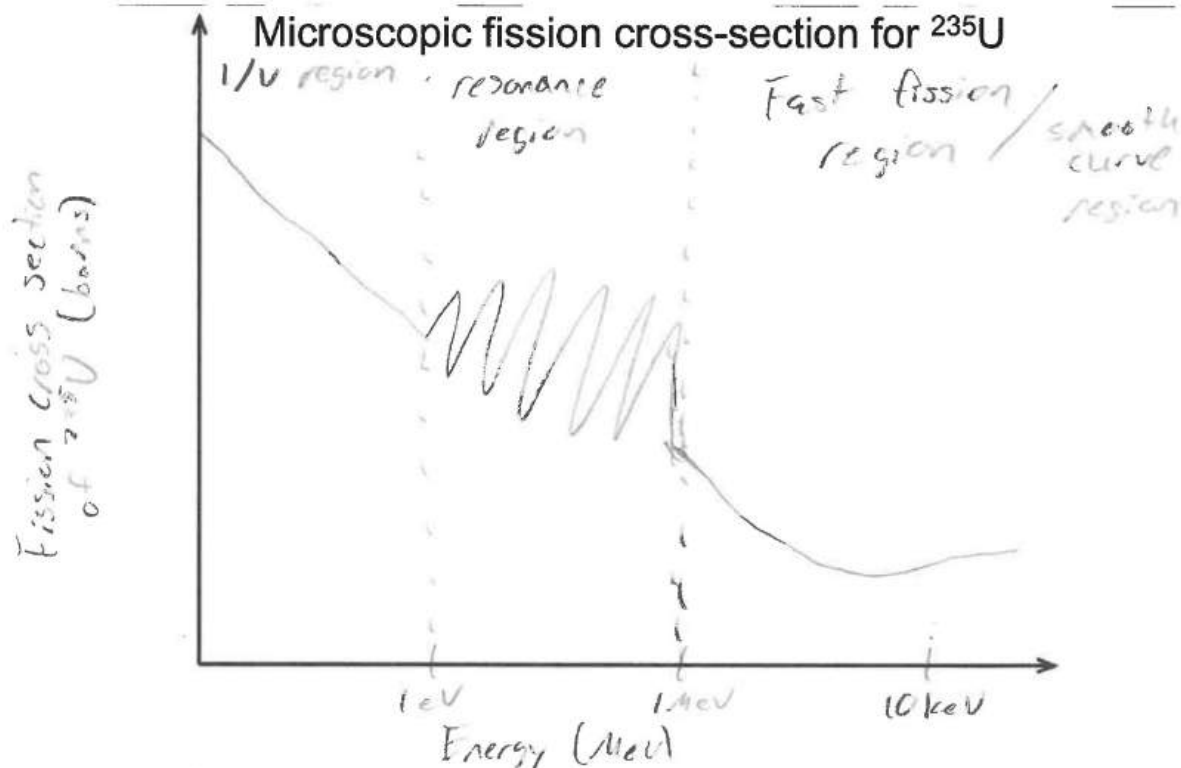
(5) c. Which of the following statements are true about the reaction above? (Choose all that apply)

- ☒ Total of (mass + energy) and charge are all conserved in the reaction.
- ☒ The reactants have less collective rest mass than the products.
- ☐ The total kinetic energy of the products must be greater than the reactants.

Wt. No.
35 11.

Cadet: [REDACTED]

Sketch the microscopic fission cross section for ^{235}U and binding energy per nucleon curve using the axes provided below. For full credit, label all axes appropriately, including units as applicable and indicate any critical values. Then, indicate significant regions of the plot.



Poor Student Work

Student

Total Points

325 / 500 pts

Question 1

Reactor Physics

72 / 90 pts

1.1 — Number Density 235U

29 / 30 pts

The rubric is hidden for this question.

1.2 — Number Density 238U

4 / 5 pts

The rubric is hidden for this question.

1.3 — Number Density O

3 / 5 pts

The rubric is hidden for this question.

1.4 — Thermal Utilization Factor

24 / 30 pts

The rubric is hidden for this question.

1.5 — Six Factor Formula

12 / 20 pts

The rubric is hidden for this question.

Question 2

Fission

24 / 40 pts

2.1 — Fission

20 / 36 pts

The rubric is hidden for this question.

2.2 — Fissionable

4 / 4 pts

The rubric is hidden for this question.

Question 3

Binding Energy

33 / 40 pts

3.1 — Binding Energy Formula

16 / 20 pts

The rubric is hidden for this question.

3.2 — Liquid Drop Model

17 / 20 pts

The rubric is hidden for this question.

Question 4

Radioactive Decay

36 / 40 pts

4.1 Activity Decay

26 / 30 pts

The rubric is hidden for this question.

4.2 Half Life

10 / 10 pts

The rubric is hidden for this question.

Question 5

Decay Equations

15 / 35 pts

5.1 Chart of Nuclides

5 / 25 pts

The rubric is hidden for this question.

5.2 Particle Penetration

10 / 10 pts

The rubric is hidden for this question.

Question 6

Thermalization

25 / 30 pts

The rubric is hidden for this question.

Question 7

Shielding

35 / 55 pts

7.1 Intensity Calculation

10 / 30 pts

The rubric is hidden for this question.

7.2 Thickness Calculation

25 / 25 pts

The rubric is hidden for this question.

Question 8

Alpha Decay

18 / 50 pts

8.1 Decay Equation

2 / 10 pts

The rubric is hidden for this question.

8.2 Kinetic Energy

16 / 40 pts

The rubric is hidden for this question.

Question 9

Binary Reaction

43 / 70 pts

9.1 Equation

8 / 10 pts

The rubric is hidden for this question.

9.2 Threshold Energy

35 / 50 pts

The rubric is hidden for this question.

9.3 Threshold Interpretation

0 / 10 pts

The rubric is hidden for this question.

Question 10

Reaction Concepts

9 / 15 pts

10.1 Reaction Type

2 / 5 pts

The rubric is hidden for this question.

10.2 Reaction Thresholds

4 / 5 pts

The rubric is hidden for this question.

10.3 Reaction Mass and Energy

3 / 5 pts

The rubric is hidden for this question.

Question 11

FNEK Curves

15 / 35 pts

11.1 Microscopic Fission XS

1 / 18 pts

The rubric is hidden for this question.

11.2 BE Per Nucleon

14 / 17 pts

The rubric is hidden for this question.

Time of completion 2116
17/19 Dec 2025

Cadet [REDACTED]
(First) (Last)

Section 62
Version 1

DEPARTMENT OF PHYSICS AND NUCLEAR ENGINEERING
NE300, Fundamentals of Nuclear Engineering
Term End Exam

1. Instructions:

- a. Bring only your TI-30XIIS or TI-36XPRO calculator, pencil(s), and a hydration source into the examination room. **No phones, cameras, or smartwatches are allowed in the examination room.**
- b. Check your exam for 11 problems worth 500 marks. Check for the instructor provided Reference Data Card and NERD-P with the examination.
- c. For multiple choice questions **fill in** your selection and choose the **one** answer that best answers each question, unless otherwise stated.
- d. For free response questions show all work; partial credit may be given for correct work shown. **Report answers in the answer boxes provided for each question.**
- e. Take up to 210 minutes (3.5 hours) to complete the examination.
- f. An instructor is in the hall or in the classroom for any questions.
- g. **When you turn in your exam check your name is written on every page.**

2. Exam Security:

- a. I will not discuss this exam with anyone until it is released from academic security on **22 December 2025 at 1700.**
- b. Cadet Signature: [REDACTED]
- c. Cadet Name (print): [REDACTED]

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Wt. No.

Cadet: [REDACTED]

- 90 1. Consider a PWR which operates at a constant temperature of 400°C and is fueled with UO_2 enriched to 3.82387 w/o ^{235}U and a mass density of 11.1538 g/cm³. The remainder of the uranium is ^{238}U , and all the oxygen is of natural assay. Fast fissions add 2.863% of neutrons relative to thermal fissions, and thermal fissions regenerate 1.86300 neutrons per absorption in the fuel. During a given lifecycle, 1 in 8 fast neutrons leak from the core and 1 in 12 neutrons are absorbed in resonance peaks as they slow. Exactly 78% of all thermal neutrons go on to be absorbed in the core. The fuel is defined to be all of the uranium.

- (40) a. Calculate the number density for ^{235}U , ^{238}U , and O. Report your answers to four significant figures.

$$\begin{aligned}
 w_{^{235}\text{U}} &= 0.0382387 & P_{\text{O}} &= ? \\
 w_{^{238}\text{U}} &= 0.9617613 \\
 P_{\text{UO}_2} &= 11.1538 \text{ g/cm}^3 & k &= \frac{0.0382387}{235.0439231} + \frac{0.9617613}{238.0507826} \\
 & & &= 237.9358042 \\
 w_{\text{U}} &= \frac{237.9358042}{237.9358042 + (2)(15.9994)} \\
 w_{\text{U}} &= 0.881457214 & N_{^{235}\text{U}} &= \frac{0.0382387(1.831597474)(4)}{235.0439231} \\
 P_{\text{U}} &= 0.881457214(11.1538) & N_{^{238}\text{U}} &= \frac{0.9617613(1.831597474)(4)}{238.0507826} \\
 P_{\text{U}} &= 9.831597474 & N_{\text{O}} &= \frac{2(11.1538)(N_{\text{U}})}{15.9994}
 \end{aligned}$$

235U
 $9.1632 \cdot 10^{20} \text{ mol}^{-1}$
 Report your answer in this box.

238U
 $2.392 \cdot 10^{22} \text{ mol}^{-1}$
 Report your answer in this box.

O
 $8.346 \cdot 10^{22} \text{ mol}^{-1}$
 Report your answer in this box.

- 90 1. Consider a PWR which operates at a constant temperature of 400°C and is fueled with UO_2 enriched to 3.82387 w/o ^{235}U and a mass density of 11.1538 g/cm³. The remainder of the uranium is ^{238}U , and all the oxygen is of natural assay. Fast fissions add 2.863% of neutrons relative to thermal fissions, and thermal fissions regenerate 1.86300 neutrons per absorption in the fuel. During a given lifecycle, 1 in 8 fast neutrons leak from the core and 1 in 12 neutrons are absorbed in resonance peaks as they slow. Exactly 78% of all thermal neutrons go on to be absorbed in the core. The fuel is defined to be all of the uranium.
- (30) b. Calculate the reactor's thermal utilization factor. Assume the listed number density in NERD-P for the moderator is valid. Report your answer to six significant figures.

$$f = \frac{\sum_a^F}{\sum_a^F + \sum_a^{NF}}$$

PWR D_2O

$$g_{235}(T) = 0.9294$$

$$g_{238}(T) = 1.0085$$

$$\sigma_{235}(E) = 687$$

$$\sigma_{238}(E) = 2.73$$

$$N_{\text{D}_2\text{O}} = 0.03323 \cdot 10^{24}$$

$$f = \frac{9.632 \cdot 10^{20} \cdot 10^{-24} (687) (0.9294) + 2.392 \cdot 10^{22} \cdot 10^{-24} (2.73) (1.0085)}{9.632 \cdot 10^{20} \cdot 10^{-24} (687) (0.9294) + 2.392 \cdot 10^{22} \cdot 10^{-24} (2.73) (1.0085) + 0.03323 \cdot 10^{24} (0.00133) (1)}$$

$$+ 0.03323 \cdot 10^{24} (0.00133) (1)$$

$$0.615001081 + 0.065856664$$

$$0.680857745$$

$$+ 0.00044126$$

$$f = 0.999935092$$

$$f = 0.999935$$

Report your answer in this box.

Consider a PWR which operates at a constant temperature of 400°C and is fueled with UO_2 enriched to 3.82387 w/o ^{235}U and a mass density of 11.1538 g/cm³. The remainder of the uranium is ^{238}U , and all the oxygen is of natural assay. Fast fissions add 2.863% of neutrons relative to thermal fissions, and thermal fissions regenerate 1.86300 neutrons per absorption in the fuel. During a given lifecycle, 1 in 8 fast neutrons leak from the core and 1 in 12 neutrons are absorbed in resonance peaks as they slow. Exactly 78% of all thermal neutrons go on to be absorbed in the core. The fuel is defined to be all of the uranium.

- (20) c. Report the following reactor kinetics terms in the chart below. Then calculate the reactor's neutron multiplication factor to six significant figures and select its criticality:

$$p_{NL}^f = \frac{7}{8}$$

$$p_{NL}^{th} = \frac{11}{12}$$

$$0.02863 \times 0.78$$

$$\eta = 1.86300$$

$$\epsilon p = 0.0223314$$

$$k_{eff} = \epsilon p_{NL}^f p_{NL}^{th} f \eta = 0.033367223$$

$$\underline{0.03337}$$

Term	Value
ϵ	Fast fission factor
p_{NL}^f	Probability of fast neutron non-leakage
p	Neutron absorption probability
p_{NL}^{th}	Probability of thermal neutron non-leakage
f	Thermal utilization factor
η	Thermal reproduction factor
k_{eff}	Neutron multiplication factor

By calculation, the reactor above is:

☒ Subcritical

☐ Critical

☐ Supercritical

Wt. 40 **No.** 2.

Cadet: _____

The relative atomic masses and critical energies of fission for several isotopes of a hypothetical element X are listed below.

Determine by appropriate calculations if ^{251}X is fissile or fissionable. If it is fissionable, calculate what neutron energy would be needed to result in fission. Report your answer to **three** significant digits if needed.

Isotope	Atomic Weight (u)	Critical Energy of Fission (MeV)
^{249}X	249.074854	5.08
^{250}X	250.076406	4.17
^{251}X	251.079587	5.02
^{252}X	252.081630	6.43

$$S_n(^A_Z\text{X}) = M(^{A-1}_Z\text{X}) + m_n - M(^A_Z\text{X})$$

$$(250.076406 + 1.00866491588 - 251.079587) \text{ u}$$

$$S_n = 5.10824722 + 5.02$$

$$10.1 \text{ MeV}$$

- ☐ Fissile —
- ☒ Fissionable

10.1 MeV

Report your answer in this box if needed.

Wt. No.
40 3.

Cadet: [REDACTED]

- (20) a. Calculate the exact binding energy of ^{112}Cd to **six** significant figures.

$$2.0141017780$$

$$BE(^A_ZX) = (ZM(^1_1\text{H}) + (A-Z)m_n - M(^A_ZX)) c^2$$

$$111.9027572$$

$$48(2.0141017780) + (64-48)1.0086649233 - 111.9027572$$

$$= 0.9127669170931444013$$

$$= 850.2369184$$

$$Z = 48$$

$$A = 64$$

$$850.237 \text{ MeV}$$

Report your answer in this box.

- (20) b. Utilize the liquid drop model to calculate the approximate binding energy of a theoretical isotope, $^{311}_{121}\text{Un}$, to **six** significant figures.

$$\text{odd/odd} \rightarrow +11.2 \text{ MeV}$$

$$E_B = a_v A - a_s A^{2/3} - a_c \frac{Z^2}{A^{1/3}} - a_a \frac{(A-2Z)^2}{A} - \frac{\pm a_p}{A^{1/2}}$$

$$15.835(311) - 16.33(311)^{2/3} - 0.714\left(\frac{121^2}{311^{1/3}}\right) - 23.2\left(\frac{(311-2(121))^2}{311}\right) \pm \frac{11.2}{\sqrt{121}}$$

$$4013.28458 - 1898.10065 + 1.018181818$$

$$2186.20 \text{ MeV}$$

$$2186.20 \text{ MeV}$$

Report your answer in this box.

Wt. No.

40

4.

Spent fuel is removed from a reactor and placed into a dry cask for storage. The cask of spent fuel contains 385.000 grams of the fission byproduct ^{131}I which has a half-life of 8.02520 days.

- (30) a. Calculate the activity in curies of the ^{131}I which remains after the cask is stored for exactly 365.000 days to six significant figures.

$$m = 385$$

$$A = \lambda N$$

$$N = 385 (6.0221419 \times 10^{23})$$

$$130,9061242$$

$$N = 1,77113536 \times 10^{24}$$

$$\uparrow 0,086371328 = \lambda$$

$$A(t) = A_0 e^{(-\lambda t)}$$

$$8,02520$$

$$T_{1/2} = \frac{\ln 2}{\lambda} \cdot \frac{1}{\ln(2)}$$

$$A = 1,529753139 \times 10^{23}$$

$$\frac{T_{1/2}}{\ln(2)} = \frac{1}{\lambda}$$

$$\lambda = \frac{\ln(2)}{T_{1/2}}$$

$$\lambda = \frac{\ln(2)}{T_{1/2}}$$

$$\lambda = 0,086371328$$

$$A(t) = 1,529753139 \times 10^{23} e^{(-0,086371328 \cdot 365)}$$

$$3,11355 \times 10^9$$

$$3,11355 \times 10^9 / 3,7 \times 10^{10} \text{ Bq}$$

$$A(t) = 3,11355 \times 10^9 \text{ Ci}$$

Report your answer in this box.

- (10) b. Consider a 1.0-Ci source of ^{137}Cs produced in the year 1995. Select the closest present-day activity of the source.

☐ $1.3 \times 10^5 \text{ Bq}$

☐ $3.7 \times 10^9 \text{ Bq}$

☒ $1.8 \times 10^{10} \text{ Bq}$

☐ $2.9 \times 10^{10} \text{ Bq}$

$$T_{1/2} = 30 \text{ yrs}$$

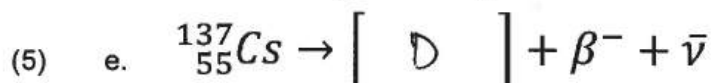
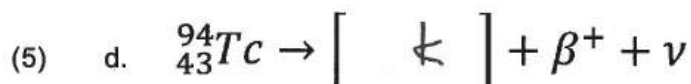
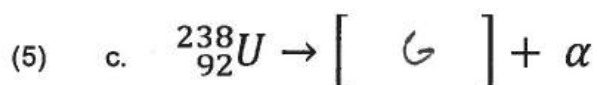
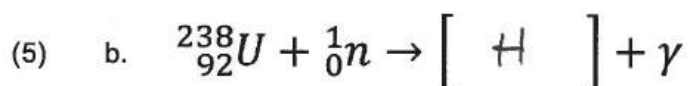
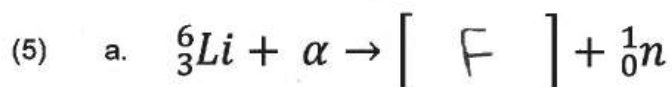
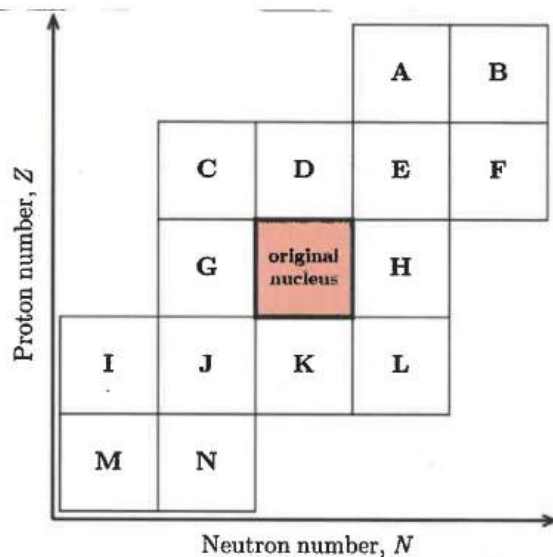
$$1 = 3,7 \times 10^{10}$$

$$0,5 (3,7 \times 10^{10})$$

$$= 3,7 \times 10^{10}$$

$$= 1,85 \times 10^{10}$$

Consider the figure below. For each decay or reaction shown, write the letter corresponding to the position of the missing nuclide in the box. Note: Some of the standard nomenclature may be omitted



- (10) f. In any given material, the range of the particle emitted in part c. is relatively _____ compared to those of other radiations.

☐ long

☐ the same

☒ short

☐ cannot be determined

Wt. No.

30

6.

Cadet: _____

A 0.62202 MeV neutron enters a target of ^{14}C . Calculate the average number of collisions between the neutron and the carbon nuclei required to slow the neutron to thermal energy (room temperature).

$$E_1 = 0.62202 \text{ MeV} \rightarrow 622.02 \text{ eV}$$

$$E_2 = 0.0253 \text{ eV}$$

↑
energy of
thermal
neutron

$$n = \frac{1}{\alpha} \ln \frac{E_1}{E_2}$$

$$\alpha = \left(\frac{14-1}{14+1} \right)^2$$

$$\left(\frac{13}{15} \right)^2$$

$$\alpha = 0.866666$$

$$E = \frac{1}{1-\alpha} \ln \alpha$$

$$1 + \frac{0.86}{1-0.86} \ln(0.86)$$

$$E = 0.06844883$$

$$n = \frac{1}{0.06844883} \ln \left(\frac{622.02}{0.0253} \right)$$

$$n = 144$$

144.75 collisions in ^{14}C

Report your answer in this box.

55

7.

An experiment produces a beam of 0.200 MeV photons with an intensity of $8.83157 \times 10^{12} \text{ cm}^{-2} \text{ s}^{-1}$ and there is a need to evaluate the feasibility of using 1.62973 cm of copper, Cu, as shielding material. The copper shield has a density of 8.95852 g/cm³. Any photon that does not collide in the shield will enter 5.28375 cm of tissue afterwards. Tissue has a density of 1.03468 g/cm³.

- (30) a. Calculate the uncollided intensity that results after passing through both the copper and the tissue to six significant figures.

$$I(x) = \frac{S_p}{4\pi r^2} e^{-\mu_t x}$$

$$\mu_t = \sum_i \left(\frac{\mu_i}{\rho} \right) \rho$$

$$I(x) = \frac{8.83157 \cdot 10^{12}}{4\pi (6.91348)^2} e^{-1.4534802(1.62973)} \mu_t = 1.4534802$$

$$1.62973 + 5.28375 = 6.91348$$

$$1.37622 \cdot 10^9$$

$$1.37622 \cdot 10^9 \text{ cm}^{-2} \text{ s}^{-1}$$

Report your answer in this box.

- (25) b. Calculate the minimum thickness of copper required for exactly 95% of the photon beam to be attenuated before reaching the tissue to six significant figures.

$$I(x) = I_0 e^{-\mu_t x}$$

$$0.05 = 1$$

$$\frac{I(x)}{I_0} = e^{-\mu_t x} (-\ln)$$

$$\frac{-\ln\left(\frac{I(x)}{I_0}\right)}{\mu_t} = x$$

$$\frac{-\ln\left(\frac{0.05}{1}\right)}{1.31690244} = 2.27483 \text{ cm}$$

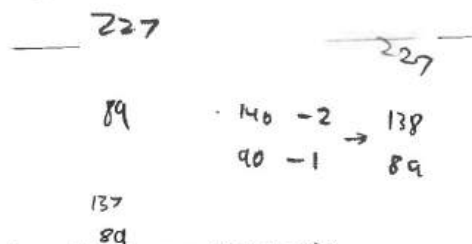
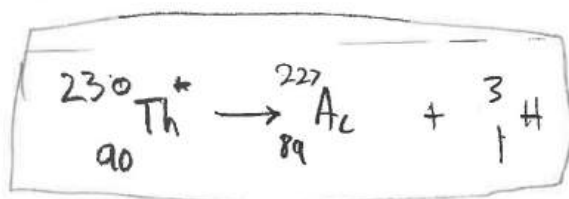
Report your answer in this box.

Wt. No.

Cadet: _____

50 8. A $^{230}_{90}\text{Th}$ nucleus undergoes alpha decay.

(10) a. Write down the complete reaction equation for this decay event using standard form.



(40) b. Calculate the kinetic energy of the alpha particle produced in this reaction to six significant figures.

$$Q = (230.0331266) - (227.0277470 + 4.0015061747) \text{ c}^2$$

$$= -1.0224504 (931.44013) = -952.4064262$$

$$Q = T_f - T_i$$

$$E_i^k \approx - \left(1 + \frac{4.0015061747}{227.0277470} \right) = -927.885405$$

$$Q + T_i = T_f$$

$$Q = (230.0331266 - (227.0277470 + 4.0015061747)) (931.44013)$$

$$= -927.885405$$

$$E_i^k \approx 944.240511 \text{ MeV}$$

944.241 MeV

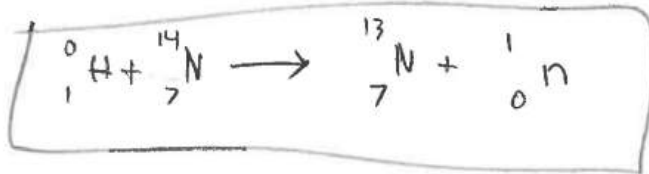
Report your answer in this box.

Wt. No.

Cadet: [REDACTED]

70 9. A 13.57945 MeV proton is absorbed by nitrogen 14, resulting in a daughter isotope and a neutron.

(10) a. Write down the complete nuclear equation in standard form.



(50) b. Calculate the threshold energy for this equation to six significant figures.

$$E_i^{\text{th}} = \max(E_i^{\text{k}}, E_i^{\text{c}})$$

$$Q = (1.0078250321 + 14.0030740052) - (13.00573858 + 1.0086649123) \cdot c^2$$

- not negative
no kinetic energy

$$= 0.00642534 \cdot 931.494013$$
$$= 6.0002296239 \text{ MeV}$$

$$E_i^{\text{c}} = \frac{1}{2} \frac{8.7}{14^{1/3} + 14^{1/3}} = 13.94$$

$$E_i^{\text{th}} = 13.94 \text{ MeV}$$

Report your answer in this box.

(10) c. Based on your calculations, this reaction

☒ can occur

☐ cannot occur

☐ not enough information given

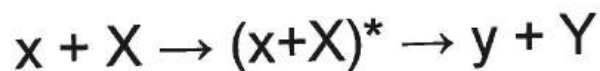
Wt. **No.**

15

10.

Cadet: _____

Consider a generic nuclear reaction with the form and conditions shown below:



Conditions:

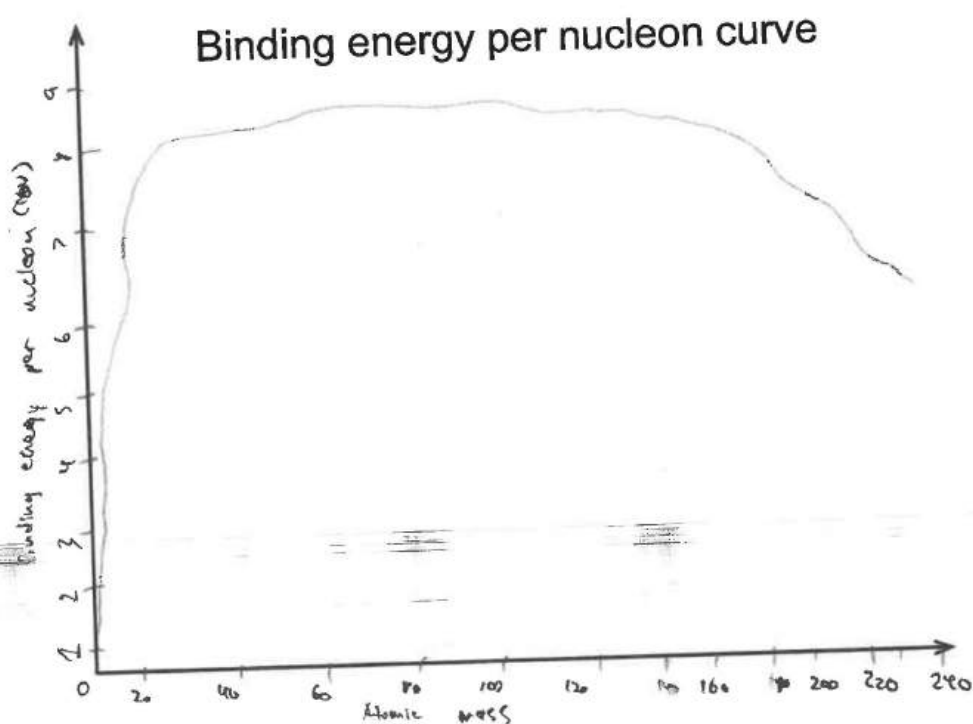
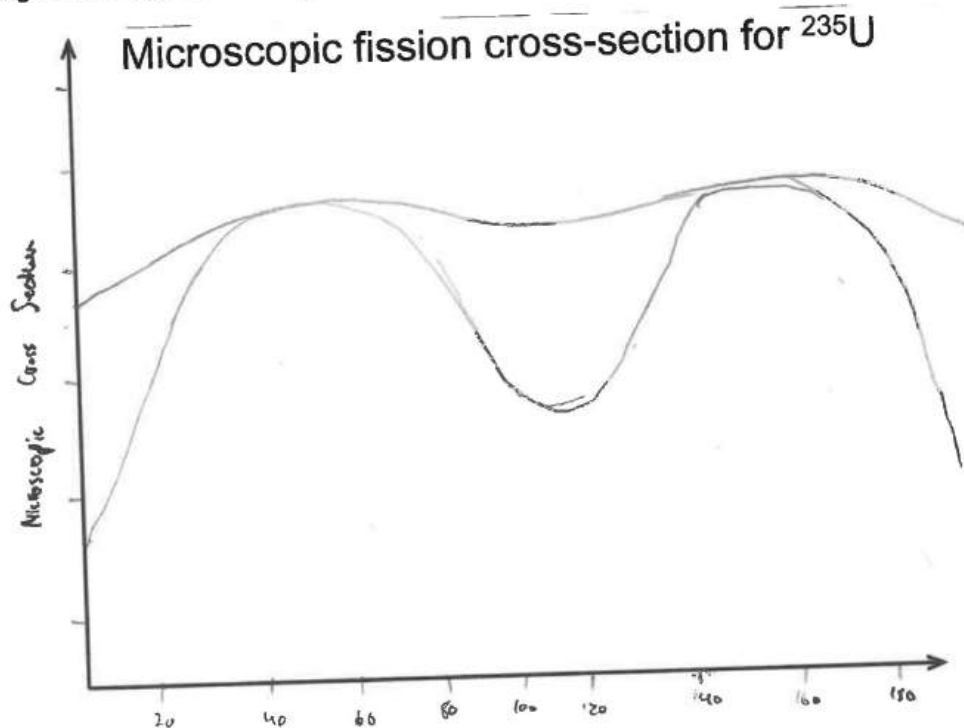
1. In this reaction, $Q = -1.19 \text{ MeV}$
2. Reactant 'x' possesses a net positive charge
3. Reactant 'X' and product 'Y' are different elements

- (5) a. The nuclear reaction indicated is an example of a(n) _____ reaction. (Choose all that apply)
- ☒ Binary
 - ☐ Radioactive Decay
 - ☐ Endoergic
 - ☐ Exoergic
 - ☒ Fission
- (5) b. Indicate all reaction threshold(s) which exist for the reaction above. (Choose all that apply).
- ☒ Coulombic Threshold
 - ☒ Critical Energy of Fission Threshold
 - ☒ Kinematic Threshold
 - ☐ Radioactive Decay Threshold
 - ☐ There are not thresholds for this reaction
 - ☐ Not enough information to determine
- (5) c. Which of the following statements are true about the reaction above? (Choose all that apply)
- ☒ Total of (mass + energy) and charge are all conserved in the reaction.
 - ☐ The reactants have less collective rest mass than the products.
 - ☐ The total kinetic energy of the products must be greater than the reactants.

Wt. 35 **No.** 11.

Cadet: XXXXXXXXXX

Sketch the microscopic fission cross section for ^{235}U and binding energy per nucleon curve using the axes provided below. For full credit, label all axes appropriately, including units as applicable and indicate any critical values. Then, indicate significant regions of the plot.





UNITED STATES MILITARY ACADEMY
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TAB J FINAL COURSE GRADES

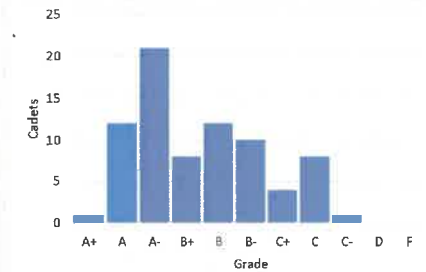
NE300, AY26-1 Final Grades

Name	Grad Year	Max Points	Points Earned	Percent	Letter Grade	Points Needed	FOS	CES	HWs	IPs	WPRs	TEE
BURNS, ANDREW N	2028	1600	1542	96.4%	A+		NEN1		99%	101%	95%	95%
YANG, BOHAN	2027	1600	1495	93.4%	A	9	BIO0		100%	113%	90%	89%
POLAND, CONNOR B	2027	1600	1493.5	93.3%	A	11	CHM2A		87%	91%	94%	96%
KIM, ELAINE S	2027	1600	1491.75	93.2%	A	12	BIO0		95%	104%	93%	90%
ESTEP, DEKLAN M	2028	1600	1485	92.8%	A	19	NEN1		95%	102%	85%	95%
LEWIS, CLAIRE C	2026	1600	1483.5	92.7%	A	21	CEN2	TST1N	93%	110%	85%	96%
HOCKENSMITH, SILAS N	2027	1600	1478	92.4%	A	26	GIS2H		94%	90%	87%	96%
CUETO, NICHOLAS C	2028	1600	1469	91.8%	A		NEN1		95%	116%	89%	87%
GORCZYNSKI, JOHN S	2028	1600	1457	91.1%	A		NEN1		93%	102%	86%	90%
FERRYING, ZANE J	2026	1600	1456.75	91.0%	A		CEN2		94%	96%	87%	92%
GERARD, ALLISON L	2027	1600	1456.75	91.0%	A		CHM2A		88%	102%	89%	91%
MATHIS, JON L	2028	1600	1456	91.0%	A		NEN1		100%	108%	79%	92%
CORBETT, RICHARD J	2027	1600	1452.25	90.8%	A		SS03		93%	84%	90%	89%
WIMER, CLAIRE E	2027	1600	1446.75	90.4%	A-	1	KIN1		97%	107%	86%	85%
JACOBS, KYRA L	2027	1600	1437.5	89.8%	A-	11	PIF0		99%	105%	86%	84%
SMITH, RILEY J	2027	1600	1433	89.6%	A-	15	LIS3		90%	97%	84%	91%
RECORDS, BENJAMIN M	2027	1600	1431	89.4%	A-	17	CHM2A		85%	94%	85%	94%
SINGH, SAMRATH	2026	1600	1430	89.4%	A-	18	CEN2		90%	116%	87%	86%
PLAGENS, ABBY E	2028	1600	1427	89.2%	A-	21	NEN1		96%	116%	79%	87%
WHITFIELD, AUDREY R	2027	1600	1423	88.9%	A-	25	BIO0		98%	113%	85%	81%
PARKES, WADE K	2026	1600	1422.25	88.9%	A-	26	MEN3		85%	99%	92%	86%
MERRIAM, CHASE A	2027	1600	1421.5	88.8%	A-	27	PHY1	SPA0N	97%	111%	81%	85%
RODRIGUEZ, ERIC J	2026	1600	1421	88.8%	A-	27	CEN2		86%	85%	87%	90%
MAALA, FRANCES ANGELO L	2027	1600	1419.75	88.7%	A-	28	NEN1		93%	99%	83%	88%
THOMSON, MICAH X	2028	1600	1414	88.4%	A-		NEN1		89%	96%	91%	81%
COX, BENJAMIN D	2026	1600	1406.25	87.9%	A-		CEN2		93%	99%	81%	91%
LATAIF, PHILIP H	2027	1600	1405.25	87.8%	A-		PAP3		95%	107%	82%	86%
COOPER, JOSHUA C	2028	1600	1404.5	87.8%	A-		NEN1		98%	96%	85%	80%
CAPRIOLA, JACK T	2027	1600	1404.25	87.8%	A-		HNT3T	RSL1N	90%	96%	80%	89%
NGUYEN, MICHAEL	2026	1600	1404	87.8%	A-		BIO0H	N	91%	104%	90%	77%
WILSON, JOHN P	2027	1600	1403.75	87.7%	A-		CHM2A		92%	91%	81%	88%
LEVENDOSKI, LAUREN E	2027	1600	1396	87.3%	A-		SS03		96%	115%	83%	77%
BARKER, BRANDON L	2028	1600	1394	87.1%	A-		NEN1		99%	108%	82%	79%
ZANG, KAITLYN P	2027	1600	1392.5	87.0%	A-		BIO0		100%	107%	75%	84%
MORRIS, ANNA M	2028	1600	1378	86.1%	B+	14	NEN1		92%	79%	85%	84%
LITRELL, COOPER A	2027	1600	1373	85.8%	B+	19	PAP3	TST1N	93%	107%	79%	85%
MCDONALD, JONATHAN B	2028	1600	1365.75	85.4%	B+	26	NEN1		91%	81%	80%	83%
GRAUBERGER, ASHLYN G	2027	1600	1354.5	84.7%	B+		EPS1	AST1N	96%	106%	76%	79%
HECKLE, CAROLYN N	2027	1600	1342.25	83.9%	B+		PHY1	N	95%	80%	79%	81%
NYHUIS, PARKER A	2027	1600	1341	83.8%	B+		HMH3T		81%	97%	79%	87%
KUMP, AUSTYN E	2027	1600	1339	83.7%	B+		LIS3		89%	99%	82%	76%
MACUNE, GRACE E	2026	1600	1336.5	83.5%	B+		CEN2		93%	110%	78%	76%
RERRAS, ROBERT	2027	1600	1334.25	83.4%	B	2			87%	96%	84%	75%
ROSS, JAMES C	2027	1600	1333.75	83.4%	B	2	CHM2A		85%	78%	80%	84%
TAYSON, REX M	2027	1600	1330.25	83.1%	B	6	SS02		92%	98%	69%	88%
DOHERTY, ROWAN E	2027	1600	1328	83.0%	B	8	PHY1	RSU1N	85%	103%	76%	83%
ADAMS, LUCAS S	2028	1590	1318.75	82.9%	B	9	NEN1		80%	75%	82%	80%
SINDLER, ALLAN Y	2028	1600	1319	82.4%	B	17	NEN1		81%	46%	81%	85%
COBB, MEGAN E	2027	1600	1318.25	82.4%	B	18	SS03		95%	107%	77%	72%
VINEYARD, JACOB L	2027	1600	1304.5	81.5%	B	32	MEN3		90%	100%	76%	73%
SHANNON, JACOB R	2028	1600	1303	81.4%	B		NEN1		81%	94%	82%	73%
HOLLAND, NOAH D	2027	1600	1288	80.5%	B		GIS2		84%	75%	78%	77%
NGUYEN, BENJAMIN H	2028	1600	1281.75	80.1%	B		NEN1		89%	99%	73%	72%
JORDAN, LILIAN R	2028	1600	1280.25	80.0%	B		NEN1		90%	105%	78%	68%
ZVIK, TAL	2027	1600	1279	79.9%	B-	1	PIF0		89%	90%	72%	80%
BLEVINS, THOMAS A	2027	1600	1276.5	79.8%	B-	4	LIS3	TST1N	89%	100%	67%	78%
BALDWIN, HUGH B	2026	1600	1254.75	78.4%	B-	25	PII0	RSE1N	83%	96%	70%	75%
RATHNAYAKE, KALANA S	2026	1600	1249	78.1%	B-	31	MEN3		83%	97%	79%	69%
GREEN, HENRY P	2027	1600	1247	77.9%	B-		ECN2		80%	97%	74%	74%
JAWORSKI, PRESTON H	2027	1600	1242.25	77.6%	B-		MGN0		76%	88%	79%	70%
PAYNE, BRENNAN N	2028	1600	1241	77.6%	B-		NEN1	RSU1N	87%	85%	71%	71%
HENG, KAYNEN M	2028	1600	1232.5	77.0%	B-		NEN1		80%	63%	70%	78%
SHIOUTAKON, CLAIRE A	2027	1600	1231	76.9%	B-		FLP2	MSM0N	93%	108%	71%	63%
DOLAN, PATRICK E	2026	1600	1226.5	76.7%	B-		LIS3		84%	113%	70%	71%
MOOK, MALACHI J	2026	1600	1210.75	75.7%	C+	13	MEN3		81%	76%	71%	77%
TRINGALI, THOMAS E	2026	1600	1206	75.4%	C+	18	CEN2		91%	91%	71%	67%
MORALES, ESTEBAN B	2027	1600	1202.25	75.1%	C+	22	CHM2A		83%	89%	59%	79%
LEE, KYLE S	2028	1600	1194	74.6%	C+	30	NEN1		93%	94%	61%	68%
HICKS, JACKSON E	2027	1600	1154.5	72.2%	C	14	ECN2	RSE1N	73%	100%	68%	61%
OLSON, ZACHARIAH D	2027	1600	1153.25	72.1%	C	15	PAP3		79%	54%	62%	72%
CANNON, KADEN R	2028	1600	1152.5	72.0%	C	16	NEN1		85%	100%	57%	68%
HALL, MICHAEL S	2027	1600	1142.25	71.4%	C	26	FLA2		90%	87%	71%	56%
BETHKE, SAMUEL J	2027	1600	1131.5	70.7%	C		PHY1		50%	52%	69%	80%
SCHERRA, JACOB R	2028	1600	1131	70.7%	C		NEN1		80%	102%	53%	70%
GRIFFITH, MATTHEW C	2027	1600	1124	70.3%	C		LIS3		88%	85%	60%	61%
BAEZ, ISAAC L	2026	1600	1116	69.8%	C		MGN0		94%	84%	59%	54%
SPAIN, JACK M	2028	1600	1100.5	68.8%	C-	11.5	NEN1		85%	79%	51%	65%
AVG				83.6%	B+							

Grading Scale	
A+	94.0%
A	90.5%
A-	87.0%
B+	83.5%
B	80.0%
B-	76.5%
C+	73.0%
C	69.5%
C-	66.0%
D	60.0%
F	<60%

Grade Distribution		
Requirement	Marks	% of Total
TEE	500	31%
WPR 1	250	16%
WPR 2	250	16%
Problem Sets (6)	300	19%
PA Brief	120	8%
Instructor Points	100	6%
FNEK Writ (1)	80	5%
Total	1600	100%

23	NEN1
4	PHY1
3	SS03
0	NSM0N
42	NE
0	Famous



Historical Data		
AY	Average	Grade
25-2	80.86%	B
25-1	83.00%	B
24-2	80.80%	B
24-1	81.61%	B
23-2	87.12%	A-
23-1	89.67%	A-
22-2	89.31%	A-
22-1		
21-2		
21-1	88.60%	A-
20-2	84.60%	B+
20-1	82.90%	B
19-2	81.20%	B

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SECTION III

ASSIGNMENTS



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TAB K

PROBLEM SETS



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PROBLEM SET #1

Problem Set #1

Problem 1

For each of the nuclides listed below, state the following:

- The number of neutrons and protons in the nucleus
- The atomic mass in amu
- Whether or not the nuclide is naturally occurring and its isotopic abundance, in decimal form, if applicable
- Whether or not the nuclide is stable and its half-life, in seconds, if applicable

(a) ^{12}C

- $Z = 6, N = 6$
- $\mathcal{A} = 12.0000000$
- Naturally occurring, $\gamma = 0.9893$
- Stable

(b) ^{26m}Al

- $Z = 13, N = 13$
- $\mathcal{A} = 25.98689166 + 228.305 \text{ keV} = 25.98713676$
Note: The excess mass comes from the fact that the metastable state is excited above the ground energy level
- Not naturally occurring
- Unstable, $T_{1/2} = 6.346 \text{ s}$

(c) ^{137}Cs

- $Z = 55, N = 82$
- $\mathcal{A} = 136.907084$
- Not naturally occurring
- Unstable, $T_{1/2} = 9.483 \times 10^8 \text{ s}$

(d) ^{206}Pb

- $Z = 82, N = 124$
- $\mathcal{A} = 205.974449$
- Naturally occurring, $\gamma = .241$
- Stable

(e) ^{235}U

- $Z = 92, N = 143$
- $\mathcal{A} = 235.0439231$
- Naturally occurring, $\gamma = .007204$
- Unstable, $T_{1/2} = 2.22 \times 10^{16} \text{ s}$

Note: some values taken from <https://www.nndc.bnl.gov/nudat3/>

Problem Set #1

Problem 2

A reactor is fueled with 5.0000 kg uranium enriched to 15.000 a/o ^{235}U with the remainder of the fuel being ^{238}U . The fuel has a mass density of 19.100 g/cm³.

- (a) What is the mass, in kg, of ^{235}U in the reactor
 - (b) Calculate the number (or atom) density of ^{235}U in the sample.
 - (c) Calculate the number (or atom) density of ^{238}U in the sample.
- (a) The mass of constituent x of object can be found by multiplying the total mass of the object, m by the weight fraction of x , w_x .

$$m_x = mw_x$$

$$\text{Using } w_x = \frac{n\mathcal{A}_x}{n\mathcal{A}_x + m\mathcal{A}_y},$$

$$m_x = m \frac{n\mathcal{A}_x}{n\mathcal{A}_x + m\mathcal{A}_y}$$

$$m_x = 5.0000 \text{ kg} \frac{(0.15)(235.0439231)}{(0.15)(235.0439231) + (0.85)(238.0507826)}$$

$$m_x = 0.741932 \text{ kg}$$

- (b) In part a) we found $w_{235\text{U}}$. We can use that information, along with PRC Equation 6, to solve for the number density of ^{235}U .

$$N_{235\text{U}} = \frac{w_{235\text{U}}\rho_{\text{fuel}}N_a}{\mathcal{A}_{235\text{U}}}$$

$$N_{235\text{U}} = \frac{(0.15)(235.0439231)}{(0.15)(235.0439231) + (0.85)(238.0507826)} \frac{(19.1 \text{ g/cm}^3)(6.02214199 \times 10^{23} \text{ atoms/mol})}{235.0439231 \text{ g/mol}}$$

$$N_{235\text{U}} = 7.2616 \times 10^{21} \text{ atoms/cm}^3$$

- (c) Similarly,

$$N_{238\text{U}} = \frac{w_{238\text{U}}\rho_{\text{fuel}}N_a}{\mathcal{A}_{238\text{U}}}$$

$$N_{238\text{U}} = \frac{(0.85)(238.0507826)}{(0.15)(235.0439231) + (0.85)(238.0507826)} \frac{(19.1 \text{ g/cm}^3)(6.02214199 \times 10^{23} \text{ atoms/mol})}{238.0507826 \text{ g/mol}}$$

$$N_{238\text{U}} = 4.1149 \times 10^{22} \text{ atoms/cm}^3$$

Problem Set #1

Problem 3

Boric Acid (BO_3H_3) is a common reactor poison (we'll talk about this in Block IV!) used to control reactivity, i.e. power generation, in a nuclear reactor. Consider a solution of boric acid with a mass density of 1.4400 g/cm^3 where the boron composition is exactly 25.0000 w/o ^{10}B , with the remainder being ^{11}B . The oxygen and hydrogen in the sample can be assumed to be naturally occurring. Calculate the number densities of ^{10}B , ^{11}B , and O in the solution.

$$\mathcal{A}_x = \left[\sum \frac{w_i}{\mathcal{A}_i} \right]^{-1} \text{ where } 25.0000 \text{ w/o } ^{10}\text{B} \rightarrow w_{10} = 0.25 \text{ and } 75.0000 \text{ w/o } ^{11}\text{B} \rightarrow w_{11} = 0.75$$

$$\mathcal{A}_B = \left[\frac{0.25}{10.0129370} + \frac{0.75}{11.0093055} \right]^{-1} \rightarrow 10.742074598$$

$$\text{Modification of } w_x = \frac{n\mathcal{A}_x}{n\mathcal{A}_x + m\mathcal{A}_y} \text{ leads to } \frac{n\mathcal{A}_x}{n\mathcal{A}_x + m\mathcal{A}_y + p\mathcal{A}_z} \text{ where } n = 1, m = 3, p = 3$$

$$w_B = \frac{(1)(10.7420746)}{(1)(10.7420746) + (3)(15.9994) + (3)(1.00797)} \rightarrow 0.173920771$$

$$w_O = \frac{(3)(15.9994)}{(3)(15.9994) + (1)(10.69017767) + (3)(1.00797)} \rightarrow 0.777120273$$

$$\text{Recall } \rho_i = w_i \rho_x$$

$$\rho_B = 0.173920771 (1.44) \rightarrow 0.250445910 \text{ g B/cm}^3$$

$$\rho_O = 0.777120273 (1.44) \rightarrow 1.119053193 \text{ g O/cm}^3$$

$$\text{Recall } N_i = \frac{w_i \rho_x N_a}{\mathcal{A}_i}$$

$$N_{B10} = \frac{(0.250000)(0.250445910)(6.02214199 \times 10^{23})}{10.0129370} \rightarrow 3.76568 \times 10^{21} \text{ atoms B10/cm}^3$$

$$N_{B11} = \frac{(0.750000)(0.250445910)(6.02214199 \times 10^{23})}{11.0093055} \rightarrow 1.02746 \times 10^{22} \text{ atoms B11/cm}^3$$

$$N_O = \frac{(1)(1.119053193)(6.02214199 \times 10^{23})}{15.9994} \rightarrow 4.21209 \times 10^{22} \text{ atoms O/cm}^3$$

Problem Set #1

Problem 4

Given a 9.000 eV photon

- (a) Determine the photon's wavelength, in nanometers.
- (b) Determine the photon's frequency, in Hertz.
- (c) Determine the photon's relativistic mass, in kilograms.

(a) Recall $\lambda = \frac{h}{p}$ and $p = \frac{E}{c}$; combine to get $\lambda = \frac{hc}{E}$

$$\lambda = \frac{(4.13566770 \times 10^{-21} \text{ MeV}\cdot\text{s})(2.99792458 \times 10^8 \text{ m/s})}{9 \times 10^{-6} \text{ MeV}} \longrightarrow 1.378 \times 10^{-7} \text{ m or } 137.8 \text{ nm}$$

(b) Definitionally, $c = \lambda\nu$ or $\nu = \frac{c}{\lambda}$

$$\nu = \frac{2.99792458 \times 10^8 \text{ m/s}}{1.378 \times 10^{-7}} \longrightarrow \nu = 2.176 \times 10^{15} \text{ Hz}$$

(c) The energy of a photon is $E = h\nu$, and mass-energy equivalence states that $E = mc^2$, so

$$m = \frac{h\nu}{c^2}$$

$$m = \frac{(6.62607015 \times 10^{-34} \text{ J}\cdot\text{s})(2.176 \times 10^{15} \text{ s}^{-1})}{(2.99792458 \times 10^8 \text{ m/s})^2}$$

$$m = 1.604 \times 10^{-35} \text{ kg}$$

Problem Set #1

Problem 5

Determine how fast a .39000 kg football must travel to result in a 1.0000% increase in mass. Report your answer as a percentage of the speed of light.

$$\text{Recall } m = \frac{m_0}{\sqrt{1 - \left(\frac{v}{c}\right)^2}}; \text{ Rearrange in terms of } v \rightarrow v = c \sqrt{-\left(\frac{m_0}{m}\right)^2 + 1}$$

$$v = c \sqrt{-\left(\frac{1}{1.01}\right)^2 + 1} \rightarrow v = 0.14037c \text{ or } 14.037\% \text{ the speed of light}$$



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PROBLEM SET #2

Problem Set #2

Problem 1

a) Using the liquid drop model, estimate the total nuclear binding energy per nucleon by quantifying each contributing term for ^{12}C , ^{56}Fe and ^{252}Cf .

b) Sketch the binding energy per nucleon curve from $A = 1$ to $A = 260$. Clearly indicate:

- The region where fusion is energetically favorable
- The region where fission is energetically favorable
- The approximate locations of ^{12}C , ^{56}Fe and ^{252}Cf

Recall $E_B = a_v A - a_s A^{2/3} - a_c \frac{Z^2}{A^{1/3}} - a_a \frac{(A - 2Z)^2}{A} - \frac{\pm a_p}{\sqrt{A}}$:

For ^{12}C , $A=12$, $Z=6$; for ^{56}Fe $A=56$ and $Z=26$; for ^{252}Cf $A=252$ and $Z=98$

Tabulation of each binding energy term for each nuclide can be found below:

Term Name	Equation	^{12}C	^{56}Fe	^{252}Cf
Volume Term	$a_v A$	190.02 MeV	886.76 MeV	3990.42 MeV
Surface Term	$a_s A^{2/3}$	96.08 MeV	268.30 MeV	731.30 MeV
Coulomb Term	$a_c \frac{Z^2}{A^{1/3}}$	11.23 MeV	126.16 MeV	1085.63 MeV
Asymmetry Term	$a_a \frac{(A-2Z)^2}{A}$	0 MeV	6.63 MeV	288.71 MeV
Pairing Term	$\frac{\pm a_p}{\sqrt{A}}$	-3.23 MeV	-1.50 MeV	-0.71 MeV
Total Binding Energy	E_B	85.95 MeV	487.17 MeV	1885.48 MeV
Binding Energy/Nucleon	B/A	7.1625 MeV	8.6995 MeV	7.4821 MeV

Problem Set #2

Problem 2

A nuclear scientist attempts to perform experiments on the stable nuclide ^{96}Mo . Determine the energy, in MeV, that the scientist will either need or get if they:

- (a) Remove a single neutron
- (b) Remove a single proton
- (c) Completely dismantle the nucleus into its individual nucleons
- (d) Fission it symmetrically into two nuclides of $^{48}_{21}\text{Sc}$

$$(a) S_n(^A_Z\text{X}) = [M(^{A-1}_Z\text{X}) + m_n - M(^A_Z\text{X})] c^2$$

$$S_n(^{96}_{42}\text{Mo}) = [M(^{95}_{42}\text{Mo}) + m_n - M(^{96}_{42}\text{Mo})] c^2$$

$$S_n(^{96}_{42}\text{Mo}) = [94.9058415 + 1.0086649233 - 95.9046789] c^2$$

$$S_n(^{96}_{42}\text{Mo}) = [0.009821808 \text{ u}] c^2 \cdot 931.494013 \frac{\text{MeV}/c^2}{\text{u}} \longrightarrow 9.154 \text{ MeV}$$

$$(b) S_p(^A_Z\text{Y}) = [M(^{A-1}_{Z-1}\text{X}) + M(^1_1\text{H}) - M(^A_Z\text{Y})] c^2$$

$$S_p(^{96}_{42}\text{Mo}) = [M(^{95}_{41}\text{Nb}) + M(^1_1\text{H}) - M(^{96}_{42}\text{Mo})] c^2$$

$$S_p(^{96}_{42}\text{Mo}) = [94.906835 + 1.0078250321 - 95.9046789] c^2$$

$$S_p(^{96}_{42}\text{Mo}) = [0.009981332 \text{ u}] c^2 \cdot 931.494013 \frac{\text{MeV}/c^2}{\text{u}} \longrightarrow 9.298 \text{ MeV}$$

$$(c) BE(^A_Z\text{X}) = [ZM(^1_1\text{H}) + (A - Z)m_n - M(^A_Z\text{X})] c^2$$

$$BE(^{96}_{42}\text{Mo}) = [(42)M(^1_1\text{H}) + (96 - 42)m_n - M(^{96}_{42}\text{Mo})] c^2$$

$$BE(^{96}_{42}\text{Mo}) = [(42)(1.0078250321) + (54)(1.0086649233) - 95.9046789] c^2$$

$$BE(^{96}_{42}\text{Mo}) = [0.891569655 \text{ u}] c^2 \cdot 931.494013 \frac{\text{MeV}/c^2}{\text{u}} \longrightarrow 830.78 \text{ MeV}$$

$$(d) F(^{96}_{42}\text{Mo}) = [(2)M(^{48}_{21}\text{Sc}) - M(^{96}_{42}\text{Mo})] c^2$$

$$F(^{96}_{42}\text{Mo}) = [(2)(47.952235) - (95.9046789)] c^2$$

$$F(^{96}_{42}\text{Mo}) = [-0.000208900 \text{ u}] c^2 \cdot 931.494013 \frac{\text{MeV}/c^2}{\text{u}} \longrightarrow -0.1946 \text{ MeV}$$

Problem Set #2

Problem 3

Calculate the Q -value for each of the following possible nuclear reactions and state whether each is exoergic, endoergic, or neither. Express your answers in MeV.

$${}^1_0\text{n} + {}^6_3\text{Li} \longrightarrow \begin{cases} {}^7_3\text{Li} \\ {}^6_2\text{He} + {}^1_1\text{p} \\ t + \alpha \\ {}^1_0\text{n} + {}^6_3\text{Li} \end{cases}$$

Begin with $Q = [\sum m_{0,i} - \sum m_{0,f}] c^2$ and then apply to each exit channel:

(a) ${}^1_0\text{n} + {}^6_3\text{Li} \longrightarrow {}^7_3\text{Li}$

$$Q = [1.0086649233 + 6.0151223] - [7.0160040]c^2$$

$$Q = [0.007778661 \text{ u}] c^2 * 931.494013 \frac{\text{MeV}/c^2}{\text{u}} \longrightarrow 7.2500 \text{ MeV (exoergic)}$$

(b) ${}^1_0\text{n} + {}^6_3\text{Li} \longrightarrow {}^6_2\text{He} + {}^1_1\text{p}$

$$Q = [1.0086649233 + 6.0151223] - [6.0188881 + 1.0078250321]c^2$$

$$Q = [-0.002928826 \text{ u}] c^2 * 931.494013 \frac{\text{MeV}/c^2}{\text{u}} \longrightarrow -2.7255 \text{ MeV (endoergic)}$$

(c) ${}^1_0\text{n} + {}^6_3\text{Li} \longrightarrow {}^3_1\text{H} + \alpha$

$$Q = [1.0086649233 + 6.0151223] - [3.0160492675 + 4.0026032497]c^2$$

$$Q = [0.005129560] c^2 * 931.494013 \frac{\text{MeV}/c^2}{\text{u}} \longrightarrow 4.7829 \text{ MeV (exoergic)}$$

(d) ${}^1_0\text{n} + {}^6_3\text{Li} \longrightarrow {}^1_0\text{n} + {}^6_3\text{Li}$

$$Q = [1.0086649233 + 6.0151223] - [1.0086649233 + 6.0151223]c^2$$

$$Q = [0.0] c^2 * 931.494013 \frac{\text{MeV}/c^2}{\text{u}} \longrightarrow 0 \text{ MeV (Neither) *Scattering*}$$

Problem Set #2

Problem 4

The average mass of potassium in a banana is about 0.435g. Using Table 5-2 from your text, estimate the activity, in Curies, of ^{40}K in the banana.

From Table 5-2 $\rightarrow T_{1/2} = 1.27 \times 10^9$ years, $\gamma_{40K} = 0.0117\%$

First, determine the mass of ^{40}K , which requires the conversion of the abundance value to a mass fraction

$$N_i = \frac{\gamma_i \rho_x N_a}{\mathcal{A}_x} = \frac{w_i \rho_x N_a}{\mathcal{A}_i} \rightarrow \frac{\gamma_i}{\mathcal{A}_x} = \frac{w_i}{\mathcal{A}_i} \rightarrow \frac{\gamma_{40K}}{\mathcal{A}_K} = \frac{w_{40K}}{\mathcal{A}_{40K}}$$

$$\frac{\gamma_{40K}}{\mathcal{A}_K} = \frac{w_{40K}}{\mathcal{A}_{40K}} \rightarrow \frac{0.000117}{39.095} = \frac{w_{40K}}{39.96399867} \rightarrow w_{40K} = 0.000119601$$

$$(0.435 \text{ g})(0.00011961) = 0.000052026 \text{ g of } ^{40}\text{K in the banana}$$

Next, the number of ^{40}K atoms must be determined

$$\text{Recall } N = \frac{mN_a}{\mathcal{A}} \rightarrow \frac{(0.000052026)(6.02214199 \times 10^{23})}{39.96399867} \rightarrow N_{40K} = 7.839798352 \times 10^{17} \text{ atoms}$$

Next, the decay constant can be found using the half-life

$$\text{Recall } T_{1/2} = \frac{\ln 2}{\lambda} \rightarrow \lambda = \frac{\ln 2}{4.005072 \times 10^{16} \text{ secs}} \text{ *Note- Half-Life needs to be in seconds*}$$

Finally, activity can now be found

$$\text{Recall } A = \lambda N \rightarrow A = (7.839798352 \times 10^{17})(1.730453968 \times 10^{-17} \text{secs}^{-1}) \rightarrow 13.566410 \text{ Bq}$$

$$13.566410 \text{ Bq} * \frac{1 \text{Ci}}{3.7 \times 10^{10} \text{ Bq}} \rightarrow 3.6666 \times 10^{-10} \text{ Ci}$$

Problem Set #2

Problem 5

A particular radionuclide decays by α -decay and emits an α -particle with energy 1.8519 MeV and leaves the daughter nucleus in the ground state. The Q-Value for this decay is 1.9049 MeV.

- (a) Determine the atomic mass of the daughter nuclide
- (b) Using your answer from Part (a), determine the most likely parent nuclide

(a) ${}^A_Z\text{P} \longrightarrow {}^{A-4}_{Z-2}\text{D} + \alpha$

$$\text{Recall } Q = T_1 \left(\frac{m_1}{m_4} - 1 \right) - \frac{2}{m_4} \sqrt{m_1 m_3 T_1 T_3} \cos \theta + T_3 \left(\frac{m_3}{m_4} + 1 \right)$$

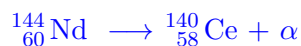
Assumption of the parent nuclide at rest ($T_1 = 0$) allows [19] to be reduced to the following:

$$Q = T_3 \left(\frac{m_3}{m_4} + 1 \right) \longrightarrow 1.9049 \text{ MeV} = (1.8519 \text{ MeV}) \left(\frac{4.0026032497}{m_4} + 1 \right)$$

Solving for m_4 yields 139.86 amu, the mass of the daughter nuclide

- (b) The mass from Part (a) is closest to 139.905 amu which is ${}^{140}\text{Ce}$

Applying this parent nuclide to the generic α -decay expression yields:





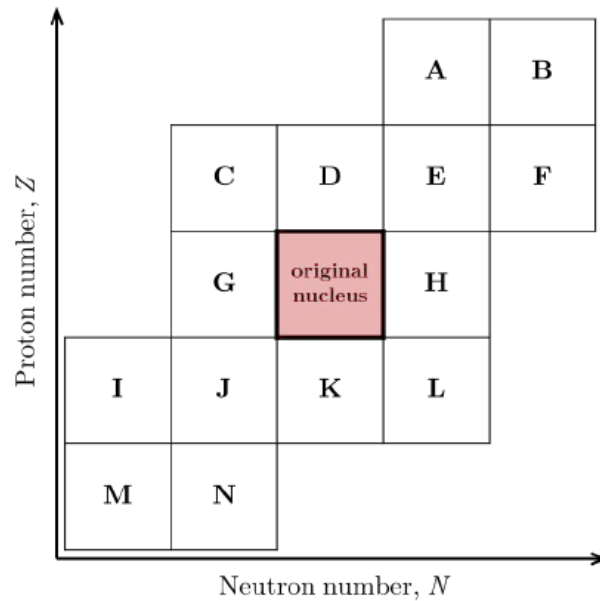
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PROBLEM SET #3

Problem Set #3

Problem 1 - 10 Points

Refer to the diagram from the Chart of the Nuclides below. For each decay process shown, fill in the boxes to complete the decay equation and then write the letter that corresponds to the position of the daughter in the blank provided.



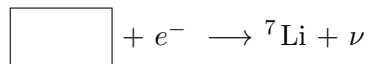
- (a) Bismuth-212 has two competing decay modes α : _____ and β^- : _____



- (b) Sodium-22 decays via β^+ decay: _____



- (c) This isotope decays via electron capture to produce Lithium-7: _____



Problem Set #3

Problem 2 - 12 Points

Consider the reaction ${}^7\text{Li}(\text{p},\text{n}){}^7\text{Be}$. This is one of the reactions used to produce quasi-monoenergetic neutrons for study.

- (a) Determine the Q-Value for this reaction
- (b) Calculate the overall threshold energy required of the proton for this reaction.
- (c) Determine the total minimum kinetic energy of the products



$$Q = [\sum m_{0,i} - \sum m_{0,f}] c^2$$

$$Q = [(7.0160040 + 1.0078250321) - (1.0086649233 + 7.0169292)] c^2$$

$$Q = [-0.0017650912 \text{ u}] c^2 \cdot 931.494013 \frac{\text{MeV}/c^2}{\text{u}} \longrightarrow -1.6442 \text{ MeV}$$

(b) Recall $E_1^K \approx -\left(1 + \frac{m_1}{m_2}\right) Q$; $m_1 = M[{}^1\text{p}]$, $m_2 = M[{}^7\text{Li}]$

$$E_1^K \cong -\left(1 + \frac{1.00727646688}{7.0160040}\right) (-1.6441718852) \longrightarrow 1.8804 \text{ MeV}$$

$$E_1^C \approx 1.2 \frac{Z_1 Z_2}{A_1^{1/3} + A_2^{1/3}}; Z_1 = 1, Z_2 = 3, A_1 = 1, A_2 = 7$$

$$E_1^C \cong 1.2 \frac{1 \cdot 3}{1^{1/3} + 7^{1/3}} \longrightarrow 1.2359 \text{ MeV}$$

$$(E_1^{th}) = \max(E_1^K, E_1^C)$$

$$(E_1^{th}) = \max(1.8804, 1.2359) \longrightarrow 1.8804 \text{ MeV}$$

(c) Modified from Example Problem 6.1 in Shultis and Faw $\longrightarrow \min(E_3 + E_4) = Q + E_1^{th}$

$$\min(E_3 + E_4) = (-1.644172073) + 1.880351781$$

$$\min(E_3 + E_4) = 0.23618 \text{ MeV}$$

Problem Set #3

Problem 3 - 12 Points

How many elastic scatters, on average, are required to slow a 2.0000 MeV neutron to below 1.0000 eV in:

(a) ${}^9\text{Be}$

(b) ${}^{208}\text{Pb}$

$$(a) \text{ Recall } \alpha = \left(\frac{A-1}{A+1} \right)^2 \rightarrow \alpha = \left(\frac{9-1}{9+1} \right)^2 \rightarrow \alpha = 0.64$$

$$\text{Recall } \xi = 1 + \frac{\alpha}{1-\alpha} \ln \alpha \rightarrow \xi = 1 + \frac{0.64}{1-0.64} \ln(0.64) \rightarrow 0.206600706$$

$$\text{Recall } n = \frac{1}{\xi} \ln \frac{E_1}{E_2} \rightarrow n = \frac{1}{0.206600706} \ln \left(\frac{2 \times 10^6}{1} \right) \rightarrow 70.226 \approx 71 \text{ scatters}$$

$$(b) \text{ Recall } \alpha = \left(\frac{A-1}{A+1} \right)^2 \rightarrow \alpha = \left(\frac{208-1}{208+1} \right)^2 \rightarrow \alpha = 0.980952817$$

$$\text{Recall } \xi = 1 + \frac{\alpha}{1-\alpha} \ln \alpha \rightarrow \xi = 1 + \frac{0.980952817}{1-0.980952817} \ln(0.980952817) \rightarrow 0.00958464$$

$$\text{Recall } n = \frac{1}{\xi} \ln \frac{E_1}{E_2} \rightarrow n = \frac{1}{0.00958464} \ln \left(\frac{2 \times 10^6}{1} \right) \rightarrow 1513.741 \approx 1514 \text{ scatters}$$

Problem Set #3

Problem 4 - 16 Points

In a particular neutron-induced fission of ^{235}U , 2 prompt neutrons are produced and one fission fragment is ^{140}Xe

- What is the other fission fragment? (Assuming only two are produced)
- How much energy is liberated promptly? (i.e before any fission fragments begin to decay)
- If the total initial kinetic energy of the fission fragments (neglecting neutrons) is 150 MeV, what is the initial kinetic energy of each fragment?
- What is the total kinetic energy shared by the 2 prompt neutrons?

$$(a) \quad {}^1_0\text{n} + {}^{235}_{92}\text{U} \longrightarrow {}^{236}_{92}\text{U}^* \longrightarrow (2){}^1_0\text{n} + {}^{140}_{54}\text{Xe} + {}^{94}_{38}\text{Sr}$$

$$(b) \quad Q = [\sum m_{0,i} - \sum m_{0,f}] c^2$$

$$Q = [(1.0086649233 + 235.0439231) - ((2)1.0086649233 + 139.921640 + 93.915360)] c^2$$

$$Q = [0.1982581767 \text{ u}] c^2 \cdot 931.494013 \frac{\text{MeV}/c^2}{\text{u}} \longrightarrow 184.676 \text{ MeV}$$

- Recall Eqn 6.41 from S&F, $\frac{E_L}{E_H} = \frac{m_H}{m_L}$, where L and H represent the light and heavy fission fragments respectively

Rearrangement of 6.41 and the definition of the total kinetic energy of the fission fragments as E_{ff} leads to the following:

$$E_H = (E_{ff} - E_L) \frac{m_L}{m_H} \longrightarrow \text{Replacement of generic variables leads to} \longrightarrow$$

$$E_{140\text{Xe}} = (E_{ff} - E_{94\text{Sr}}) \frac{M(^{94}\text{Sr})}{M(^{140}\text{Xe})} \longrightarrow \text{Further algebraic expansion leads to} \longrightarrow$$

$$E_{140\text{Xe}} = (E_{ff}) \frac{M(^{94}\text{Sr})}{M(^{140}\text{Xe}) + M(^{94}\text{Sr})} \longrightarrow E_{140\text{Xe}} = (150\text{MeV}) \frac{93.915360}{93.915360 + 139.921640}$$

$$E_{140\text{Xe}} = 60.244 \text{ MeV}$$

$$E_{94\text{Sr}} = (E_{ff}) \frac{M(^{140}\text{Xe})}{M(^{140}\text{Xe}) + M(^{94}\text{Sr})} \longrightarrow E_{94\text{Sr}} = (150\text{MeV}) \frac{139.921640}{93.915360 + 139.921640}$$

$$E_{94\text{Sr}} = 89.756 \text{ MeV}$$

- By COLM $E_N = Q - E_{ff} \longrightarrow E_N = 184.676304624 - 150 \longrightarrow E_N = 34.676 \text{ MeV}$



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PROBLEM SET #4

Problem Set #4

Problem 1

An Ar-41 source of unknown strength is placed under a hemispherical cloche comprised of magnesium oxide (MgO) with a uniform wall thickness of 1.5000 cm and a diameter of 8.0000 cm. Furthermore, the MgO cloche has a physical density of 5.8600 g/cm³ and a composition of 60.3041 w/o magnesium with the remainder being naturally occurring oxygen.

- Determine the remaining intensity, as a percentage, one would expect on the outer wall of the cloche. You can neglect any attenuation effects from air.
- Determine if increasing the diameter of the cloche or increasing its thickness will have a more appreciable attenuation effect, justify your answer.
- Determine the predominant photon interaction mechanism in this scenario, justify your answer.

- Recall the point source equation, $I(r) = \frac{S_p}{4\pi r^2} e^{-\mu_t x}$; from the problem statement we know $r = 5.5$ cm (4 cm + 1.5 cm), $x = 1.5$ cm, and $S_p = 1$. We now must determine our linear attenuation coefficient, μ_t

Recall $\mu_t = \left(\frac{\mu_t}{\rho}\right) \rho$ where $\rho = 5.8600$ g/cm³. Now we need to determine our mass attenuation coefficients, (μ_t/ρ)

Enter Table 2-1 of the NERD-P and apply linear interpolation (x2) to determine correct mass attenuation coefficients for each element. Note our photon energy was determined to be 1.29364 MeV per Tbl 1-3 of the NERD-P.

Two-point linear interpolation $\rightarrow y = y_1 + (x - x_1) \frac{y_2 - y_1}{x_2 - x_1}$

	Energy [MeV]		$(\mu_t/\rho)_{\text{Mg}}$ [cm ² /g]
x_1	1.25	y_1	0.0560
x	1.29364	y	0.055162112
x_2	1.50	y_2	0.0512

	Energy [MeV]		$(\mu_t/\rho)_{\text{O}}$ [cm ² /g]
x_1	1.25	y_1	0.0568
x	1.29364	y	0.0559272
x_2	1.50	y_2	0.0518

As we have multiple elements present we need to apply $\left(\frac{\mu_t}{\rho}\right) = \sum_j w_j \left(\frac{\mu_i}{\rho}\right)^j$ to determine our overall mass attenuation coefficient

$$\left(\frac{\mu_t}{\rho}\right) = \sum_i \left(\frac{\mu_i}{\rho}\right) \rightarrow w_{\text{Mg}} \left(\frac{\mu_t}{\rho}\right)_{\text{Mg}} + w_{\text{O}} \left(\frac{\mu_t}{\rho}\right)_{\text{O}} \rightarrow$$

$$(0.60304)(0.055162112 \text{ cm}^2/\text{g}) + (0.396959)(0.0559272 \text{ cm}^2/\text{g}) \rightarrow 0.055465821 \text{ cm}^2/\text{g}$$

Now we can determine the linear attenuation coefficient:

$$\mu_t = (0.055465821 \text{ cm}^2/\text{g})(5.86 \text{ g/cm}^3) \rightarrow 0.325029709 \text{ cm}^{-1}$$

Problem Set #4

$$I(5.5\text{cm}) = \frac{1}{4\pi(5.5\text{ cm})^2} e^{-(1.5\text{ cm})(0.325029709\text{ cm}^{-1})} \longrightarrow 0.00161557396 \text{ or } \approx 0.16\% \text{ remaining on the outer wall}$$

- (b) To determine which change will have a more consequential effect on the Intensity we need to calculate the geometric attenuation ($\frac{S_p}{4\pi r^2}$) and material attenuation ($e^{-\mu_t x}$) terms separately.

$$I(5.5\text{cm}) = \frac{1}{4\pi(5.5\text{ cm})^2} \longrightarrow 0.00263066022$$

$$e^{-(1.5\text{ cm})(0.3250293829\text{ cm}^{-1})} \longrightarrow 0.614132508$$

The lower of the two numbers indicates the more consequential effect, i.e. the distance of the source from the detector is far more impactful on the measured activity than the shield itself is.

- (c) The energy of the Ar-41 γ -emission is 1.29364 MeV and the effective shielding Z is 10 making **Compton Scattering** the predominate interaction

Problem Set #4

Problem 2

A neutron flux of 3.14×10^{15} neutrons/cm²/s emitted from a beam port at the NIST research reactor is directed at 0.50000 cm thick sample of cadmium nitrate glass ($\text{CdN}_2\text{O}_{10}\text{H}_8$) with a density of 3.6500 g/cm³. The ambient temperature can be assumed to be 100°C and the elemental disposition of cadmium nitrate can be found below:

Element	Weight Fraction (%)
Hydrogen (H)	2.6140
Nitrogen (N)	9.0810
Oxygen (O)	51.864
Cadmium (Cd)	36.441

- Determine the neutron flux, in neutrons/cm²/s, immediately after the cadmium nitrate glass sample.
- Determine which element has the most impact on neutron attenuation, justify your answer.

- Recall $\phi(x) = \phi_0 e^{-\Sigma_t x}$; $\phi_0 = 3.14 \times 10^{15}$ neutrons/cm²/s and we are given x , 0.50000 cm, so have to determine Σ_t in order to calculate $\phi(0.50 \text{ cm})$

Recall $\Sigma_t = \Sigma_s + \Sigma_a$ where Σ_s and Σ_a can be defined as $\Sigma_i = N\sigma_i(E)g_i(T)$. Expansion of both interactions (scattering and absorption) for each element present (H, N, O and Cd) leads to the following:

$$\Sigma_t = N_H\sigma_{s,H}(0.0253 \text{ eV})g_{s,H}(100^\circ) + N_N\sigma_{s,N}(0.0253 \text{ eV})g_{s,N}(100^\circ) + N_O\sigma_{s,O}(0.0253 \text{ eV})g_{s,O}(100^\circ) + N_{Cd}\sigma_{s,Cd}(0.0253 \text{ eV})g_{s,Cd}(100^\circ) + N_H\sigma_{a,H}(0.0253 \text{ eV})g_{a,H}(100^\circ) + N_N\sigma_{a,N}(0.0253 \text{ eV})g_{a,N}(100^\circ) + N_O\sigma_{a,O}(0.0253 \text{ eV})g_{a,O}(100^\circ) + N_{Cd}\sigma_{a,Cd}(0.0253 \text{ eV})g_{a,Cd}(100^\circ)$$

From Tbl 2.7 of the NERDP;

Element	$\sigma_s(0.0253 \text{ eV})$ [b]	$\sigma_a(0.0253 \text{ eV})$ [b]	$g_s(100^\circ)$ [-]	$g_a(100^\circ)$ [-]
Hydrogen (H)	0	0.332	1	1
Nitrogen (N)	10.6	1.85	1	1
Oxygen (O)	3.76	0.00027	1	1
Cadmium (Cd)	5.6	2450	1	1.5990

The equation above can now be reduced to the following

$$\Sigma_t = N_H\sigma_{a,H} + N_N(\sigma_{s,N} + \sigma_{a,N}) + N_O(\sigma_{s,O} + \sigma_{a,O}) + N_{Cd}(\sigma_{s,Cd} + \sigma_{a,Cd}g_a)$$

Now the number densities must be determined (N_H, N_N, N_O, N_{Cd})

Problem Set #4

Assuming natural atomic compositions, we can calculate the number densities directly using the equations below. Note, atomic masses were determined from Tbl 2.7 values ($\mathcal{A}_H = 1.00797$ g/mol, $\mathcal{A}_N = 14.0067$ g/mol, $\mathcal{A}_O = 15.9994$ g/mol, $\mathcal{A}_{Cd} = 112.40$ g/mol)

$$N_H = \frac{w_H \rho N_A}{\mathcal{A}_H} \longrightarrow \frac{(0.026140)(3.65)(6.02214199 \times 10^{23})}{1.00797} \longrightarrow 5.7004 \times 10^{22} \text{ atoms H/cm}^3$$

$$N_N = \frac{w_N \rho N_A}{\mathcal{A}_N} \longrightarrow \frac{(0.090810)(3.65)(6.02214199 \times 10^{23})}{14.0067} \longrightarrow 1.4251 \times 10^{22} \text{ atoms N/cm}^3$$

$$N_O = \frac{w_O \rho N_A}{\mathcal{A}_O} \longrightarrow \frac{(0.51864)(3.65)(6.02214199 \times 10^{23})}{15.9994} \longrightarrow 7.1253 \times 10^{22} \text{ atoms O/cm}^3$$

$$N_{Cd} = \frac{w_{Cd} \rho N_A}{\mathcal{A}_{Cd}} \longrightarrow \frac{(0.36441)(3.65)(6.02214199 \times 10^{23})}{112.40} \longrightarrow 7.1264 \times 10^{21} \text{ atoms Cd/cm}^3$$

Now can calculate the effective macroscopic cross section, Σ_t . Note all micorscopic cross sections need to be converted from barns to cm^2 to arrive at the desired final units of cm^{-1}

$$\Sigma_t = (5.700354072 \times 10^{22})(0.332 \times 10^{-24}) + (1.42508807 \times 10^{22})(12.45 \times 10^{-24}) + (7.125349441 \times 10^{22})(3.76027 \times 10^{-24}) + (7.126361195 \times 10^{21})[(5.6 \times 10^{-24}) + (2450 \times 10^{-24})(1.5990)]$$

$$\Sigma_t = 28.42206494 \text{ cm}^{-1}$$

With the effective macroscopic cross section, we can now solve for the fraction of neutrons able to traverse the glass

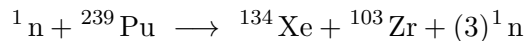
$$\phi(x) = \phi_0 e^{-\Sigma_t x} \longrightarrow (3.14 \times 10^{15}) e^{-(28.42206494 \text{ cm}^{-1})(0.5 \text{ cm})} \longrightarrow 2.114251581 \times 10^9 \text{ neutrons/cm}^2/\text{s}$$

- (b) Comparison of the elemental macroscopic cross sections shows that the cadmium cross section is over 100x greater than any other elements cross section, making it the most impactful on neutron attenuation.

Problem Set #4

Problem 3

Estimate the range, in cm, of the fission fragments, in air, produced by the fission reaction outlined below. You can assume 15.000% of available kinetic energy is carried away by the free neutrons.



Enter Equation 7.48 from textbook, $\rho R = CE^{2/3}$, where $C=0.14$ for a fission fragment in air

Need E for ^{134}Xe and ^{103}Zr

$$Q = [\sum m_{0,i} - \sum m_{0,f}] c^2$$

$$Q = [(1.0086649233 + 239.0521565) - ((3)1.0086649233 + 133.9053945 + 102.926600)] c^2$$

$$Q = [0.202832153 \text{ u}] c^2 \cdot 931.494013 \frac{\text{MeV}/c^2}{\text{u}} \longrightarrow 188.93693653598 \text{ MeV}$$

Recall Eqn 6.41 from S&F, $\frac{E_L}{E_H} = \frac{m_H}{m_L}$, where L and H represent the light and heavy fission fragments respectively

Rearrangement of 6.41 and the application of Q value, with 15% reduction, leads to the following:

$$E_H = (.85Q) \frac{m_L}{m_H} \longrightarrow \text{Replacement of generic variables leads to} \longrightarrow$$

$$E_{^{134}\text{Xe}} = (.85Q - E_{^{103}\text{Zr}}) \frac{M(^{103}\text{Zr})}{M(^{134}\text{Xe})} \longrightarrow \text{Further algebraic expansion leads to} \longrightarrow$$

$$E_{^{134}\text{Xe}} = (.85Q) \frac{M(^{103}\text{Zr})}{M(^{134}\text{Xe}) + M(^{103}\text{Zr})} \longrightarrow E_{^{134}\text{Xe}} = (160.59639 \text{ MeV}) \frac{102.926600}{102.926600 + 133.9053945}$$

$$E_{^{134}\text{Xe}} = 69.794797 \text{ MeV}$$

$$E_{^{103}\text{Zr}} = (.85Q) \frac{M(^{134}\text{Xe})}{M(^{134}\text{Xe}) + M(^{103}\text{Zr})} \longrightarrow E_{^{103}\text{Zr}} = (160.59639 \text{ MeV}) \frac{133.9053945}{133.9053945 + 102.926600}$$

$$E_{^{103}\text{Zr}} = 90.80159 \text{ MeV}$$

$$R_{^{103}\text{Zr}} = \frac{CE^{2/3}}{\rho} \longrightarrow \frac{(0.14)(90.80159 \text{ MeV})^{2/3}}{1.20479 \text{ mg/cm}^3} \longrightarrow 2.34753 \text{ cm}$$

$$R_{^{134}\text{Xe}} = \frac{CE^{2/3}}{\rho} \longrightarrow \frac{(0.14)(69.794797 \text{ MeV})^{2/3}}{1.20479 \text{ mg/cm}^3} \longrightarrow 1.96984 \text{ cm}$$



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PROBLEM SET #5

Problem Set #5

Problem 1

An experimental thermal reactor is being constructed of a homogeneous mixture of uranium and reactor grade graphite (i.e. carbon) with a physical density of 10.58 g/cm^3 . Designers are forecasting a uranium composition of 18.3478 w/o ^{235}U . Furthermore, this reactor is a 4.0000 m cuboid with an operating temperature of 20°C . Finally, with these materials, the resonance escape probability and fast fission factor can be estimated at 1.0800 collectively.

- Determine the number density of ^{235}U , ^{238}U , and the graphite.
- Calculate the thermal utilization factor for this geometry.
- Calculate the thermal fission factor for this geometry.
- Calculate the effective multiplication factor for this geometry and determine if the reactor is supercritical, critical, or subcritical
- Determine the mass of ^{235}U required for this reactor.
- List two options available to reactor designers if they want to reduce the effective multiplication factor and explain how these options affect criticality.

$$(a) \quad 81.6522 \text{ w/o } ^{238}\text{U} \rightarrow w_{U238} = 0.816522 \text{ and } 18.3478 \text{ w/o } ^{235}\text{U} \rightarrow w_{U235} = 0.183478$$

$$\mathcal{A}_x = \left[\sum \frac{w_i}{\mathcal{A}_i} \right]^{-1} \rightarrow \mathcal{A}_U = \left[\frac{0.816522}{238.0507826} + \frac{0.183478}{235.0439231} \right]^{-1} \rightarrow 237.4933408$$

$$w_x = \frac{n\mathcal{A}_x}{n\mathcal{A}_x + m\mathcal{A}_y} \text{ where } n = 1, m = 1$$

$$w_U = \frac{(1)(237.4933408)}{(1)(237.4933408) + (1)(12.01115)} \rightarrow 0.9518599847$$

$$\text{Recall } \rho_i = w_i \rho_x$$

$$\rho_U = 0.9518599847 (10.58) \rightarrow 10.07067864 \text{ g U/cm}^3$$

$$\text{Recall } N_i = \frac{w_i \rho_x N_a}{\mathcal{A}_i}$$

$$N_{U235} = \frac{(0.183478)(10.07067864)(6.02214199 \times 10^{23})}{235.0439231} \rightarrow 4.734179265 \times 10^{21} \text{ atoms U235/cm}^3$$

$$N_{U238} = \frac{(0.816522)(10.07067864)(6.02214199 \times 10^{23})}{238.0507826} \rightarrow 2.080213957 \times 10^{22} \text{ atoms U238/cm}^3$$

$$N_C = N_{U235} + N_{U238} \rightarrow 2.553631883 \times 10^{22} \text{ atoms C/cm}^3$$

$$(b) \quad \text{The thermal utilization factor, } f \text{ can be determined via } f = \frac{\overline{\Sigma}_a^F}{\overline{\Sigma}_a^F + \overline{\Sigma}_a^{NF}}.$$

Problem Set #5

Expansion of this expression using $\Sigma_i = N\sigma_i(E)g_i(T)$ leads to:

$$\frac{N^{U235}\sigma_a^{U235}(0.0253 \text{ eV})g_a^{U235}(20 \text{ C}) + N^{U238}\sigma_a^{U238}(0.0253 \text{ eV})g_a^{U238}(20 \text{ C})}{N^{U235}\sigma_a^{U235}(0.0253 \text{ eV})g_a^{U235}(20 \text{ C}) + N^{U238}\sigma_a^{U238}(0.0253 \text{ eV})g_a^{U238}(20 \text{ C}) + N^C\sigma_a^C}$$

From Tbl 2.6/2.7 of the NERDP;

Isotope/Element	$\sigma_a(0.0253 \text{ eV})$ [b]	$g_a(20^\circ)$ [-]
²³⁵ U	687	0.9780
²³⁸ U	2.73	1.0017
Carbon	0.0034	1

$$f = \frac{(4.734179265 \times 10^{21})(687 \times 10^{-24})(0.9780) + (2.080213957 \times 10^{22})(2.73 \times 10^{-24})(1.0017)}{+ (2.553631883 \times 10^{22})(0.0034 \times 10^{-24})}$$

$$f = 0.999973184436648$$

- (c) The thermal fission factor, η , can be determined by $\eta = \nu \frac{\bar{\Sigma}_f^F}{\bar{\Sigma}_a^F}$

Expansion of this expression using $\Sigma_i = N\sigma_i(E)g_i(T)$ leads to:

$$\nu^{U235} \frac{N^{U235}\sigma_f^{U235}(0.0253 \text{ eV})g_f^{U235}(20 \text{ C})}{N^{U235}\sigma_a^{U235}(0.0253 \text{ eV})g_a^{U235}(20 \text{ C}) + N^{U238}\sigma_a^{U238}(0.0253 \text{ eV})g_a^{U238}(20 \text{ C})} \text{ Per Table 2.4/2.6/2.7 of the NERD-P}$$

Isotope/Element	$\sigma_f(0.0253 \text{ eV})$ [b]	$g_f(20^\circ)$ [-]	ν
²³⁵ U	587	0.9759	2.436

$$\eta = (2.436) \frac{(4.734179265 \times 10^{21})(587 \times 10^{-24})(0.9759)}{(4.734179265 \times 10^{21})(687 \times 10^{-24})(0.9780) + (2.080213957 \times 10^{22})(2.73 \times 10^{-24})(1.0017)}$$

$$\eta = 2.04045$$

- (d) The fast non-leakage probability can be calculated via $P_{NL}^f = e^{-B_c^2 \tau_T}$ where τ_T is 368 cm² per Tbl 3.2 of the NERD-P. B_c^2 has to be calculated using Table 3-5

For a cuboid, $B_c^2 = (\frac{\pi}{a})^2 + (\frac{\pi}{b})^2 + (\frac{\pi}{c})^2$ where a, b, c are equal to cube's length, width, and height

$$B_c^2 = (\frac{\pi}{400})^2 + (\frac{\pi}{400})^2 + (\frac{\pi}{400})^2 \rightarrow 0.000185055 \text{ cm}^2$$

$$P_{NL}^f = e^{-(0.000185055 \text{ cm}^{-2})(368 \text{ cm}^2)} \rightarrow 0.9341667996$$

Problem Set #5

The thermal non-leakage probability can be determined via $P_{NL}^{th} = \frac{1}{1 + L_T^2 B_C^2}$ where $L_T^2 = 3500 \text{ cm}^2$ per Table 3.1 of the NERD-P (Note: the same buckling expression was also applied to this term)

$$P_{NL}^{th} = \frac{1}{1 + (3500 \text{ cm}^2)(0.000185055 \text{ cm}^{-2})} \rightarrow 0.606909253$$

Recall $k_{eff} = \epsilon P_{NL}^f p P_{NL}^{th} f \eta$. Combination of the two non-leakage terms (P_{NL}^{th}, P_{NL}^f) above with previously calculated η, f values yields

$$k_{eff} = (1.08)(0.9341667996)(0.606909253)(0.999973184436648)(2.04045)$$

$$k_{eff} = 1.249358458 \text{ (Supercritical)}$$

(e) Recall the volume of a cuboid, $V_{cube} = l \times w \times h$

$$V_{cube} = (400 \text{ cm})^3 \rightarrow 6.4 \times 10^7 \text{ cm}^3$$

Multiplication of this volume by the calculated ^{235}U number density yields the total number of ^{235}U atoms

$$(4.734179265 \times 10^{21} \text{ atoms/cm}^3)(6.4 \times 10^7 \text{ cm}^3) \rightarrow 3.029874729 \times 10^{29} \text{ atoms of } ^{235}\text{U}$$

$$\text{Recall } N = \frac{m N_a}{A} \rightarrow m = \frac{N_{U235} A}{N_a} \rightarrow \frac{(3.029874729 \times 10^{29})(235.0439231)}{6.02214199 \times 10^{23}}$$

$$m = 1.182558704 \times 10^8 \text{ grams of U-235}$$

- (f) 1) Change the geometry - Reducing the surface area to volume ratio or altering the shape, to an infinite plane for instance, would lower the non-leakage terms and thus reduce the effective multiplication factors
 2) Change the moderator - Changing the moderator material to light or heavy water would alter the material buckling constants (i.e. Fermi Age and Diffusion Area) thus reducing the multiplication factor
 3) Reduce the fuel concentration - Reducing the ratio of fissile U-235 to U-238 would lower the fuel utilization factor



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PROBLEM SET #6

Problem Set #6

Problem 1

A spherical tank located in a university laboratory is filled with a solution of heavy water and ^{239}Pu . The ratio of heavy water molecules to ^{239}Pu atoms is 1600:1. The resonance escape probability and fast fission factor can be estimated at 1.0800 collectively. Thermal leakage is negligible. What is the maximum safe radius of this sphere?

- a. 'Safe' can be characterized by a subcritical configuration (i.e. $k_{eff} < 1$)

Recall $k_{eff} = \epsilon P_{NL}^f p P_{NL}^{th} f \eta$, we now need to determine each of six-factors to calculate our effective multiplication factor

- **ϵ Determination** - Given as 1.08

- **f Determination** Recall $f = \frac{\bar{\Sigma}_a^F}{\bar{\Sigma}_a^F + \bar{\Sigma}_a^{NF}}$. Expansion of this expression using $\Sigma_i = N\sigma_i(E)g_i(T)$ leads to $\rightarrow \frac{N^{Pu239}\sigma_a^{Pu239}(0.0253 \text{ eV})g_a^{Pu239}(20 \text{ C})}{N^{Pu239}\sigma_a^{Pu239}(0.0253 \text{ eV})g_a^{Pu239}(20 \text{ C}) + N^{D2O}\sigma_a^{D2O}}$

Note, Assumption of naturally occurring non-fuel elements and room temperature conditions We now need to simplify the number density expressions by dividing the numerator and denominator by N^{Pu239} leading to $\rightarrow$

$$\frac{\sigma_a^{Pu239}(0.0253 \text{ eV})g_a^{Pu239}(20 \text{ C})}{\sigma_a^{Pu239}(0.0253 \text{ eV})g_a^{Pu239}(20 \text{ C}) + \frac{N^{Pu239}}{N^{D2O}}\sigma_a^{D2O}}$$

Per Table 2.6/2.7 of the NERD-P; $\sigma_a^{Pu239} = 1020 \text{ b}$, $\sigma_a^{D2O} = 0.00133 \text{ b}$, and $g_a^{Pu239}(20 \text{ C}) = 1.0723$

$$f = \frac{(1020)(1.0723)}{(1020)(1.0723) + (1600)(0.00133)} \rightarrow 0.998058171$$

- **η Determination** The thermal fission factor, η , can be determined by $\eta = \nu \frac{\bar{\Sigma}_f^F}{\bar{\Sigma}_a^F}$

Expansion of this expression using $\Sigma_i = N\sigma_i(E)g_i(T)$ leads to:

$$\nu^{Pu239} \frac{N^{Pu239}\sigma_f^{Pu239}(0.0253 \text{ eV})g_f^{Pu239}(20 \text{ C})}{N^{Pu239}\sigma_a^{Pu239}(0.0253 \text{ eV})g_a^{Pu239}(20 \text{ C})} \rightarrow \nu^{Pu239} \frac{\sigma_f^{Pu239}(0.0253 \text{ eV})g_f^{Pu239}(20 \text{ C})}{\sigma_a^{Pu239}(0.0253 \text{ eV})g_a^{Pu239}(20 \text{ C})}$$

Per Table 2.4/2.6/2.7 of the NERD-P

Isotope/Element	$\sigma_f(0.0253 \text{ eV})$ [b]	$g_f(20^\circ)$ [-]	ν
^{239}Pu	749	1.0487	2.884

$$\eta = (2.884) \frac{(749)(1.0487)}{(1020)(1.0723)} \rightarrow 2.07115$$

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- P_{NL}^f **Determination** The fast non-leakage factor, P_{NL}^f , can be determined by $P_{NL}^f = e^{-B_C^2 \tau_T}$

Per Table 3.2/3.5 of the NERD-P

Medium	τ_T [cm ²]	B_T^2 [cm ⁻²]
D ₂ O	131	$(\frac{\pi}{R})^2$

$$P_{NL}^f = e^{-(\frac{\pi}{R})^2 * 131}$$

$$k_{eff} = (1.08)(.998058171)(2.07115)(e^{-(\frac{\pi}{R})^2 * 131})$$

Where $k_{eff} = 1$ for critical configuration. Solving for R leads to R = 40.0123 cm

Problem Set #6

Problem 2

List and describe each of the major processes in the once through fuel cycle. Ensure you identify the chemical forms of uranium at each step.

1. Mining - Removal of the uranium from the ground by various mining methods
2. Milling - Removal of the uranium element from the ore and production of 'yellow cake' (U_3O_8)
3. Conversion - Change of yellow cake into green salt (UF_6) to facilitate follow on enrichment activities
4. Enrichment - Increase of U-235 concentration in uranium by any number of enrichment processes
5. Fuel Fabrication - Conversion of enriched green salt into UO_2 for use as nuclear fuel
6. Spent Fuel Storage - The 'interim' storage of used nuclear fuel until such a time as it can be reprocessed or stored permanently.

Problem Set #6

Problem 3

Sketch a thermal, light-water pressurized water reactor (PWR) and boiling water reactor (BWR). Ensure you label the following components, as appropriate: Primary loop, secondary loop, condenser, heat exchanger, pressurizer, recirculation pumps, control rods, core, fuel, reactor pressure vessel, and steam separators.



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TAB L PRACTICAL APPLICATIONS PRESENTATION

NE300: FUNDAMENTALS OF NUCLEAR ENGINEERING						
Practical Applications Group Lesson (AY 25-2)						
Lesson Topic:		Cadet Names:			Date:	
Content: (90 Pts)						
Physics (50)		Poor (0 - 60%)	Marginal (72%)	Good (85%)	Excellent (94 - 100%)	Score
Content Depth (20)	Depth / Accuracy in presentation and discussion of Physics which govern the application	- Little or no discussion of the Physics concepts and how they govern the application - Significant amount of incorrect information or demonstrates poor understanding of concepts	- Includes only surface level discussion of Physics concepts - Noticeable amount of incorrect or overly simplified information	- Provides reasonable depth on core Physics concepts involved - Little to no incorrect information	- Demonstrates a thorough understanding of many of the Physics concepts involved - All information is correct and scientifically credible	
Concept Identification (10)	Clearly identifies core Physics at play	- Fails to identify any of 5 major concepts: Decay, Binary Reactions, Particle Interactions, Mass / Energy, Reactor Core Physics	- Poor or incorrect identification of core Physics topics	- Correctly IDs major Physics topics but offers little elaboration on how it applies	- Clearly identifies all major concepts and articulates how each is involved	
Conceptual Understanding (20)	Group demonstrates a consistent foundational knowledge base of the base topic	- Some members appear to lack understanding of topic or offer only superficial answers to reasonable questions	- All members of group appear to have a basic understanding of the topic as a whole, but one or two members carry the bulk of the lesson		- All group members demonstrate a consistent baseline level of understanding and share evenly in content delivery and answering questions	
Notes:						
Relevance (20)		Poor (0 - 60%)	Marginal (72%)	Good (85%)	Excellent (94 - 100%)	Score
Societal Relevance (10)	Clearly articulates the relevance of the application to society	- Fails to discuss how the NE Application contributes to advancement of society as a whole	- Indicates the ways this application contributes to society but fails to elaborate or capture its significance	- Identifies relevance to society and offers some discussion on how it contributes to progress	- Clearly articulates the relevance & importance - Offers perspective on how the application fits, historically, with similar or competing ideas	
NE300 Course Relevance (10)	References specific NE300 course content which applies	- Fails to link to specific course concepts or merely lists those which apply	- Mentions sub-topics from NE300 which apply but offers only surface level connection	- Discusses specific NE300 sub-topics in depth and identifies individual lessons which contribute to topic	- Detailed discussion of NE300 sub-topics and lessons which apply - Clearly connects the Physics of the application to course material	
Notes:						
Research (20)		Poor (0 - 60%)	Acceptable (80 %)		Excellent (90 - 100%)	Score
Sources (10)	Indicates thorough research using reputable, academic sources	- Indicates the lesson is sparsely researched - Multiple non-reputable or superficial sources used	- Lesson is sufficiently researched using multiple reputable resources - Lacks high quality academic style sources or relies solely on information compiled from various websites		- Lesson is thoroughly researched using a broad variety of credible resources, including academic writings	
Citation (10)	Properly credits sources and corresponding information used	- Lists sources only	- Lists sources used and generally indicates which material or ideas were derived from each		- Detailed documentation on any/all products and clearly identifies how specific information was sourced	
Notes:						Subtotal:
Delivery: (20 Pts)						
Content Delivery (20)		Poor (0 - 60%)	Acceptable (80 %)		Excellent (90 - 100%)	Score
Group Collaboration (10)	Indicates a consolidated group effort from development through execution	- Preparation and delivery appear disjointed as if broken into individually assigned pieces - Indicates minimal or no rehearsal time as a group	- Preparation and delivery indicate that group members are generally on the same page - Generally appears to be a consolidated effort with significant portions of the work done collectively - Demonstrates sufficient group rehearsal time to deliver a seamless product		- Preparation and delivery indicate a single, cohesive group effort with consistent contribution from all members in all phases - Demonstrates clear, extensive individual and group rehearsal time	
Content Quality (10)	Content delivery and any supplemental products demonstrate professional quality	- Relies solely on a single product or delivery method	- Incorporates a variety of content delivery methods to keep the lesson engaging and approachable		- Exhibits a high level of creativity and professional quality in lesson planning and content delivery	
Notes:						Subtotal:
Preparation / Time Management: (10 Pts)						
Preparation / Time Management (10)		Poor (0%)	Marginal (50%)	Acceptable (80%)	Excellent (100%)	Score
Time Management (10)	Lesson meets 15-minute time target for content	- Lesson contains < 8 mins of content on topic - Indicates minimal or no rehearsal time as a group	- Lesson contains < 10 mins of content on topic - Group preparation and rehearsal time is lacking	- Lesson contains < 13 mins of content on topic - Demonstrates sufficient group rehearsal time to deliver a seamless product	- Lesson contains 15 (+/- 2) mins of content on topic - Demonstrates clear, extensive individual and group rehearsal time	
Notes:						Subtotal:
Instructor Feedback:					Total Points:	
					Group Score:	

Historical Figures and Advancement:

- Henri Becquerel
- Leo Szilard
- Radium Girls
- Otto Hahn & Lise Meitner
- Marie and Pierre Curie
- Ernest and John Lawrence
- Ernest Rutherford
- James Chadwick
- J. J. Thomson
- Enrico Fermi
- Robert Oppenheimer
- Chicago Pile 1
- Manhattan Project
- West Point Silver Repository
- Quantum Chromodynamics
- The Standard Model
- Role in the Grand Unified Theory

History and Viability:

- Chicago Pile 1
- Modern Fission Reactors
- Viability of Fusion Power
- ITER Fusion Reactor

- Shoreham Nuclear Power Plant

Reactor Types and Models:

- Boiling Water Reactors
- Pressurized Water Reactors
- AP-1000 Reactors
- RBMK Reactor Design
- Liquid Metal Fast Breeding Reactors
- Experimental Breeder Reactors
- Molten Salt Reactors
- Heavy Water Reactors (CANDU)
- Gas Cooled Reactors
- GEN III & IV Reactor Designs

Nuclear Fuel Types and Cycle:

- Nuclear Fuel Sources (Fission and Breeding)
- Nuclear Fuel Cycle
- Enrichment Methods and Levels
- Conversion and Breeding
- Y-12, K-25, and S-50 Plants
- Spent Nuclear Fuel and Waste Management
- Yucca Mountain Repository
- Tritium Production for Defense Applications

Diagnostic Imaging and Tracing:

- X-Rays

- Mammography
- Bone Scans
- Positron Emission Tomography
- Magnetic Resonance Imaging (MRI)
- Diagnostic Radiotracers
- Radioimmunoassay

Radiation Therapy and Treatment:

- Proton / Electron Beam Therapy
- Brachytherapy
- Boron Neutron Capture Therapy
- 3D Conformal Radiation Therapy
- Heavy Charged Particle Therapy

Radiation Doses, Hazards, and Protection:

- Measuring and monitoring dose
- Short- and long-term health effects
- Sources of dangerous radiation
- Radiation exposure mitigation and protection measures

Nuclear Accidents and Effects:

- Chernobyl meltdown
- Three-mile Island
- Fukushima Daiichi
- Louis Slotin Criticality Accident
- SL-1 Explosion

- Windscale fire

Military Applications:

- Radiation Detection (Robotics, Drones, Cargo-Scans, etc)
- Military Radiation Detection Methods
- Dual-axis Radiographic Hydrodynamic Test (DAHRT) Facility
- Stockpile Stewardship

Propulsion and Conversion Devices:

- Naval Propulsion
- Thermal & Ion Propulsion
- Multi-mission Radioisotope Thermoelectric Generators (MMRTGs)
- Space Reactor Program (U.S. and Russia)

Agencies and Policy:

- Defense Threat Reduction Agency
- Nuclear Regulatory Commission
- Department of Energy
- United Nations Security Council
- Joint Comprehensive Plan of Action
- Comprehensive Nuclear-Test-Ban Treaty

Industry and Research:

- Radiotracer Applications in Engineering
- Material Effects on Radiation
- Radiation Effects on Matter
- Food Irradiation

- Radiometric dating
- Research Reactors
- Particle Accelerators

Applications Involving Neutrons:

- Discovery of the Neutron (Chadwick)
- Neutron Spectroscopy
- Ultra-Cold Neutrons
- Spallation Neutron Source
- Neutron Activation Analysis

Nuclear Policy:

- Nuclear material/fuel recycling
- Nuclear domestic disposal
- Current obstacles to new reactor construction

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